



GE Medical Systems

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GE Medical Systems **LightSpeed 3.X Installation Manual**

Chapters 4 to End **(Electrical Installation Procedures)**

The information in this service manual applies to the following
LightSpeed 3.X CT systems:

- LightSpeed Ultra (8-slice MDAS)
- LightSpeed Plus (4-slice MDAS)
- LightSpeed QX/i (4-slice MDAS)

***Retain this installation manual permanently with your CT
System. Any other revisions of this installation manual
may not exactly match your system.***

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Chapter 4

Electrical Introduction



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 14](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Introduction

Use the continuity and ground checks to verify the system power connections have not shorted to ground and that the ground and neutral connections are intact.

Section 2.0 Review Mechanical Hand Off Material

Complete the Mechanical Hand Off checklist:

- ☐ All options were installed. If not, contact your install specialist.
- ☐ Check for short ships.
- ☐ Review cable connections with mechanical team.
- ☐ Complete mechanical vendor evaluation.

Section 3.0 Electrical Procedures Block Diagram

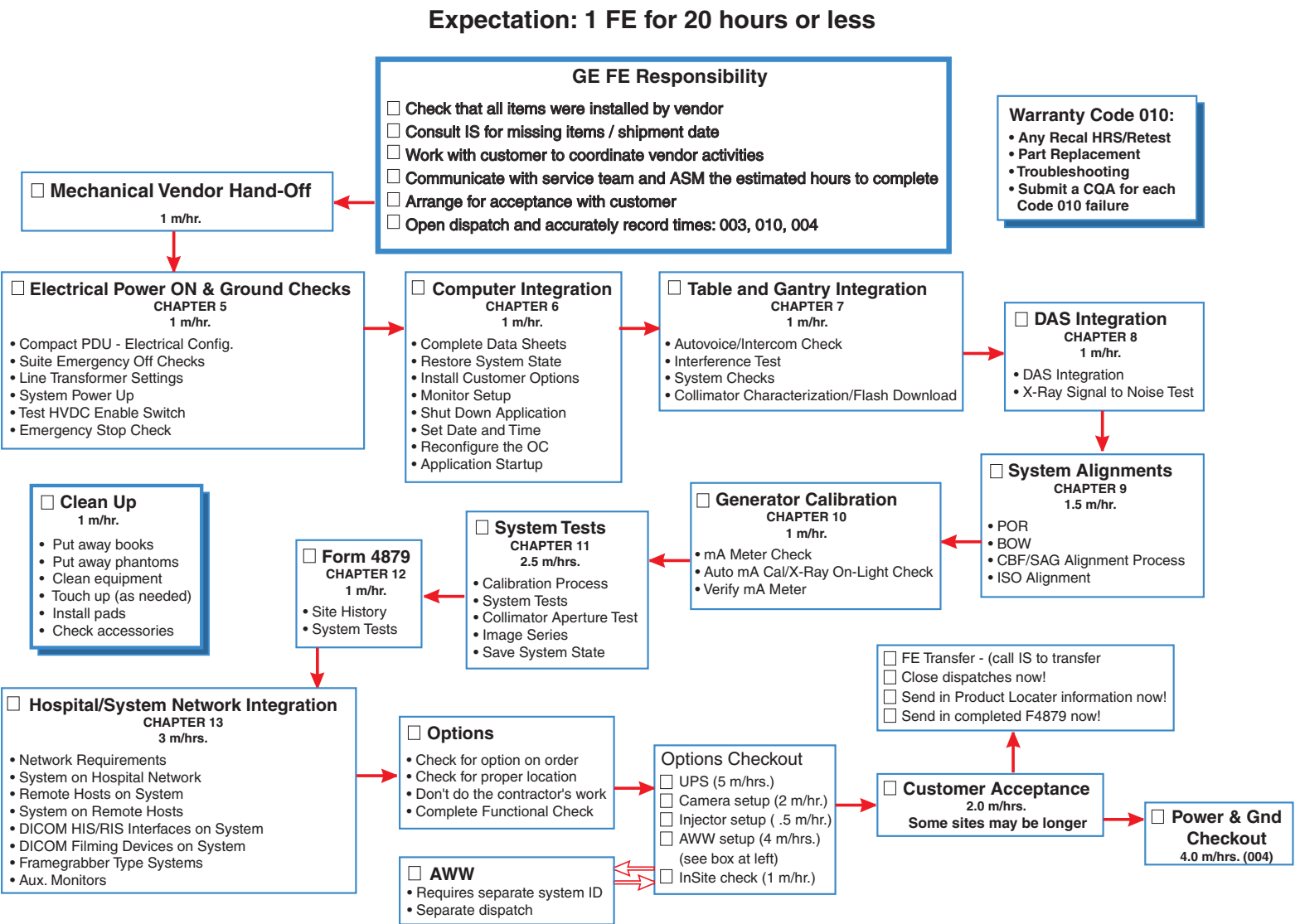


Figure 4-1 Electrical Installation Block Diagram

Section 4.0

Required FE Common Tools and Supplies

4.1 FE Calibration and Service Tool List

Special CT Tools used for mechanical alignments (found in the install support kit):

- 5mm hex bit for 3/8" drive 6" long (Snap-On FAML5E or equivalent)
- 10mm hex bit for ratchet wrench (ball end) (Snap-On FABM10E or equivalent)
- 2.5mm Allen hex bit for 1/4"
- Standard FE Tool Kit (not included in install support kit)
- 10mm open-end thin wrench (Snap-On SRSM10 or equivalent)
- 21mm open-end thin wrench (Snap-On LTAM2124 or equivalent)
- 14mm ball hex socket bit for 3/8" drive
- 14 mm hex socket bit for 1/2" drive

4.2 Electrical Tools

These tools must be calibrated yearly.

- Fluke 87 DVM or equivalent
- Clamp on amp meter

4.3 Image Quality Calibration Tools

- GE Performance Phantom
- QA Phantom (2206352)
 - 20cc (2144721)
 - 40cc (2144721-2)
 - Phantom Holder

4.4 Detector Service Tools

These tools can be found in the install support kit.

- 2.5 mm hex bit
- Anti-Static Bags
- Electric Screw Driver
- Canned Air (Grounded) with metal nozzle (Miller Stephenson Aero Duster MS-222N or equivalent)
- Ground Wrist Strap
- Grounded Mat
- Dental Pick

4.5 Optional Tools

Needed for warranty recalibration:

- GE HV Bleeder
- Scope with 10X probes
- Beckman CT231 clamp-on Amp probe: 46-194427P228 (Fits up to #2 size wire)
- Beckman CT232 clamp-on Amp probe: 46-194427P270 (Fits #1 size wire and larger)
- Scope Probe-to-bleeder Cable 46-219921G1 (Belden #8422 cable, 30 ft)
- Polaroid type 52 film and developer
- Radial dial indicator (mm or inches) and mounting bracket
- Caliper Dial indicators (mm or inches)
- Ground Rod Tester Clamp on Amp Probe
- Torque Wrench kit

4.6 Safety Materials

- Lockout/Tagout kit, or equivalent
- Safety Glasses
- Nitrile Rubber Gloves

4.7 Cleanliness

Any dirt on the surface increases leakage current on the filter or converter cards and causes the DAS to fail the drift spec. Wear Nitrile gloves (part number 2207303-6 [large] and 2207303-7 [extra large]) when you handle the DAS, because fingerprints on the board can cause problems during humid conditions. Use only clean, new Nitrile gloves. Do not use hospital grade gloves.



NOTICE NEVER USE AN ERASER TO CLEAN ANY PART OF THE DAS.
DUST COVERS REQUIRED for installations on construction sites.

Section 5.0 Requirements/Assumptions

- All stations in a suite must have the same SUITE ID.
- You need the name of all hosts in the suite and their corresponding Internet/Ethernet numbers.
- You need the Internet (IP) addresses the first time you execute a reconfig on the system.
 - When you connect the system to a network, contact the system administrator to obtain the IP addresses for all the computers in the suite.
 - You also need an IP address for each gateway (second) ethernet board in any OC or IC.
 - You can use the default internet number on stand-alone systems (stand-alone = not connected to any network).
- For more detailed information and instructions regarding Network Integration, see [Chapter 12 on page 251](#).

Chapter 5

Electrical Power ON & Ground Checks

NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 14](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Compact PDU - Electrical Configuration

WARNING


THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

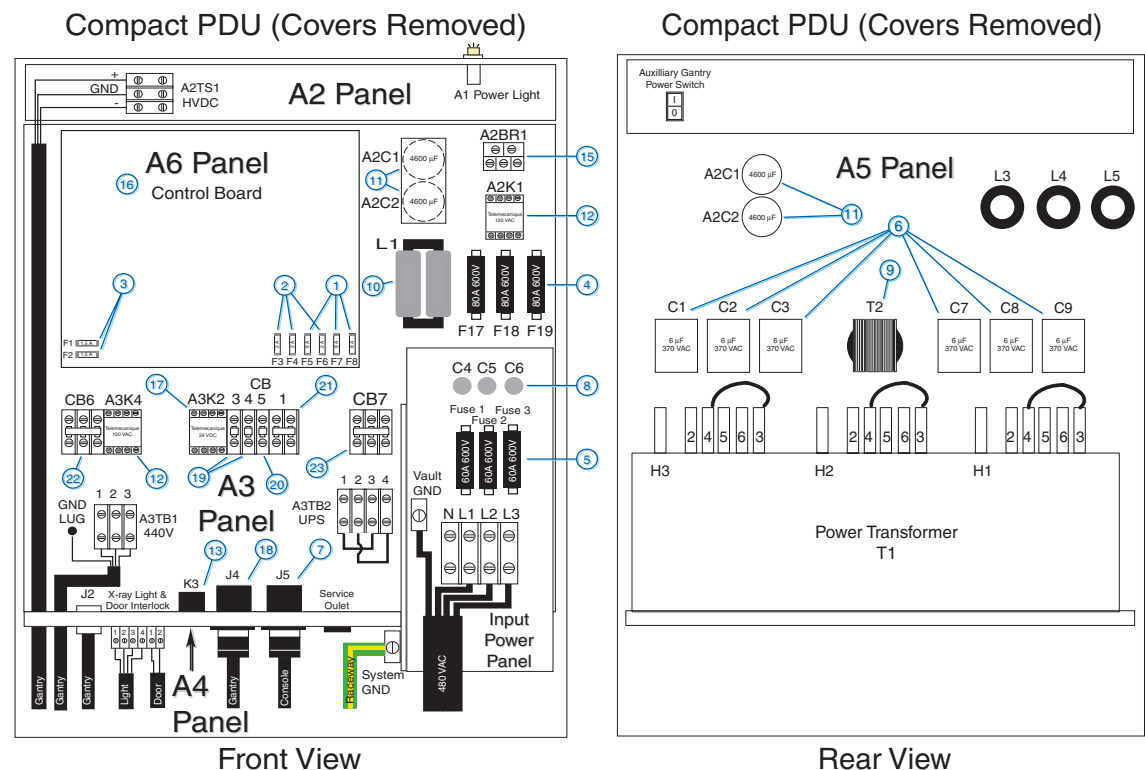


Figure 5-1 Compact PDU

1.1 Required Tools

- Multimeter with a rating of at least 1000 volts
- Multimeter leads with a rating of at least 1000 volts

1.2 Initial PDU Configuration



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

1.2.1 Circuit Breakers

Set all circuit breakers to OFF

1.2.2 Relay Board

- 1.) Set SW1 and SW2 to the normal position.
- 2.) Set the Room Light jumper (JP1) to meet hospital requirements (refer to [Figure 5-2](#)).
 - Position A enables the room warning light during X-Ray only.
 - Position B enables the room warning light as soon as HVDC turns on.

Note: **Use position B unless the hospital requires otherwise.**

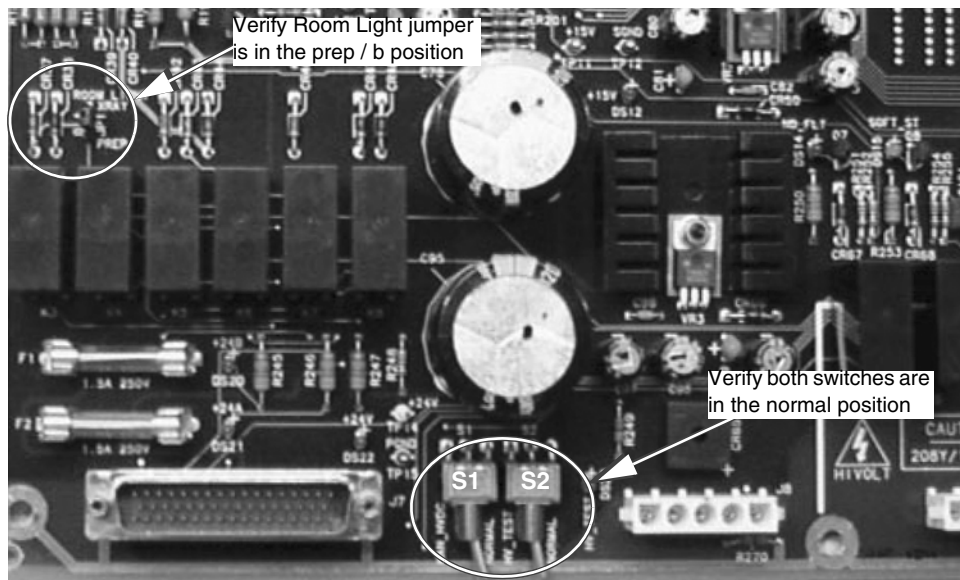


Figure 5-2 PDU Control Board (A6 Panel)

1.2.3 Power Switches

Turn OFF all power switches on all subsystems (gantry status display box, STC, DAS, table, console, computer, and SCU).

1.2.4 Hardware and Connection Check

Use this step to check mechanical connections and tighten anything that may have shaken loose during shipment. Verify **all** hardware and connections in the PDU are securely fastened.

1.2.5 Covers

Install, or verify the presence of, all the lexan safety covers.

Section 2.0 NGPDU - Electrical Configuration



WARNING

THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

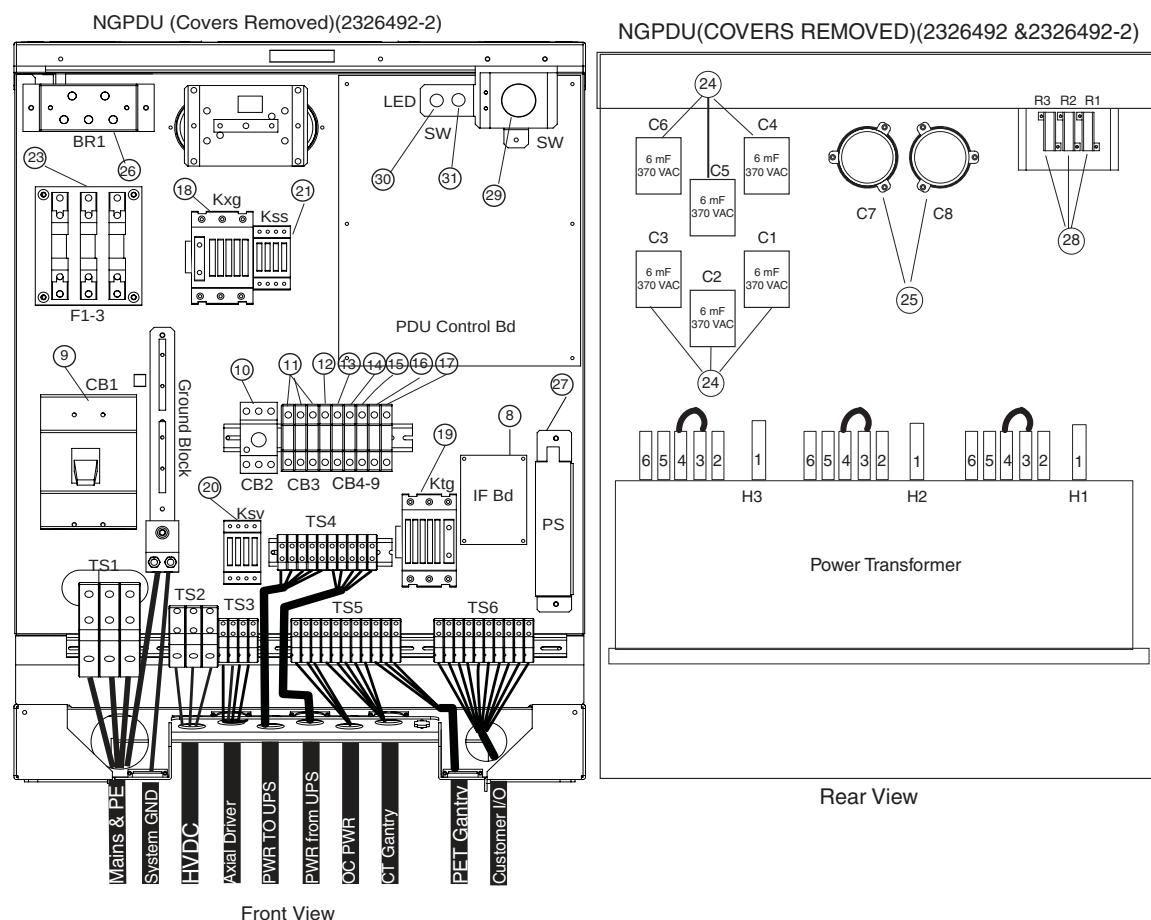


Figure 5-3 NGPDU

2.1 Required Tools

- Multimeter with a rating of at least 1000 volts
- Multimeter leads with a rating of at least 1000 volts

2.2 Initial PDU Configuration



WARNING

THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

2.2.1 Circuit Breakers

Set all circuit breakers to OFF

2.2.2 Relay Board

- 1.) Set SW to the normal position.
- 2.) When system is already, three lamps are both lighting (refer to [Figure 5-4](#)).

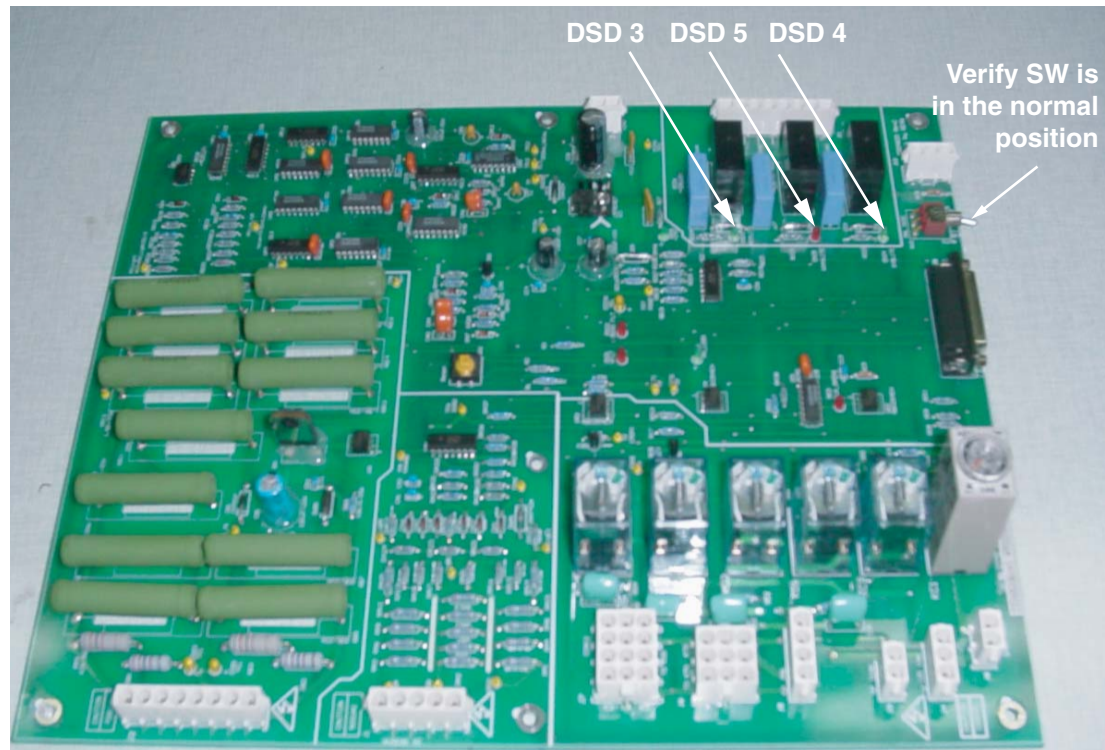


Figure 5-4 NGPDU Control Board

2.2.3 Power Switches

Turn OFF all power switches on all subsystems (gantry status display box, STC, DAS, table, console, computer, and SCU).

2.2.4 Hardware and Connection Check

Use this step to check mechanical connections and tighten anything that may have shaken loose during shipment. Verify **all** hardware and connections in the PDU are securely fastened.

2.2.5 Covers

Install, or verify the presence of, all the lexan safety covers.

Section 3.0

Suite Emergency Off Checks

This procedure is for systems with PDU 2133533 and 2326492-2 (NGPDU).



WARNING



VERIFY ALL PERSONNEL HAVE CLEARED THE SYSTEM BEFORE YOU TURN ON WALL POWER.

- 1.) Turn wall power ON to the PDU.
- 2.) **Press the suite emergency off button and verify it turns off wall power to the PDU.**
[Typically, this red palm button is located on the wall close to the console, within the scan suite.]
- 3.) Verify that all "Emergency Off" button are working properly.
- 4.) Leave power "OFF".

Section 4.0

Line Transformer Settings

4.1 Requirements

- 1.) The PDU is shipped configured for 480VAC.
- 2.) Complete only if your site uses a voltage other than 480VAC.
- 3.) If PDU is configured for 480VAC, go to Section 5.0. Otherwise, proceed to [Section 4.2](#).



WARNING

TURN OFF, TAG AND LOCK THE MAIN WALL POWER BEFORE CHANGING TAPS. FAILURE TO DISCONNECT POWER AT MAIN INPUT MAY RESULT IN ELECTROCUTION. TURN OFF WALL POWER TO CONNECT OR MOVE METER LEADS, OR TO REMOVE OR INSTALL COVERS.

4.2 Line Input Conditions

4.2.1 Compact PDU

- 1.) Remove metal cover over right A3 panel.
- 2.) Monitor the No Load Line to Line Voltage at A3 L1, L2, L3, during the workday. Do not record this data during “brown out” conditions.
- 3.) After you determine the nearest nominal line, verify the tap connections match (refer to [Table 5-1](#) and [Figure 5-5](#) for tap locations).

Transformer Taps and Jumpers

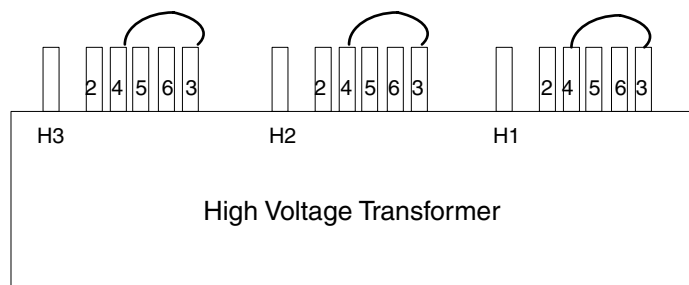


Figure 5-5 PDU Tap Positions (Rear)

Note: Taps should be shipped as shown for 480 VAC only. For all others, you must move the taps. The tap check should be completed by the mechanical installer.

- 4.) Verify that the No Load Line to Line Voltage never falls outside the corresponding minimum and maximum values listed in [Table 5-1](#).
- 5.) Use a 0-750 AC voltmeter of 3/4% accuracy to measure the line-to-line voltages at A3 L1, L2, and L3.
 - Verify the highest line-to-line voltage does not exceed 1.02 times the lowest voltage.
 - **Example:** If the lowest voltage equals 474, the highest voltage should not exceed $474 \times 1.02 = 483.5$ volts.



WARNING

THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

6.) Replace the metal cover over the right A3 panel.

NO LOAD		TAP CONNECTIONS		
Line to Line Voltages		(All 3 phases must have same the configuration)		
Nominal	Maximum Range (8%)	Phase A Connection	Phase B Connection	Phase C Connection
480V*	442 to 518*	3-4*	3-4*	3-4*
460V	423 to 497	3-5	3-5	3-5
440V	405 to 475	3-6	3-6	3-6
420V	386 to 454	2-4	2-4	2-4
400V	368 to 432	2-5	2-5	2-5
380V	350 to 410	2-6	2-6	2-6
* Factory Default				

Table 5-1 PDU 2269902 Line Tap Connections

4.2.2 NGPDU

- 1.) Monitor the No Load Line to Line Voltage at L1, L2, L3, during the workday. Do not record this data during “brown out” conditions.
- 2.) After you determine the nearest nominal line, verify the tap connections match (refer to [Table 5-2](#) and [Figure 5-6](#) for tap locations).

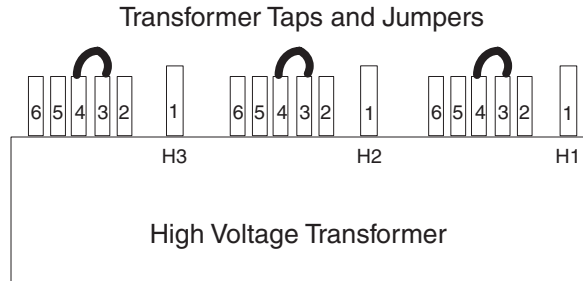


Figure 5-6 PDU Tap Positions (Rear)

Note: Taps should be shipped as shown for 480 VAC only. For all others, you must move the taps. The tap check should be completed by the mechanical installer.

- 3.) Verify that the No Load Line to Line Voltage never falls outside the corresponding minimum and maximum values listed in [Table 5-2](#).
- 4.) Use a 0-750 AC voltmeter of 3/4% accuracy to measure the line-to-line voltages at L1, L2, and L3.
 - Verify the highest line-to-line voltage does not exceed 1.02 times the lowest voltage.
 - **Example:** If the lowest voltage equals 474, the highest voltage should not exceed $474 \times 1.02 = 483.5$ volts.



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

NO LOAD Line to Line Voltages		TAP CONNECTIONS (All 3 phases must have same the configuration)		
Nominal	Maximum Range (8%)	Phase A Connection	Phase B Connection	Phase C Connection
480V*	442 to 518*	3-4*	3-4*	3-4*
460V	423 to 497	3-5	3-5	3-5
440V	405 to 475	3-6	3-6	3-6
420V	386 to 454	2-4	2-4	2-4
400V	368 to 432	2-5	2-5	2-5
380V	350 to 410	2-6	2-6	2-6
240V	221 to 259	1-4	1-4	1-4
220V	202 to 238	1-5	1-5	1-5
200V	184 to 216	1-6	1-6	1-6
* Factory Default				

Table 5-2 PDU 2326492-2 Line Tap Connections

Section 5.0

System Power Up

- 1.) Turn OFF wall power.
- 2.) Turn ON all PDU A3 circuit breakers.



CAUTION Verify all personnel have cleared the system before you turn on wall power.

- 3.) Turn ON wall power.
- 4.) Check A1 Emergency Off push-button.

SUB-SYSTEM POWER-UP

- 1.) Turn ON switch S3 in the table (120vac 24hr power).
- 2.) Turn ON the gantry 120VAC enable switch on the STC. Tube fans turn on.
- 3.) Leave AXIAL DRIVE ENABLE OFF.
- 4.) Leave HV DC ENABLE OFF.
- 5.) Turn ON the DAS power switch located on the outside of the DAS power supply assembly.
- 6.) Turn ON Table switches S1 and S2.
- 7.) Turn ON console power.
- 8.) Make certain that the switch on the back of the PDU is ON.
- 9.) Turn ON HEMRC Power ON.
- 10.) Verify that the gantry 210VAC is ON at the Gantry Power Pan.

Section 6.0

Test HVDC Enable Switch



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



2.) Click on the UTILITIES icon.

3.) Select INSTALL.

4.) Select TABLE/GANTRY INTEGRATION.

5.) Select HV DC ENABLE SWITCH.

6.) Select START.

7.) Select the following parameters in the Rotor Functional window:

PARAMETER	MX200
Test Selection:	HEMRC
Rotor Speed:	LOW
Rotor Loop:	OPEN
Rail Diff Voltage:	Normal
Test Duration:	10 seconds
OBC Service Input:	Disabled

Table 5-3 Rotor Functional Window Parameters

8.) Verify all personnel have cleared the system, *before* you enable the HVDC.

9.) Verify the following:

- a.) HVDC enable switch on the STC is ON
- b.) "HVDC ENABLE ON" light on the STC is ON

10.) Select RUN. If the system asks for a diagnostic download at this time, answer YES, and wait 90 seconds.

11.) Verify the "HVDC BACKUP CONTACTOR CLOSED LIGHT" on the STC status display turns ON.

12.) When the test finishes, verify the system displays the following message along the bottom of the screen: Diagnostics have completed with zero errors reported.



Figure 5-7 STC Service Switches

VERIFY/PERFORM

- 1.) Turn OFF the HVDC enable switch on the gantry service switch box.
- 2.) Select RUN to run the test again.
- 3.) Verify the test aborts.
- 4.) Verify the message log contains the following error: *Gentry I/O backup contactor error*.
- 5.) Select FILE → EXIT.
- 6.) Select DISMISS.
- 7.) Select CLEANUP.
- 8.) Wait 90 seconds for the message "Scanning Hardware Reset Successful" to appear on the screen.

AXIAL ENABLE SWITCH TEST

- Note: For the initial condition, do not leave the tube at the 2:30 position.
- 1.) Turn OFF axial drive enable switch S2 on the STC backplane.
 - 2.) Clear the gantry are for rotation.
 - 3.) Press the alignment light push button.
 - 4.) Verify that the gantry did not rotate.
 - 5.) Turn ON axial drive enable switch S2 on the STC backplane.



WARNING

MAKE SURE THAT THERE IS NOT OBSTRUCTIONS AROUND THE GANTRY. PRESSING THE ALIGNMENT LIGHT PUSHBUTTON WILL CAUSE THE GANTRY TO ROTATE.

- 6.) Press the alignment light push button.
- 7.) Verify that the gantry rotates.

ROTATION SAFETY CHECKLIST

- 1.) Turn OFF enable switch S2 on the gantry.
- 2.) Turn OFF HV enable switch S3.
- 3.) Press red E-STOP button.
- 4.) Manually rotate the gantry 360 degrees. (Keep one finger at the console E-STOP button.)
 - Listen for any interference between the rotating and stationary parts.
(Correct any interference problems.)
 - Listen for any loose parts.
(Tighten parts as needed.)
- 5.) Turn ON the enable switch S2 on the gantry.
- 6.) Turn ON the Drive power switch S1.
- 7.) Leave the large gantry bay door half open. (This will make it easier to hear for any loose or interfering parts.)
- 8.) Using DDC, prescribe a 4 second X-ray OFF scan. (Keep one finger at the console E-STOP button.)
 - Listen for any interference between the rotating and stationary parts.
(Correct any interference problems.)
 - Listen for any loose parts.
(Tighten parts as needed.)
- 9.) After completing the 4 second scan, repeat Step 8 with 2.0, 1.0, 0.7, and 0.5 second scans

Section 7.0

Emergency Stop Check

- 1.) Use the gantry push-buttons to advance the cradle about 0.5m (2ft) from the home position.
- 2.) Press one of the E-STOP buttons on the gantry.
- 3.) Make sure the TABLE POWER shuts off, and the green LED flashes.
- 4.) Depress one of the table elevation buttons, to verify the emergency stop disabled table elevation.
- 5.) Depress one of the cradle drive buttons, to verify the emergency stop disabled the cradle drive.
- 6.) Press one of the **RESET** buttons to turn on X-RAY DRIVES POWER.
- 7.) Press the other E-STOP button on the gantry.
 - a.) Make sure the Table Power shuts off.
 - b.) Manually move the cradle to the home position to make sure the cradle clutch released.
 - c.) Make sure the cradle latches securely in the home position.
- 8.) Press one of the **RESET** buttons to turn on X-RAY DRIVES POWER.
- 9.) Press one of the four table tape switches to make sure the X-RAY DRIVES POWER shuts off.
 - a.) Turn on TABLE POWER.
 - b.) Repeat with the three remaining table tape switches.
- 10.) Press the console emergency stop switch; make sure the Table Power shuts off.
- 11.) Turn on STC 120 Power.
- 12.) Exit.
- 13.) Return to Service Desktop home page.

Chapter 6

Computer Integration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 14](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

This chapter describes the reconfiguration, system state restore, options, and monitor adjustment procedures.

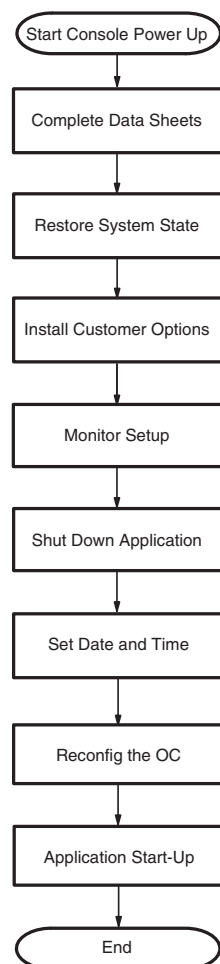


Figure 6-1 Computer Integration Process Overview

Section 2.0

System Configuration Data Sheets

2.1 Requirement

Record valuable system information in the data sheets that follow. Consult with your customer or network administrator to obtain the information. Understanding how the customer plans to use their LightSpeed 3.X scanner, and their network and filming expectation reduces the time required to reconfigure the system.

- [Table 6-1 Manual Film Composer Options, on page 171](#)
- [Table 6-2 System Network Configuration, on page 172](#)
- [Table 6-3 Networking Application \(Image transfer\) Configuration, on page 173](#)
- [Table 6-4 DASM Laser Camera Configuration, on page 173](#)
- [Table 6-5 DICOM Print Camera Configuration, on page 174](#)
- [Table 6-6 DICOM Print Camera Advanced Configuration, on page 174](#)

2.2 Manual Film Composer Options

MANUAL FILM COMPOSER OPTIONS	
Slide Format (if available):	
Greyscale:	
Auto Printing:	
Auto Clear Page:	
Icon Labels:	
Expose Order:	
No. of Copies:	

Table 6-1 Manual Film Composer Options

2.3 System Network Configuration

SYSTEM NETWORK CONFIGURATION			
	FIELD NAME:	SETENV NAME:	FIELD VALUE:
System Settings:	Service ID	SERVICE_ID	
	Hospital Name	HOSPITAL_NAME	
	Exam Number *	* ask customer or checklog	
	DAS Type	DASTYPE	
	PDU Type	PDUTYPE	
Network Settings:	Gateway Host Name	GATEWAY_HOSTNAME	
	Gateway IP	GATEWAY_IP	
	Gateway Net Mask	GATEWAY_NETMASK	
	Gateway Broadcast Mask	GATEWAY_BROADCAST	
	Suite Name	SUITEID	
Option	Network Printer IP Address		
Option	HIS Server IP Address		
Option	HIS Server AE Title		
Option	HIS server AE Port		
Option	CT Server AE Title		
Option	Connect Pro IP Address		

Table 6-2 System Network Configuration

2.4 Host Ethernet Address

_____ : _____ : _____ : _____ : _____

2.5 Network Application (Image Transfer) Configuration

Record the network application (image transfer) configuration.

NETWORKING APPLICATION (IMAGE TRANSFER) CONFIGURATION				
AE TITLE OR HOST NAME	NETWORK ADDRESS	NETWORK PROTOCOL	PORT NUMBER	COMMENTS

Table 6-3 Networking Application (Image transfer) Configuration

2.6 Camera Application Configuration

Record the camera application configuration for the DASM or DICOM print camera.

DASM LASER CAMERA CONFIGURATION	
Camera Type:	
DASM Type:	
Film Smooth/Sharp Setting:	
Options:	
Valid Film Formats:	
Default Film Formats:	

Table 6-4 DASM Laser Camera Configuration

DICOM PRINT CAMERA CONFIGURATION

Camera Type:	
Host Name:	
IP Address:	
AE Title:	
TCP/IP Listen Port:	
Comments (Optional):	
Valid Film Formats:	
Default Film Formats:	
Destination:	
Orientation:	
Medium Type:	
Magnification Type:	

Table 6-5 DICOM Print Camera Configuration

DICOM PRINT CAMERA ADVANCE CONFIGURATION

Smoothing Type:	
Configuration:	
Minimum Density:	
Maximum Density:	
Empty Density:	
Border Density:	
Association Timeout:	
Session Timeout:	
N-Set Timeout:	
N-Action Timeout:	
N-Create Timeout:	
N-Delete Timeout:	
N-Get Timeout:	

Table 6-6 DICOM Print Camera Advanced Configuration

Section 3.0 Configuration

Note: Type the text shown in **boldface**, and press the ENTER key on the keyboard.

3.1 Restore System State

Tools Required: None

Your system should have a system state MOD, located in the software collector box.
The system state MOD contains:

- Collimator Characterization
- Phantom Calibrations

The installation process uses all the system state files. At this time, use the system state MOD to restore the phantom calibrations and system characterization files.

If you cannot locate an existing system state MOD, manually characterize the system and perform phantom calibrations.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the PM icon.
- 3.) Select SYSTEM STATE.
- 4.) Insert the MOD in the MOD drive.
- 5.) Select CHARACT.
- 6.) Select CALS.
- 7.) Select RESTORE to restore the system characterization and phantom calibration files to the system.

Note: **Restore State can take as long as ten minutes, although typical times average about three minutes. When Restore State completes, dismiss the tool, and proceed to the next section.**

If any error should occur during the restore process, see the Software Load Procedure manual (2211220-100) for information regarding possible error messages and their recovery.

- 8.) Click NO for Reset Scan Hardware popup message.
- 9.) Select DISMISS.

3.2 Install Customer Options

Tools Required: None

3.2.1 Software

Note: **Your system may have a MOD that contains customer purchased options. If your system has an options MOD, install it at this time.**

Ensure that the options MOD is NOT write protected at this time. The initial install requires that the MOD be write enabled; subsequent installs can be done with the MOD write protected.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select INSTALL OPTIONS.

An Options Window appears (Figure 6-2):

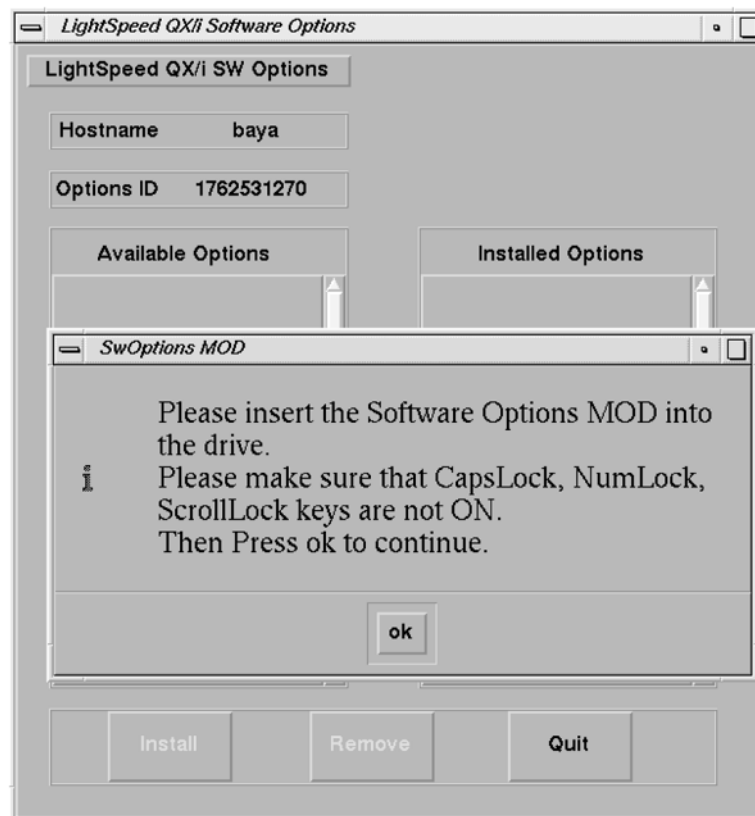


Figure 6-2 Options Window when First Selected

- 5.) Insert the options MOD into the MOD drive and click on OK. (If you do not have an options MOD, click on OK anyway, wait for the abort pop up, then abort the process.)

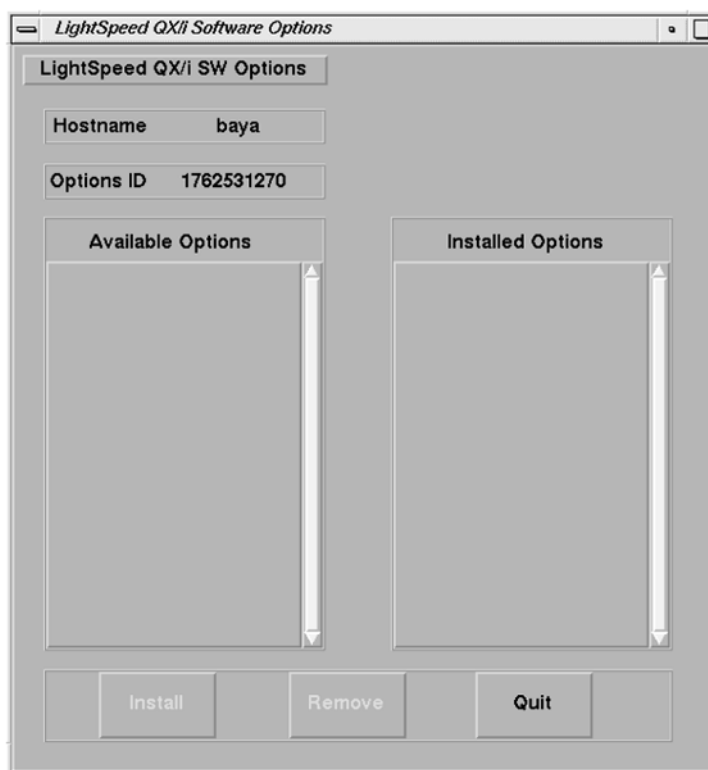


Figure 6-3 Example: Options Window

- 6.) Select all of the options in the left-hand column to install the corresponding software.
- 7.) Select INSTALL. A box may appear while the options are loading. When an option is displayed in the Installed Options list, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half minute (due to the fact that they are installing software).
- 8.) After the options are installed, select QUIT.
- 9.) Select OK.
- 10.) Remove the MOD and write protect the side with options.
- 11.) When the system prompts to Reboot, click YES, and reboot the system to complete the installation.

3.2.2 Camera

Tools Required: Small flat blade screw driver

Note: DASM hardware must be installed before proceeding.

For details on camera configuration, refer to the Software Load Procedures manual.

For details on troubleshooting the camera, refer to the System Service manual.

- 1.) Click on the SERVICE DESKTOP icon.
- 2.) Select UTILITIES.
- 3.) Select INSTALL CAMERA.
- 4.) Select ADD.
- 5.) Select DASM or DICOM.
- 6.) Follow procedures on the screen.
- 7.) Return to Home Page.

3.3 Monitor Setup

Tools Required: Sony Service Manual (located in monitor box)

3.3.1 Overview

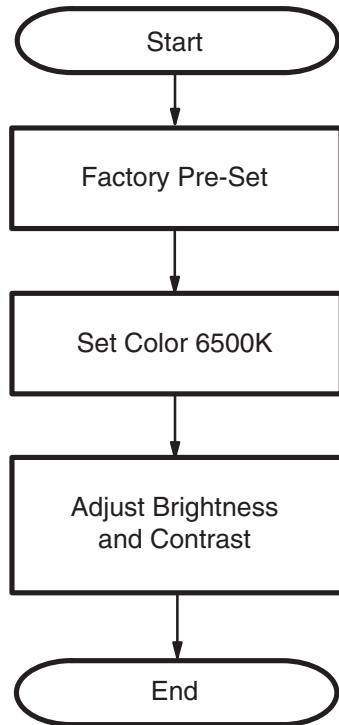


Figure 6-4 Monitor Setup Flowchart

3.3.2 Color Adjustments

- 1.) Adjust the ambient room light to the level used by the operator during image review.
- 2.) Select the COLOR menu to adjust the picture's color temperature. (You can use this to match the monitor's colors to a printed picture's colors.

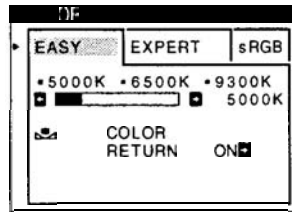


Figure 6-5 Color Menu

- 3.) Set monitor color temperature to 6500K. Refer to the system service manual for this procedure, if necessary.
- 4.) Use one of the following methods:



- a.) Click the IMAGEWORKS icon and proceed to step 3.

OR

- b.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- c.) Click on the IMAGE QUALITY icon.
- d.) Select INSTALL SMPTE IMAGE and proceed to step 3. (Approximately 3-4 minutes.)
- 5.) Display the SMPTE pattern. Use the browser to select Exam 1000, which contains the SMPTE pattern, and enlarge the image to full screen display.
- 6.) Increase the Contrast to maximum.

Note: Adjust monitor contrast until the operator sees the anatomical structure (window width) at the expected window levels.

- 7.) Increase the Brightness to maximum.

Note: Adjust the monitor brightness until the operator sees the anatomical structure (window width) at the expected window levels.

- 8.) Decrease the Brightness, until the raster just fades into, and matches, the monitor screen background. At this point, the 5% and 95% patches should be just visible.

If additional tweaking is required to attempt to match the monitor image to the filmed image, use only the brightness control.

If the CRT image exhibits any tearing or smearing of the alphanumeric characters, then reduce the contrast setting slightly until the tearing/smearing is just eliminated. The optimum setting for contrast is the highest setting that does not cause tearing/smearing of the alphanumeric characters.

You should always finish up by displaying and filming images of anatomy (typical heads and bodies), and asking the technologist to compare the CRT image to the film image.

- 9.) To view SMPTE on text monitor, swap video cable.

3.3.3 Using Menu Buttons

- 1.) Press the Main Menu button to display the main menu on your screen.



Figure 6-6 Main Menu Button

- 2.) Press the / buttons to highlight the menu you want to adjust. Press the Main Menu button to select the menu item.

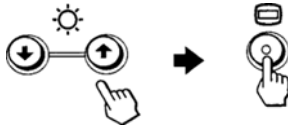


Figure 6-7 Select Menu Item

- 3.) Press the / buttons to select the desired adjustment item. Press the / buttons to make the adjustment.



Figure 6-8 Adjust the Menu

- 4.) Press the Main Menu button once to return to the main menu, and twice to return to normal viewing. If no buttons are pressed, the menu closes automatically after about 30 seconds.



Figure 6-9 Main Menu Button

- 5.) Press the reset button to reset the adjustments back to the factory settings.



Figure 6-10 Reset Button

3.3.4 Adjusting Brightness and Contrast

Brightness and contrast adjustments are made using a separate BRIGHTNESS/CONTRAST menu. These settings are stored in memory for the signals from the currently selected input connector.

- 1.) Press either one of the brightness or contrast buttons. The BRIGHTNESS/CONTRAST menu appears on the screen.

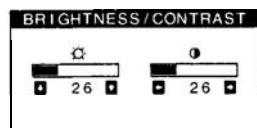


Figure 6-11 Brightness/Contrast Menu

- 2.) Press the / buttons to adjust the brightness and the / buttons to adjust the contrast.

3.4 Shut Down Application

- 1.) If Applications is currently running, you must shutdown system applications to the CT user desktop.
 - a.) Click on the SERVICE DESKTOP button.
 - b.) On the desktop toolbar select UTILITIES.
 - c.) Select APPLICATIONS SHUTDOWN (to bring down applications only).
- 2.) Open a UNIX Shell window.
 - a.) Type: `su -` ENTER at the prompt.
 - b.) Enter the root (super user) password: #bigguy

3.5 Check/Set Date and Time

Check the date and time.

- 1.) Type/enter `setdate` ENTER.
- 2.) Follow the instructions to set the correct time and date.

3.6 Reconfig the OC

Note: The document collector box that arrived with your system contains the LightSpeed 3.X Software Load Procedure manual, which documents the reconfiguration procedure in more detail.

3.6.1 Overview

On the following screens, you should make the changes necessary, pressing the corresponding button at the top of the screen to move from screen to screen. When you are done, you can either press the ACCEPT button to start the reconfiguration process, or press the QUIT button to exit without changing the system configuration.

While the reconfiguration is going on, messages are displayed in a shell window that closes when reconfiguration is complete. Should you later want to review the reconfiguration output, it is logged to the following file:

```
/var/adm/install.log.YYYYMMDDWWHHMMSS
```

Where

YYYYMMDDWWHHMMSS is the Date/Time that the reconfiguration was started.

To view the file, type: `more /var/adm/install.log.YYYYMMDDWWHHMMSS`

It is possible to abort the reconfiguration while entering information on the reconfiguration screens. Simply press the QUIT button at the top of the screen. There is NO safe way to abort the reconfiguration after pressing the ACCEPT button. If the entries made in the screens were incorrect, DO NOT try to stop the reconfiguration, instead wait for it to complete, and rerun reconfig, entering the correct parameters.

3.6.2 Procedure

- 1.) Change directory to scripts:
Type: `cd /usr/g/scripts` ENTER at the prompt.
- 2.) Launch the LightSpeed Install utility:
Type: `reconfig` ENTER at the prompt.
The OC displays the LightSpeed Install Utility Window as shown in [Figure 6-12](#).



Figure 6-12 Install Utility Window

- 3.) Using the mouse, click on the CONFIG button.
The OC displays the LightSpeed 3.X System Configuration - System Settings Screen as shown in [Figure 6-13](#).

Comment: The following pages show the screens that are used to change the configuration of the system. These screens are the same as those used for the Software Configuration during Load From Cold. The actual screens will vary depending on the current configuration of your system.

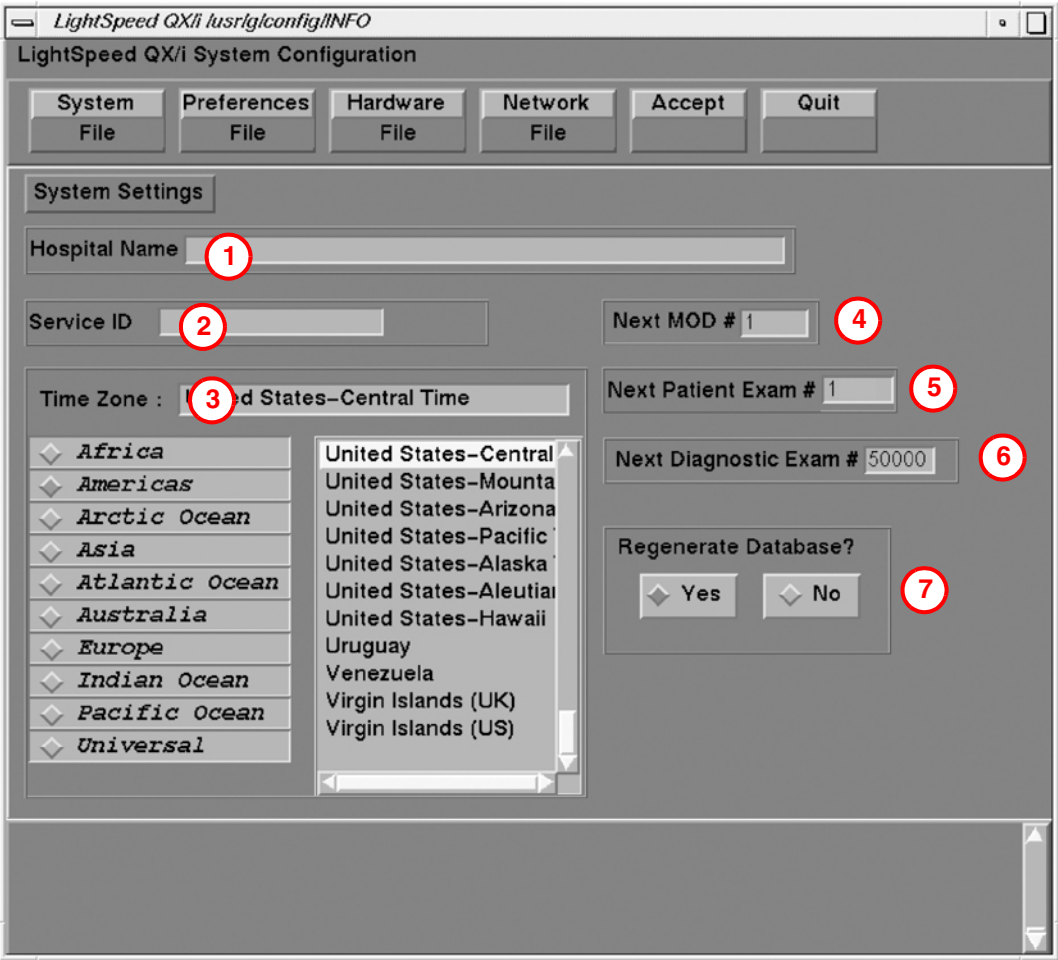


Figure 6-13 System Settings Screen

4.) Configure System Settings

- a.) Hospital Name (Figure 6-13, item 1) configures the name that will show up on images produced by this scanner. *Example:* ST MARYS HOSPITAL
- b.) Service ID (Figure 6-13, item 2) is issued by the Service organization.
Example: 262785CT2 (no spaces)
- c.) Select the Time Zone for the site.

Note: Use the scrollbar at the bottom of the time-zone selection list to view the entire description of the time-zone you are about to select, to ensure that you are selecting the correct time-zone for your location.

If the time-zone of your location is not in the list above, select one of the universal times in the selection menu. In this case, automatic changes for daylight savings time will not take effect. See Load from Cold manual, if you require more information regarding time-zone setting & selection.

- d.) Next MOD # *Customer selected.*
 - e.) Next Patient Exam # configures the next Exam number the scan user interface will use. At initial system installation, type: 1
 - f.) Next Diagnostic Exam # *Customer Selected.*
- 5.) Select the PREFERENCES button to display the Preference Settings Screen as shown in Figure 6-14.
- 6.) Regenerate Database (refer to Figure 6-13) determines whether the Scan Database will be regenerated during reconfiguration.

Note: If YES is selected, then all scan data will be deleted on the Scan Data Disk. To avoid accidental regen, NO is selected by default each time reconfig is run. Regenerating the Scan Database destroys all the scan and calibration files on the Scan Data Disk. Make sure that all images have been reconstructed before regenerating the Scan Database.

This procedure is normally only needed when a Scan Data Disk is replaced or reformatted. See Load from Cold manual, if you require more information regarding Scan Database Regeneration.

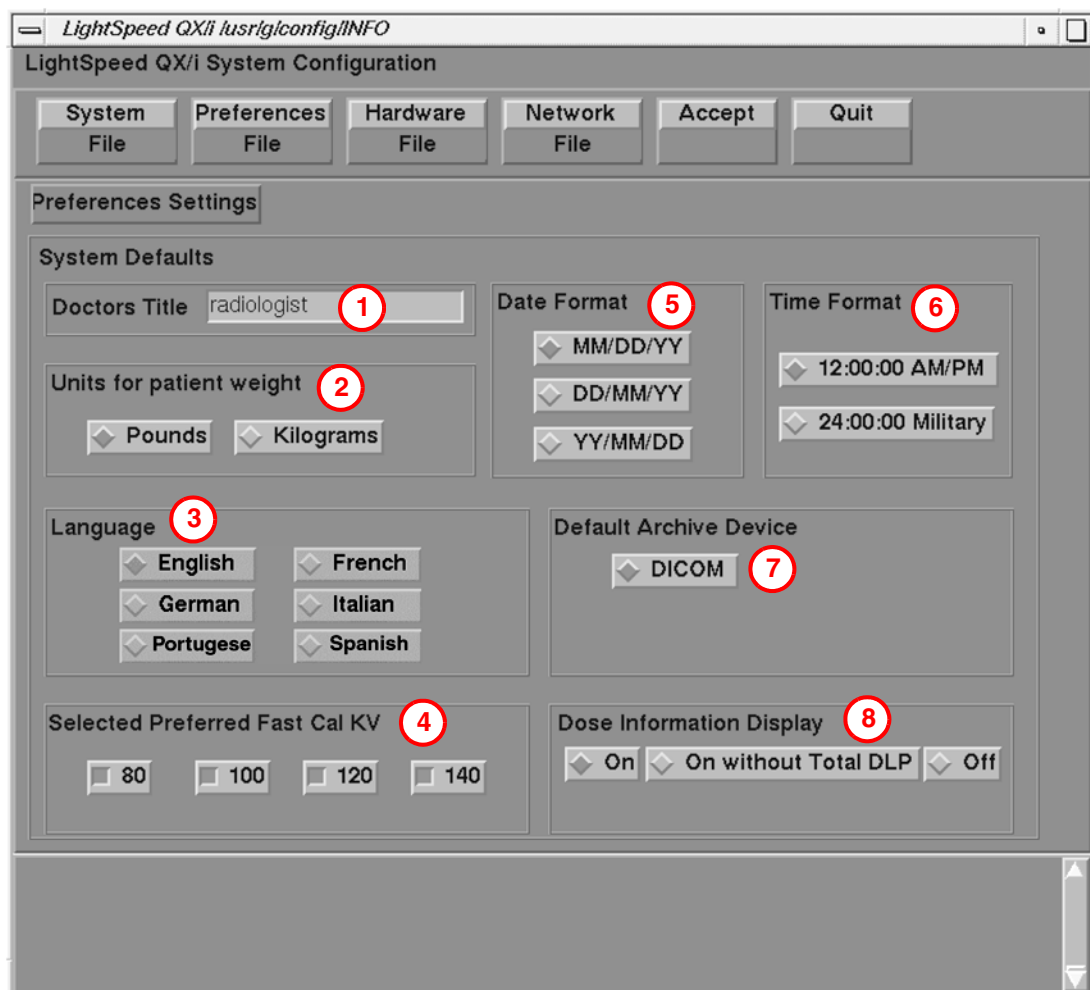


Figure 6-14 Preferences Setup Screen

7.) Configure Preferences Settings

a.) Language configures the language to be displayed on the Application screens (ENGLISH, FRENCH, GERMAN, ITALIAN, PORTUGUESE, SPANISH).

b.) Preferred Fast Cal KV configures the preferred kV that the Fast Cal Routine will calibrate (80, 100, 120, 140 in the Selected Preferred Fast Cal KV field). The default selections are 120 and 140.

Comment:

These kVs should include all kVs that the site uses for patient scanning. Deselecting All Preferred FastCal KVs is the same as selecting ALL the Preferred FastCal KVs

c.) Date Format and Time Format configure the format in which the date and time will be displayed on the images.

d.) Select the site preferred Dose Information Display option for the site to use in monitoring calculated Patient Dose:

- * Select ON (full CTDiw Display)
- * Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
- * Select OFF (no CTDiw Display)

- 8.) Select the **HARDWARE** button to display the Hardware Settings Screen (Figure 6-15).

Figure 6-15 Hardware Settings Screen

- 9.) Configure Hardware Settings
- DAS Type configures which DAS the software should expect.
 - Tube Type configures which X-Ray Tube the software should expect.
 - PDU Type configures which PDU the software should expect.
 - Collimator/Filter Type configures which Collimator/Filter the software should expect. Presently, LightSpeed 3.X systems only support HELIOS G1.
 - Gantry Type selects the type of Gantry connected to this console.
 - Select **Network Image Printer** only if the site plans to use a network connected postscript printer (such as Codonics printers) with this system. When selected:
 - * Enter the printer Hostname in the **Name** field.
 - * Enter the printer IP Address in the **IP Address** field.

Note: A "PRINT SERVER" IS NOT SUPPORTED (i.e., a computer with a printer attached); a Network Image Printer (which is supported) is a printer attached directly to the network.

- g.) Camera Setup:

Comment: Camera Setup is not performed using Reconfig, but rather is performed from the Service Desktop once Applications have been started. See "Declaring DICOM Filming Devices on the System" on page 264.

- 10.) Select the **NETWORK** button to display the Network Settings Screen as shown in [Figure 6-16](#).

Figure 6-16 Network Settings Screen

- 11.) Configure Network Settings

Comment: This screen provides the ability to declare the LightSpeed system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for LightSpeed systems on a subnet) must be obtained from the hospital network administrator.

See Chapter 12 for more information and complete details of setting the Hospital/System Network Configuration.

- a.) Enter the Suite Name.

The Suite Name must start with a letter, followed by three alphanumeric characters. Total must be four characters long. The name of the OC interface will be <Suite Name>, within the scanner's subnet.

Typically, you should use su01 or ct01 ("su" and "ct" must be lowercase), unless the customer prefers a different suite name.

- b.) Enter the hospital provided Host Name.

The Host Name identifies the network hostname and AE Title of the LightSpeed system to the hospital's network. The Host Name:

- * **MUST NOT** exceed 16 Characters.
- * **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _.

Comment: The Host Name is typically stmary or ct01.

- c.) Enter the hospital provided IP Address for the system.
d.) Enter the hospital provided Net Mask (if the LightSpeed system is on a subnet).
e.) Enter the Broadcast Address.

Comment: The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1's or 0's depending on the configuration of the network. The standard default is 1's, but older Sun OS machines used 0's.

For example:

If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the Broadcast Address.

If the network contains genesis based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.

- f.) Enter the hospital provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.
- g.) Select NIS (a.k.a. Yellow Pages database) Advanced Option—only if requested by the hospital network administrator—as follows:
 - * Select ADVANCED OPTIONS button on the Network Settings Screen.
 - * Select USE NIS button.
 - * Enter the hospital provided Domain Name (from the hospital network administrator).
- 12.) Select ACCEPT on the LightSpeed 3.X System Configuration Screen.

Comment: The system loads the CT Application Software, OS patches, kernel changes and configures the system on the OC.

The loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window that closes when the loading process is complete. All the window output is logged to a file named:

- 13.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES. (See [Figure 6-17](#).)



Figure 6-17 Reboot Screen

- 14.) The system will automatically login as ctuser after the reboot. Select OK on the Autostart Disabled popup message.

3.7 Applications Start-Up

Open the Console shell window, and type: `st` ENTER. The applications desktop appears on the monitor.

Chapter 7

Table and Gantry Integration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

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- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 14](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Introduction

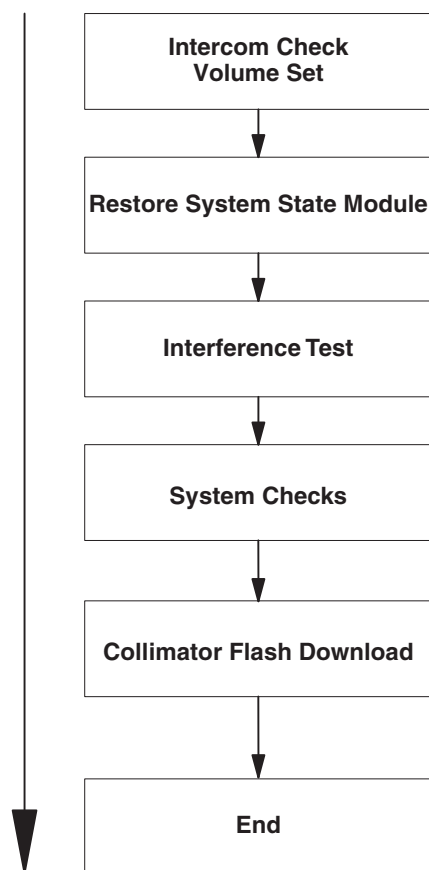


Figure 7-1 Table/Gantry Integration Flow Chart

REQUIRED TOOLS

- Multimeter
- Scope (or Multimeter with Frequency Measurement Capability, such as the FLUKE 87.)

Section 2.0

Autovoice/Intercom Check

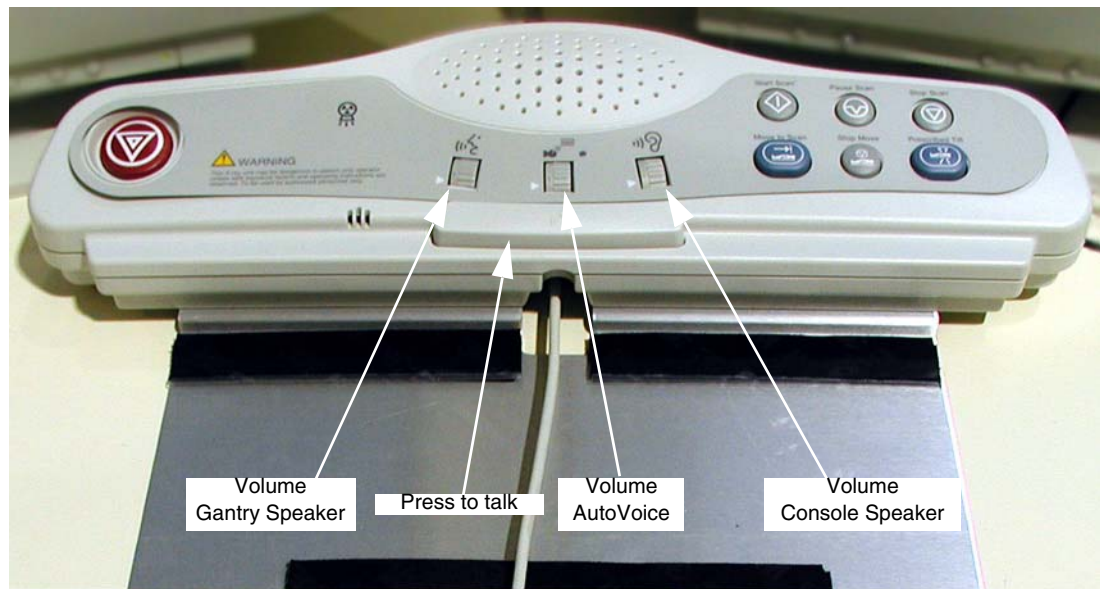


Figure 7-2 SCIM Volume Controls

2.1 Gantry Speaker

To adjust the gantry speaker volume, adjust the left-most volume thumb wheel while speaking into the console microphone (press the bar on the SCIM).

2.2 Console Speaker

To adjust the console speaker volume, adjust the right volume thumb wheel until a person speaking near the gantry can be clearly heard.

2.3 Autovoice Volume

- 1.) On the Scan Desktop, select PROTOCOL MANAGEMENT.
- 2.) Select AUTO VOICE RECORD.
- 3.) Click on the 3.4 button, to the right of "FF2. Inspiration".
- 4.) Click on the PLAY button, to play the Inspiration AutoVoice message.
- 5.) Adjust the center volume thumb wheel, while Autovoice is playing, to set the volume for the gantry speaker.
- 6.) Repeat steps 4 and 5, as necessary to achieve satisfactory volume.
- 7.) Select DONE, then Select QUIT.

Section 3.0

Table/Gantry Characterizations

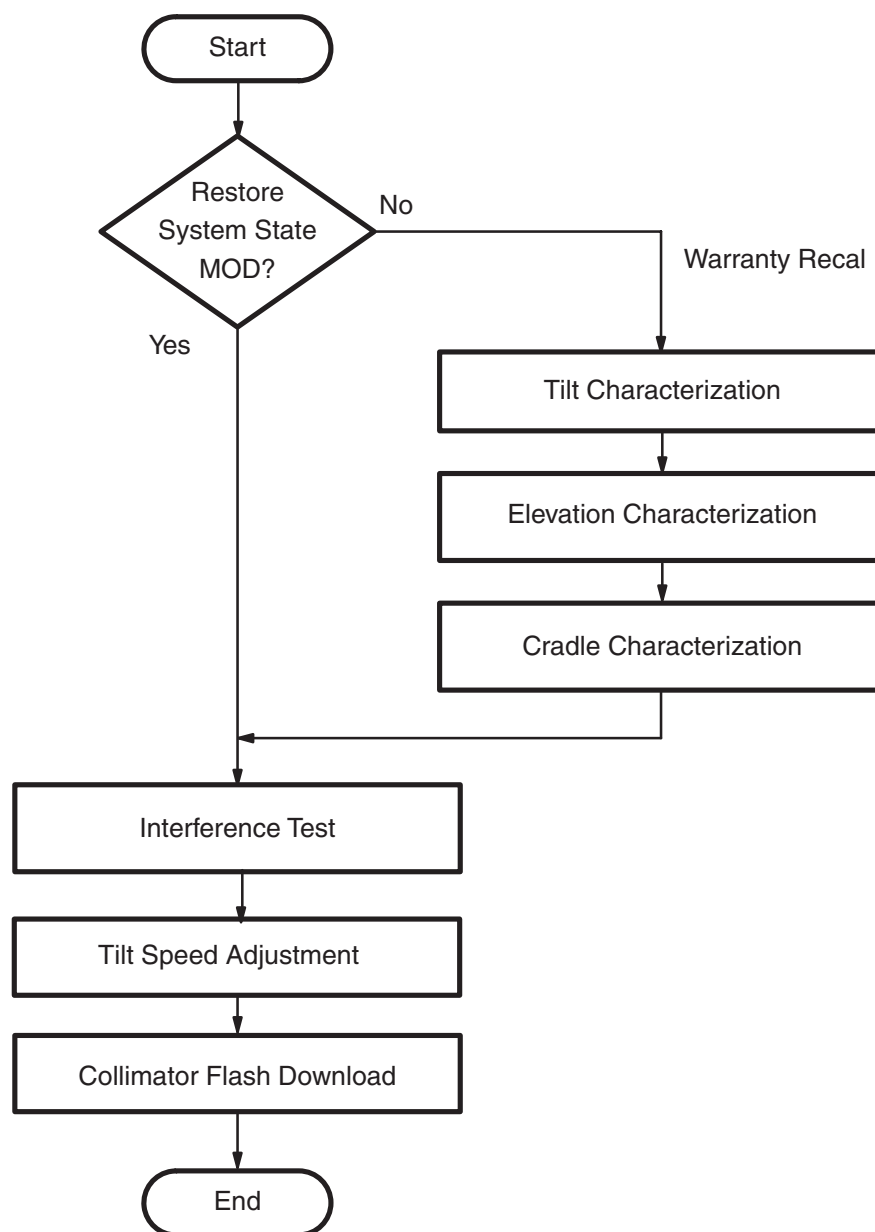


Figure 7-3 Characterization Flowchart

Section 4.0

Interference Test

Reset the hardware to download the new characterization values to the ETC, ICE, and STC before proceeding to the table/gantry interference test on this page.

If the following elevation and/or tilt display readings fail to meet specifications, DO NOT adjust the limit switches. Instead, re-characterize elevation and/or tilt.



NOTICE **Interference Test Requirement: No tilt motion shall cause the gantry tilting frame to touch the stationary base covers for any tilt angle.**

- 1.) Select HARDWARE RESET.

Press only the **fast** push-button to display the elevation limits for the present elevation/tilt combination.

- a.) Move the cradle to the home position.
- b.) Use the DOWN pushbutton to lower the table to the minimum height.
- c.) Make sure the elevation display reads 500.0 $\pm$ 3 mm.
- d.) Use the UP pushbutton to raise the table to the maximum height.
- e.) Make sure the elevation display reads 25.0 $\pm$ 3 mm.

Note: If this step fails, check table/gantry level, and run table elevation characterization.

“I” tilt means the top of the gantry tilts away from the table. “S” tilt means the top of the gantry tilts toward the table.

- 2.) Advance the cradle about 36 inches from the home position.
 - a.) Tilt the gantry to I23 degrees.
 - b.) Use the DOWN pushbutton to lower the table until it automatically stops.
 - c.) Make sure the elevation display reads 115.5 $\pm$ 3 mm.
 - d.) Make sure the table clears the gantry covers by at least ½ inch.
- 3.) Tilt the gantry to I20 degrees.
 - a.) Use the DOWN pushbutton to lower the table until it automatically stops.
 - b.) Make sure the elevation display reads 210.0 $\pm$ 3 mm.
 - c.) Make sure the table clears the gantry covers by at least ½ inch.
- 4.) Use the UP push-button to raise the table until the elevation display reads between 115.5mm and 210mm.
 - a.) Tilt the gantry to I10 degrees.
 - b.) Tilt the gantry away from the table; make sure it stops at I22.0 $\pm$ 0.5 degrees.
 - c.) Tilt the gantry toward the table; make sure it stops at S18.5 $\pm$ 0.5 degrees.
- 5.) Return the cradle to home position.
 - a.) Make sure you can tilt the gantry to 30.0 $\pm$ 0 degrees.
 - b.) Leave gantry tilted to I30 degrees for now.
- 6.) Raise the table until you can move the cradle.
 - a.) Advance the cradle about 12 inches from the home position.
 - b.) Use the HOME Button to return the cradle to the home position, tilt the gantry to 0 degrees, and lower the table to minimum height.

Make sure you cannot move the cradle at table heights below 210.0.
 - c.) Verify that the *cradle lock* indicator on the gantry display is lit.
- 7.) Select FILE —> EXIT from the pull-down menu.
- 8.) Select CLEANUP.

Section 5.0

Tilt Speed Adjustment Procedure

- 1.) Remove rear gantry base covers.
- 2.) Using a stop watch or suitable time piece adjust for one (1) degree per second motion.



NOTICE Adjustments are very sensitive. Improper speed adjustments will result in overspeed error generation.

Note: Tilt speed will vary based upon hydraulic fluid temperature. Adjustments should be made at normal scan room temperature settings.

- 3.) From zero (0) degrees tilt, press and hold the tilt forward button,  until gantry display reads S25. Observe tilt speed using time piece.
- 4.) Press and hold backward tilt button,  until gantry display reads I25. Observe tilt speed using time piece.
- 5.) Adjust forward speed control valve for one (1) degree per second motion. See [Figure 7-4](#).
- 6.) Adjust backward speed control valve for one (1) degree per second motion. See [Figure 7-4](#).
- 7.) Repeat steps 3 through 6 until tilt speed for both directions is correct.
- 8.) Now tilt the gantry forward to S30. Observe the S25 to S30 speed. Do this several times. If a noticeable difference is observed, then check the hydraulic fluid levels.

Note: This is a self bleeding system. Trapped air can cause slowed or limited tilt range. Exercising the full range of motion several times should purge any trapped air from the hydraulic system.

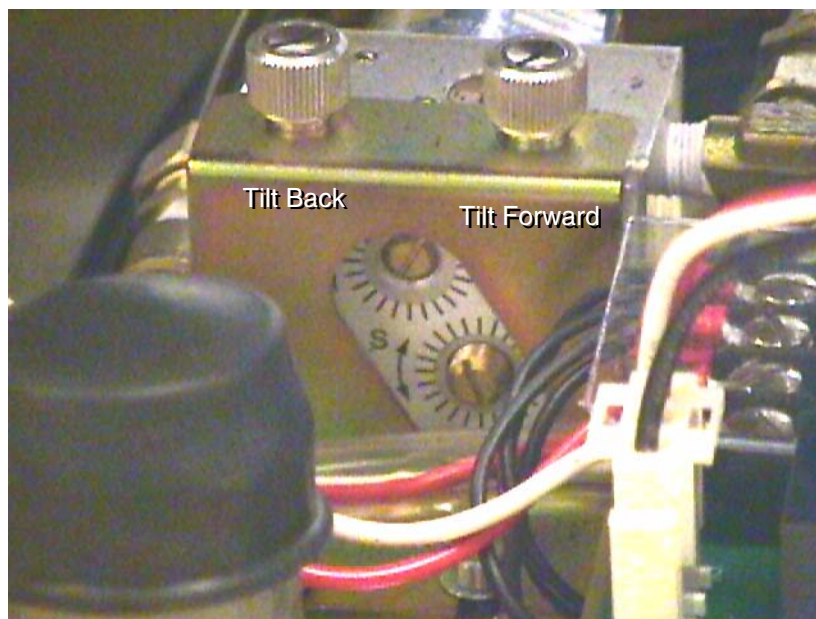


Figure 7-4 Tilt Speed Adjustment Screws

Section 6.0 System Checks

6.1 Alignment Lights

6.1.1 Turn Alignment Lights ON

-  **CAUTION** Verify all personnel have cleared the system. The Gantry rotates during this check.
- 
- 1.) Turn ON the Axial Drive Enable and HVDC Enable switches (located on the STC backplane – refer to [Figure 5-7, on page 166](#)).
 - 2.) Press the alignment light button on the gantry control panel.
The gantry will rotate and the lights will turn ON.

-  **WARNING** **NEVER STARE DIRECTLY INTO THE LASER BEAMS WHEN YOU OPERATE THE ALIGNMENT LIGHTS. STARING INTO THE BEAMS CAN CAUSE PERMANENT EYE INJURY DAMAGE.**

6.1.2 Internal Axial Lights

- 1.) Place a sheet of plain white paper over the output port of each light.
 - 2.) Verify that the two laser lines coincide and appear as a single line.
- Note: **GE designed the internal axial lasers on the current CT system to shine *down* on the collimator. Do NOT adjust the internal alignment lights at this time. The tomographic plane tests use the QA phantom to check the internal axial lasers alignment to the collimator.**

6.1.3 External Axial to Internal Axial Distance

- 1.) Raise the table to its highest elevation.
- 2.) Extend the cradle until you see both the internal and external laser lights shining on the cradle.
- 3.) Place a metric rule on the right edge of the cradle, and measure the distance from the internal axial laser line to the external axial line. Verify this distance equals 240.0 mm $\pm$ 1.0 mm.
- 4.) Place the rule on the left edge of the cradle, and measure again.
- 5.) Leave the cradle in its current position, and lower the table to the minimum elevation.
- 6.) Measure the distance between the internal and external lights on both edges of the cradle, as above. Verify the distance remains equal to 240.0 mm $\pm$ 1.0 mm.

6.1.4 Coronal Lights

- 1.) Place a sheet of plain white paper at the left side of the patient opening, in front of the coronal laser light. Verify the two coronal lines coincide.
- 2.) Move the paper to the right side of the patient opening. Verify the two coronal lines coincide.
- 3.) Place the paper in the center of the gantry opening. Use a level to verify that the coronal lines are horizontal.

6.1.5 Turn Lights OFF

- 1.) Press the alignment light button on the gantry control panel, again, to turn the lights off.
- 2.) Turn OFF the axial drive enable and HVDC enable switches.

6.2 Laser Warning Labels

- 1.) Make sure all laser warning labels appear in the correct location on the outside of the gantry.
- 2.) Obtain and install replacements for any missing labels.



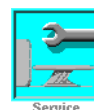
Figure 7-5 Laser Warnings and Precautions

Section 7.0 Collimator Characterization & Flash Download

The Collimator Flash Download diagnostic compares the values in flash memory on the collimator control board with the values stored on the system, and updates if necessary. If the memory values don't match, the diagnostic prompts you to type/enter the collimator serial number to complete the update.

To prevent invalid test results, remove any obstructions from the gantry opening.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon,



- 2.) Click on the UTILITIES icon,
- 3.) Select INSTALL.
- 4.) Select FLASH DOWNLOAD TOOL.
- 5.) Select UPDATE.
- 6.) Look for "Download Successful" message.
- 7.) Select DISMISS when finished.



Note: New installations upload values from FLASH.

Chapter 8

DAS Integration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 14](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.
- Refer to DAS Cleaning in the GE Medical Systems *LightSpeed 3.X System Service Manual*

Section 1.0 Introduction

1.1 Failure Protocol

Follow standard troubleshooting protocol to verify all failures.

- If you are moving boards, make sure you follow special handling procedures (e.g., use ESD protection).
- Record all board moves on the DAS board tracking sheet.
- When you verify a failure, identify the defective part and remove it from the system.

Before running any verification tests, turn on the DAS and let it warm up. It takes two (2) minutes of warm-up time for every minute that the DAS was turned off, up to two (2) hours.

If you encounter failures in any section, restart the procedure at the DASTools.

1.2 Special Handling Precautions



NOTICE Potential for Static Electricity Damage



DAS circuit cards are more sensitive to both STATIC DAMAGE and DIRT than any other circuit cards in the LightSpeed system. Please read Sections [1.3](#) and [1.5](#), and observe all precautions.

1.3 Anti-static Handling

You **MUST** wear a grounding wrist strap anytime you remove the boards from the DAS or a conductive bag. **NEVER** place the DAS boards on non-conductive surfaces, such as white bench tops, plastic covered books, or non-conductive plastic bags.

When removing the covers from the DAS, put on a wrist strap, and push all the DAS boards firmly into their slots on the DAS backplane. Do **NOT** remove and re-seat any boards at this time.

1.4 Cleanliness

Any dirt on the surface increases leakage current on the converter cards and causes the DAS to fail the drift spec. Dirt also causes microphonic failures, which cause image artifacts. Wear anti-static finger cots (part number 2221235 [small] and 2221235-2 [large]) when you handle the DAS, because fingerprints on the board can cause problems during humid conditions. Use only clean, new anti-static finger cots.

Follow cleaning and de-ionizing procedures in the System Manual.



NOTICE NEVER USE AN ERASER TO CLEAN ANY PART OF THE DAS.

1.5 Required Tools

TOOLS REQUIRED

- Ground Wrist Strap
- Aero Duster, Part #MS-222N (if cleaning is needed) with static-free metal nozzle inserted
- Lint-free towels, Part #46-307237P1
- Amax Contact and Circuit Cleaner, Part #2274104
- DAS/DET Interface Kit 2227089 (if cleaning is needed)

1.6 Process Overview

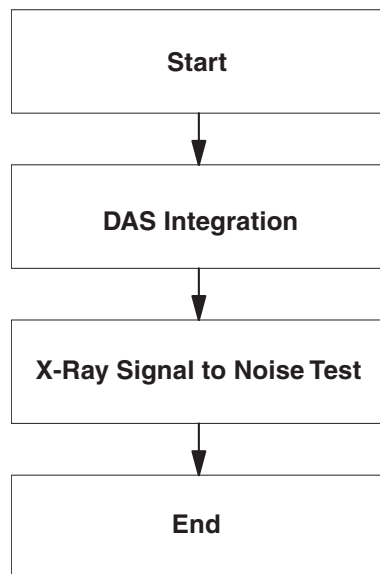


Figure 8-1 DAS Integration Flow Chart

Section 2.0

DAS Tools



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select DAS INTEGRATION.
- 5.) Select DAS TOOLS to open the corresponding menu window.
- 6.) Select ACCEPT.

DAS Tests

Auto Test

Manual Test

Interconnect Test

Pop Noise Microphonics Test

Auxillary Channel Test

X-ray Verification Test

DAS Viewers

DAS Architecture

View Log

View Specification

Dismiss

Test Name	Number of Iterations	Remaining Iterations	Total Successes	Total Failures
DC CAL Absolute	1	1	0	0
Offset Drift	1	1	0	0
Pop Noise & microphonics	10	10	0	0
DC Noise & DC Offsets	1	1	0	0

Options

Number of repetitions for Group

Delay between groups (msecs.)

Total Failure before stopping

Remaining

1

1

200

128

128

Accept

Stop Tests

Figure 8-2 DAS Tools Screen (Auto Test Selected)

- 7.) Select AUTO TEST to run all available DAS Tools.
- Use the default iteration values.
 - The auto test displays **comfort dots** to show activity, and displays iteration pass/fail information on the screen.

-or-

Manually select the following tests:

- DCCal: 1 iteration
- Offset Drift: 1 iteration
- Pop Noise: 10 iterations
- Microphonics: 10 iterations (See [Figure 8-3.](#))
- Offset Means: 1 iteration
- Offset Noise: 1 iteration

Auto Test takes approximately 35 minutes to complete 1 pass.

- 8.) After DAS Tools successfully completes one pass of the auto test, or the designated number of manual test iterations, select DISMISS to exit the test. If the Total Failures column lists any failures, troubleshoot before proceeding.

Test Name	Number of Iterations	Remaining Iterations	Total Successes	Total Failures
Pop Noise & microphonics	10	10	0	0

Options

Pop Noise & microphonics Options

☒ Rotor On ☐ Rotating
☐ Rotor Off ☐ Non-Rotating

Number of repetitions for Group: 1 Remaining: 1 Accept
 Delay between groups (msecs.): 200
 Total Failure before stopping: 128 128 Stop Tests

Figure 8-3 DAS Microphonics Test screen

Section 3.0

Verify X-Ray/Signal to Noise Test

This section uses X-Ray to test the DAS for opens, lows, and shorts.

System Configuration:

- All covers ON
- Nothing in detector FOV

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon,



- 2.) Click on the UTILITIES icon,



- 3.) Select INSTALL.

- 4.) Select DAS INTEGRATION.

- 5.) Select START.

- 6.) Select X-RAY VERIFICATION.

- 7.) Select X-RAY VERIFICATION (DDC).

- 8.) Follow on-screen instructions to acquire each set of X-ray verification scans, for a total of four sets.

- 9.) Select ACCEPT RX.

- 10.) Press **START SCAN** when it flashes.

- 11.) Select DISMISS to close the X-ray verification window.

- 12.) Select X-RAY VERIFICATION ANALYSIS.

- 13.) If X-ray verification scans do not pass, troubleshoot with DAS Tools and System Service Manual before continuing.

- 14.) Complete the F4879 Tube Signal/Output check.

Chapter 9

System Alignments



NOTICE Potential for Data Loss and/ or Equipment Damage

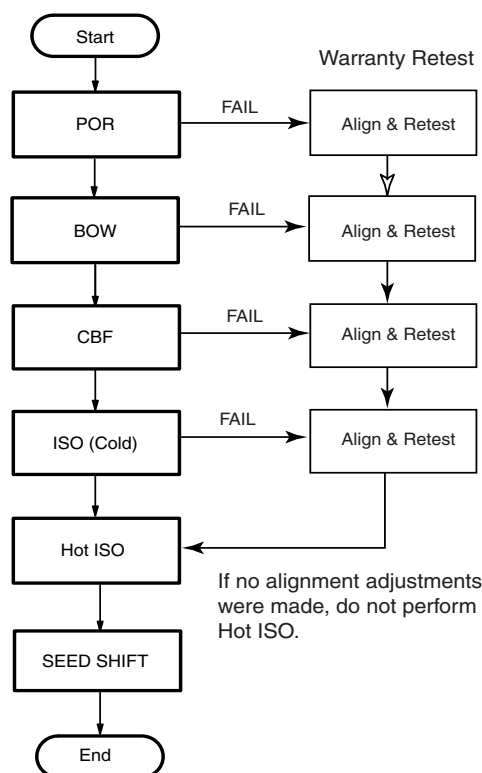
To prevent potential data loss or equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 14](#) of this book.
- Locate the HHS data sent with your LightSpeed 3.X scanner. It can be found in the software collector carton. Retain this data with the HHS data collected at the site in a safe place.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Note: For each adjustment procedure, do not loosen anything until after you have run your first scan and calculated your first adjustment. Before starting this chapter, the tube must have cooled at least two hours.

Section 1.0 Introduction

Use the procedures in this chapter to make sure the x-ray beam focuses directly upon the detector window. The flowchart in [Figure 9-1](#) shows an overview of this chapter.



IMPORTANT:

If alignment adjustments are made during this procedure, a FULL SET of ALL detail calibrations must be completed before proceeding with the system tests in Chapter 11.

The procedure for completing detail calibrations can be found in the *System Manual*, in the *Tube Change* section.

This should also be Code 10: Install Warranty.

Figure 9-1 System Alignment Overview

1.1 Required Tools

- Polaroid type 52 film
- Film developer
- Caliper Dial indicators
- Masking tape
- Socket set
- Torque wrench
- 48cm Phantom
- Dial indicator mount
- Radial dial indicator

1.2 Gantry Geometry Definitions

Theta: (Also known as ISO) Along the tangent to gantry rotation, on the rotating structure. Move the X-Ray tube to the 180° (six o'clock) position to equal 0° theta. You can adjust the tube unit and collimator in the theta direction.

Z-Axis: Parallel to the axis of gantry rotation, in the direction of cradle motion.

- You can adjust the tube unit and detector in the Z-Axis direction.
- Positive Z-Axis direction: Move the tube/detector toward the table.
- Negative Z-Axis direction: Move the tube/detector away from the table.

Radial: Along the radius, from the center of gantry rotation, in and out from gantry isocenter. You cannot adjust the LightSpeed detector in the radial direction.



Figure 9-2 Gantry Geometry

1.3 Tube Heat Definitions

Cold Tube:	No exposures for more than 6 hr. (overnight)
Cool Tube:	No exposures for more than 2 hr., but less than 6 hr.
Warm Tube:	No exposures for more than 1 hr., but less than 2 hr.
Hot Tube:	Exposures in less than 1 hr.

Section 2.0

Plane of Rotation (POR)

2.1 Overview

The purpose of Plane-of-Rotation measurement/alignment is to put the X-ray tube in the correct physical relationship to the detector and verify it. This is normally only necessary when X-ray tube has been replaced or moved.

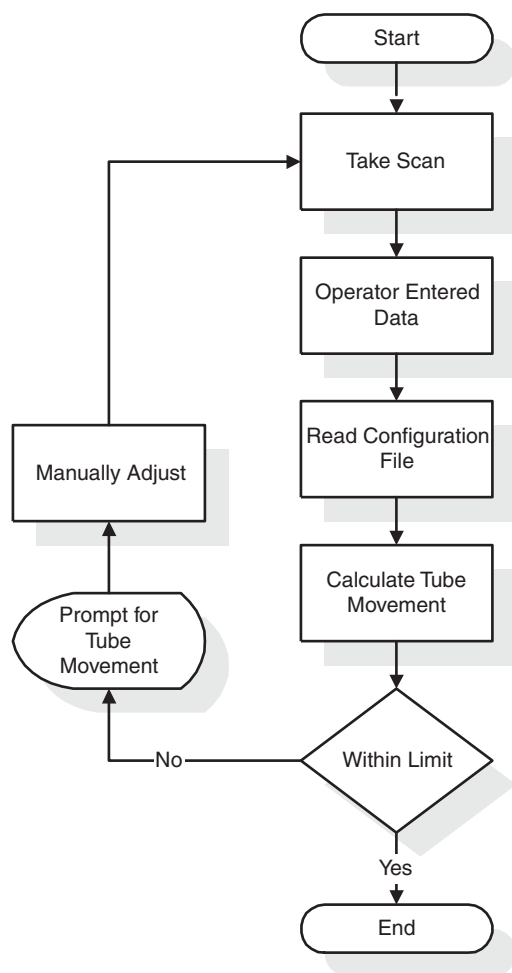


Figure 9-3 POR Procedural Flow

2.2 Tools Required



Figure 9-4 Vernier Caliper & Dial Gauge Indicator

- Calibrated Vernier Caliper (millimeters or inches).
- Dial Gauge Indicator (millimeters or inches)
- Type 52 Polaroid Film (see [Figure 9-7](#))
- Type 52 Polaroid Film Developer

2.3 Procedure

Remember to always perform POR measurements when the X-Ray tube unit is cool.

- If you have acquired any low technique scans prior to beginning the Plane-of-Rotation (POR) alignments, wait at least 5 minutes before starting the following procedure.
- If you recently acquired image scans, and want to re-check the Plane-of-Rotation, wait at least 90 minutes to allow the tube unit to cool. (*Wait 90 minutes, if the tube unit had more than 25 Kilo joules exposure, $[KV \times mA \times Sec \div 1000]$ within the last 30 minutes.*)

2.3.1 Start the POR Software

Note: Correct tube temperature is critical.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Click on INSTALL.
- 4.) Click on the SYSTEM ALIGNMENTS option.
- 5.) Click on the PLANE OF ROTATION option.



Plane of Rotation		System Messages Instructions Log
Scan Execute PLANE-OFROTATION scan		
Calculate Calculate the amount of tube movement		
△ ▽ Prior Next Restart Dismiss		Mount the phantom holder and 48cm phantom on the cradle Tape a type 52 film at 3:00 o'clock position on the outside edge of the 48 cm phantom, so the film projects into the Gantry. Advance the cradle and use the alignment lights to position the film for an exposure near the image center mark on the film pack.
		Save Log

Figure 9-5 Plane Of Rotation Alignment Screen

2.3.2 Measure Tube Alignment

- 1.) Get the system's phantom holder and attach the 48cm phantom to it. Refer to [Figure 9-6](#).

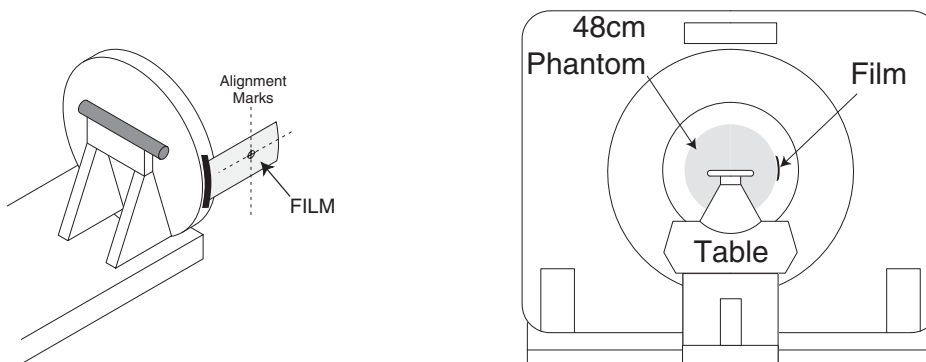


Figure 9-6 48CM Phantom and Film on Table

- 2.) Attach one *Polaroid type 52* film on the outside edge of the 48cm phantom, at the 3 o'clock position.

Note:

Only the film should be projecting into the Gantry bore when complete. The phantom is used only to position and hold the film in the gantry bore. Refer to [Figure 9-6](#).

Orient the side of the film side marked “**This side toward lens**” towards iso center (see [Figure 9-7](#)). When exposed and developed later, the film will show the alignment of the x-ray beam with respect to the table, as viewed from the X-Ray tube in the 3:00 o'clock position.

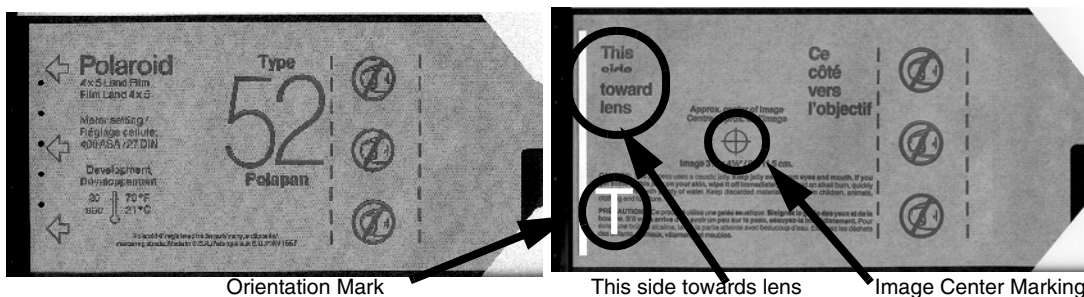


Figure 9-7 Polaroid Film, Type 52

- 3.) Advance the cradle, while using the alignment lights, to position the film's center marking on the alignment light marks.
- 4.) Take a scan.
- 5.) Remove the exposed film and immediately mark the outside of the film nearest the table. For example, use a pen and print the letter **T** on the film nearest the table ([Figure 9-8](#)). Go and develop the exposed film.



Figure 9-8 Placement of “T” on Film

- 6.) After the film is developed, transfer the table marking to the film itself while keeping the orientation correct (Figure 9-9).

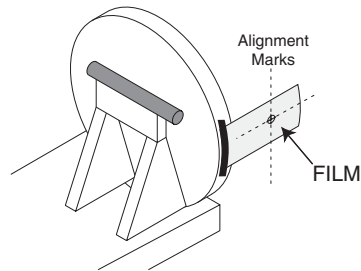


Figure 9-9 Alignment Marks on Film

- 7.) Verify the following (refer to Figure 9-10):
- Film's narrow (white beam) slit lies within (between) the wider (gray) X-Ray slit.
 - Film's edges in both Z direction are equally well defined by the exit slit of the collimator. The edges of the narrow beam should be much sharper than the wide beam.

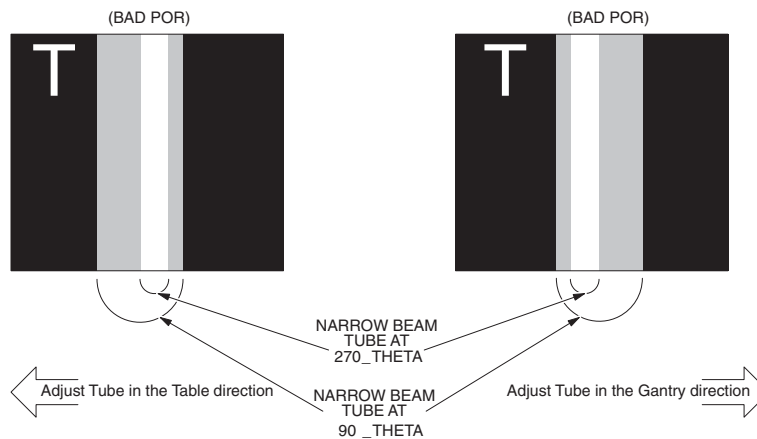


Figure 9-10 Plane of Rotation Film Interpretation

- 8.) Refer to Figure 9-10 and Figure 9-11 during the following instructions. Use a vernier calipers to measure the width of the two wider (dark gray) slits. They're the dark gray slits that extend past the edges of the narrower (off-white) slit and to the blackest part of the film.

- a.) Take three measurements to obtain an average value for X_F . X_f is the side of the film closest to the table. Using the same film measurement tool, take one X_f measurement at the top of the film, another near the middle and another near the bottom. Add these three X_f distances together and divide this sum by 3 or n . Where,

$$X_F = (X_{f1} + X_{f2} + X_{f3}) / (n)$$

Note:

It is important that you take multiple measurements, to reduce the likelihood of a measurement error and consequently increase the accuracy of the alignment.

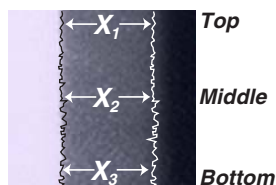


Figure 9-11 XF and XR measurement points

- b.) Now take three measurements to obtain an average value for X_R . Take one X_R measurement at the top of the film, another near the middle and another near the bottom of the film. Add these three values together and divide the sum by 3 or n to obtain an average for X_R .

$$X_R = (X_{r1} + X_{r2} + X_{r3}) / (n)$$

- c.) Use the values obtained for X_F (front distance) and X_R (rear distance) in the calculation that follows.

- 9.) Click on the CALCULATE button (refer to [Figure 9-5](#)). Enter the values for X_F and X_R obtained in the steps above. The software will do the distance calculation.

Note: **The specification limits are 0.059 - 0.082 inches or 1.50 - 2.083 millimeters.**

- 10.) Check the results
- If the tube is out of specification, perform steps 1 and 2 ([page 207](#)), and move the tube. See [Figure 9-12](#) for more details.
 - If adjustment is necessary, clockwise rotation (in) of adjustment bolt moves the tube towards the table side. Repeat Steps 1 through 9 again if you have moved the tube, to check accuracy of adjustment. See [Figure 9-10](#) for more details.

Note: Mount the dial indicator onto the collimator/tube assembly as shown [Figure 9-12](#). Make sure you zero the dial indicator when it's securely in place.

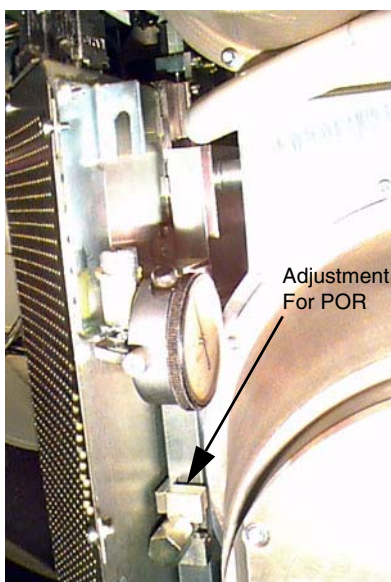


Figure 9-12 Dial Indicator Attached to Collimator/Tube Assembly

- When the tube is within specification, tighten the (4) M-12 Bolts on the tube and torque them to 50 ft-lbs (69 kg-cm) and you're done.
- 11.) Enter the results in the HHS table.

Section 3.0

Beam on Window Alignment (BOW)

3.1 Overview



NOTICE
Potential for
Image
Artifacts

Make sure the tube is cool before continuing. Wait 15-30 minutes after the completion of Plane-of-Rotation.

Purpose: Bring the detector into alignment with the x-ray beam of the now correctly positioned tube. There are two specifications for BOW Alignment. The software expects the user to evaluate the raw vectors to determine if it meets the requirements.

- The first specification is for the initial check of BOW. The limits are $-1.5 \pm 0.25\text{mm}$.
- The second specification is for verification scans when the system is warmer. The limit for verification scans is $-1.5 \pm 0.5\text{mm}$.

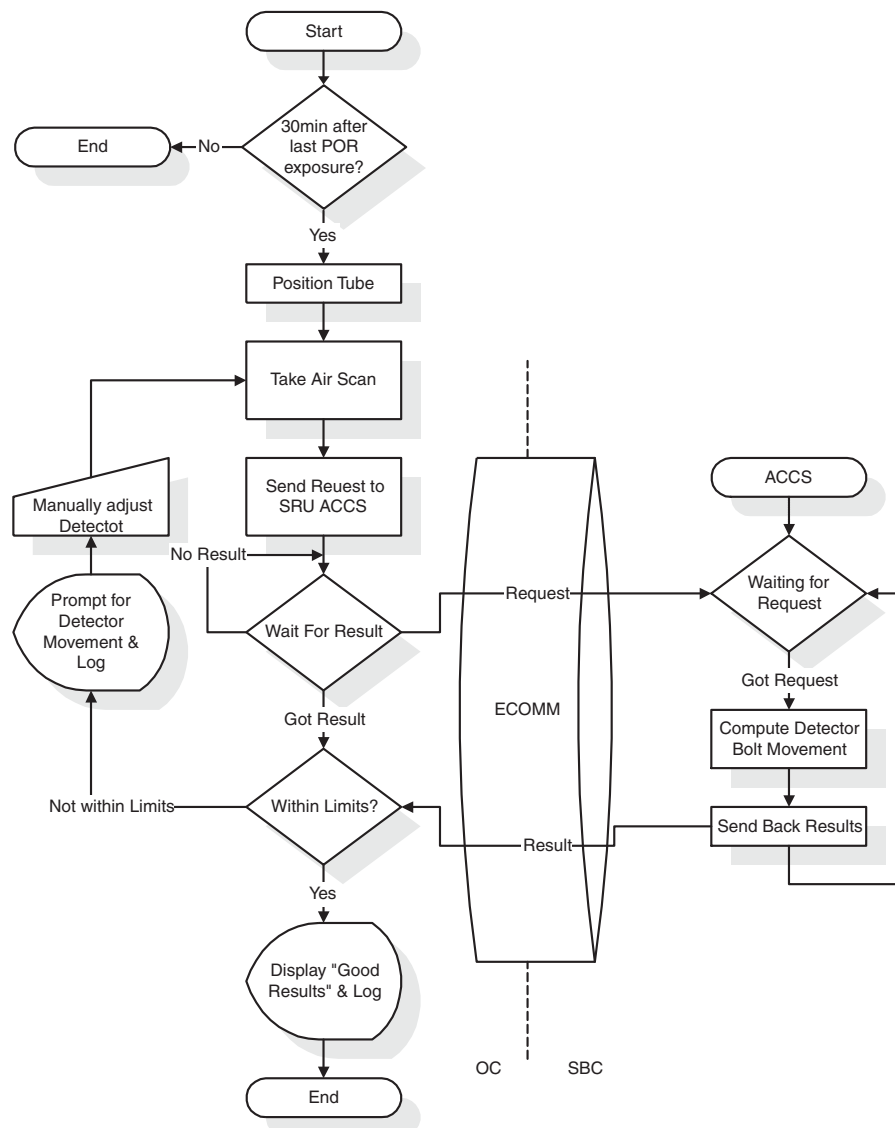


Figure 9-13 BOW Alignment

3.2 Cool Tube

Make sure the tube is cool before continuing. Wait 10 minutes per shot taken after the completion of Plane-of-Rotation.

3.3 Access the Software

Never position the 1/8 inch metal shaft directly on the ISO center - always offset it. The software calculates tube movement for IOS center and looks for a sinusoidal waveform generated by the off center of the metal shaft. A metal shaft, positioned at IOSO center, generates a waveform consisting of a straight line.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select SYSTEM ALIGNMENTS.
- 5.) Select BEAM ON WINDOW.

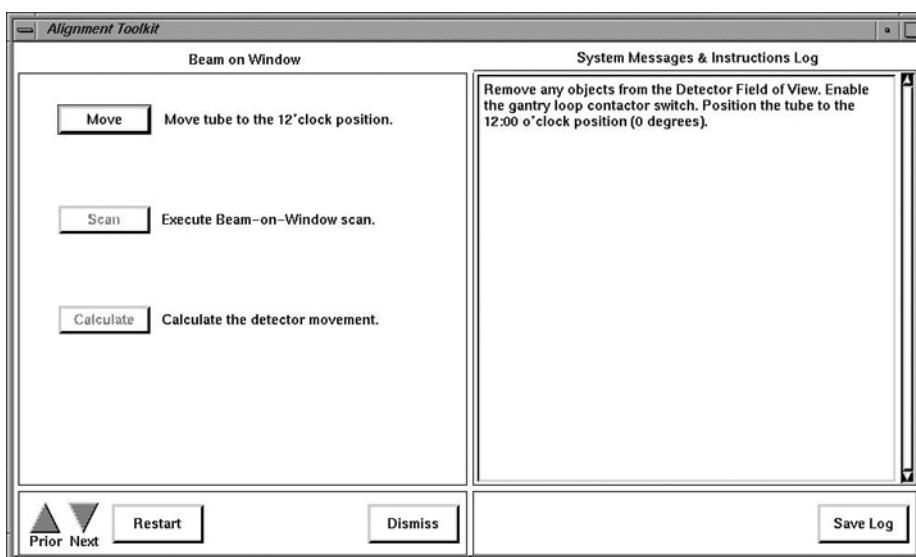


Figure 9-14 BOW screen

3.4 Adjustment Procedure

Tools Required: 21mm wrench
5mm 6"long hex bit
5mm thin wrench

Note: To make the adjustment, you must remove the front and rear covers (step 4).

Note: **Do not loosen anything until after you have run your first scan and calculated your first adjustment.**

- 1.) Select MOVE to send the tube to the twelve o'clock position.
- 2.) Acquire a Beam on Window Scan.

- 3.) Select CALCULATE and make adjustments as indicated by the program.
You'll get screen output similar to the following:
Upon initial inspection: spec = -1.5 +- 0.25mm
Upon verification scan: spec = -1.5 +- 0.50mm
Compare the raw alignment vectors to these specifications, make alignments suggested if specification is not met.
- 4.) Remove the front and rear covers.
- 5.) Refer to [Figure 9-15](#). Loosen the M-6 cap head screws, located at each end of the detector. (total of two cap head screws).
- 6.) Loosen the middle nut (jam nut) with a 21 mm thin profile wrench. Make adjustments, using a 21mm thin open end wrench, as requested by software. Tighten jam nut and cap screws after adjustment.
- 7.) Acquire a Beam on Window Scan.
- 8.) Select CALCULATE.
- 9.) If the adjustments pass the calculation, proceed to step 10, otherwise return to step 4.
- 10.) Tighten all three M-6 cap head screws, using a 5mm x 6" long hex bit, for 3/8" drive. (Torque to 10 ft-lb., or 14 N/m).

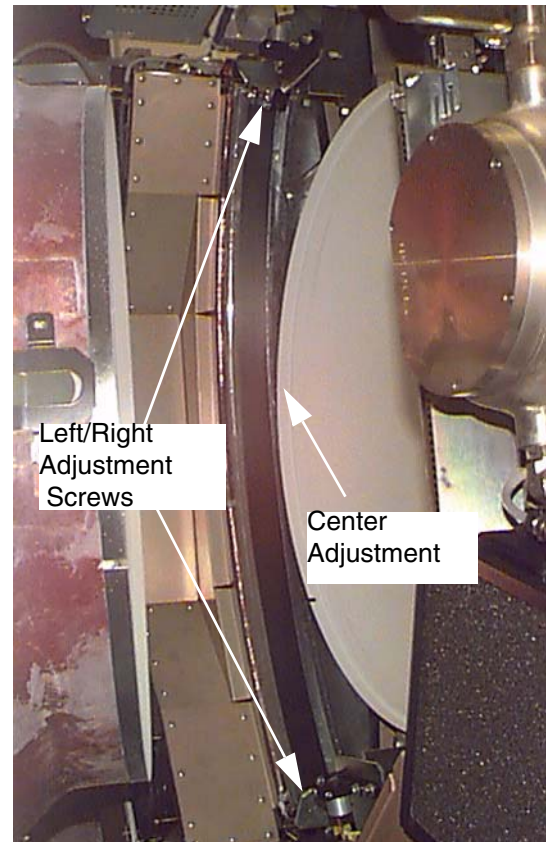
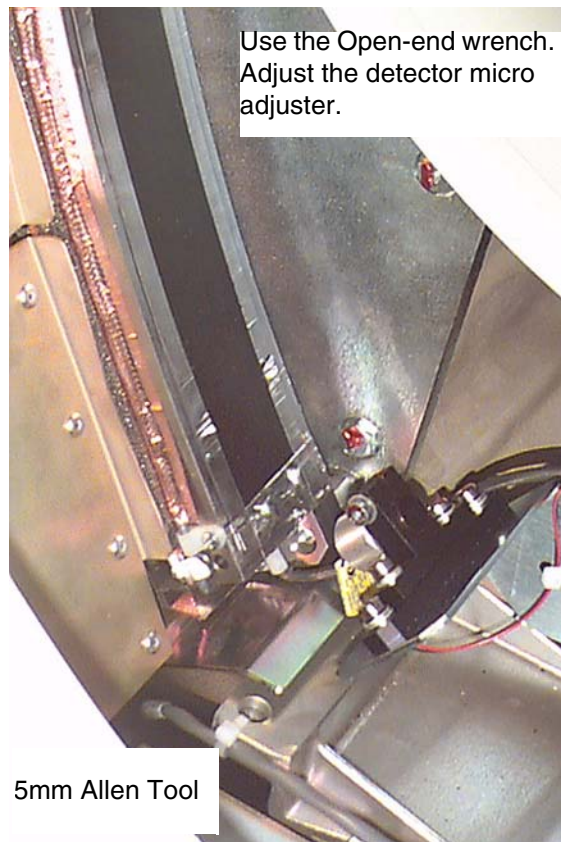


Figure 9-15 Detector Cap Head Screws

Note: **CW (Clockwise) moves the detector towards the mounting plate. CCW (Counter-clockwise) moves the detector away from the mounting plate. Right=Low Chnl, Center=Medium Chnl, Left=High Chnl**

- 11.) Enter results in the HHS table.

Section 4.0

CBF/SAG Alignment Process

Purpose: CBF/SAG Alignment ensures the focal spot is accurate, the bowtie filter is centered, and center of rotation is in a straight line.

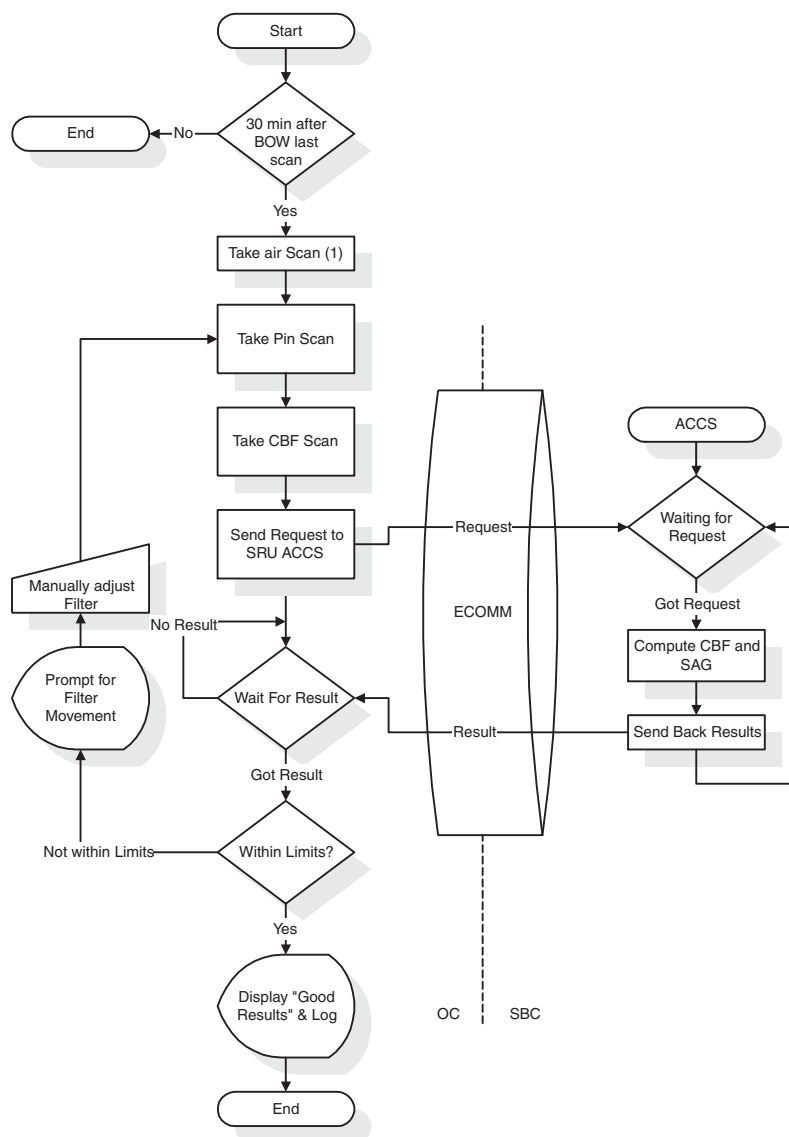


Figure 9-16 CBF Procedure

Never position the 1/8 inch metal shaft directly on the ISO center - always offset it. The software calculates tube movement for IOS center and looks for a sinusoidal waveform generated by the off center of the metal shaft. A metal shaft, positioned at IOSO center, generates a waveform consisting of a straight line.

4.1 Cool Tube

Make sure the tube is cool before continuing. Wait 10 minutes per shot taken after the completion of Beam On Window.

4.2 Access the Software

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select SYSTEM ALIGNMENT.
- 5.) Select CBF/SAG ALIGNMENT.

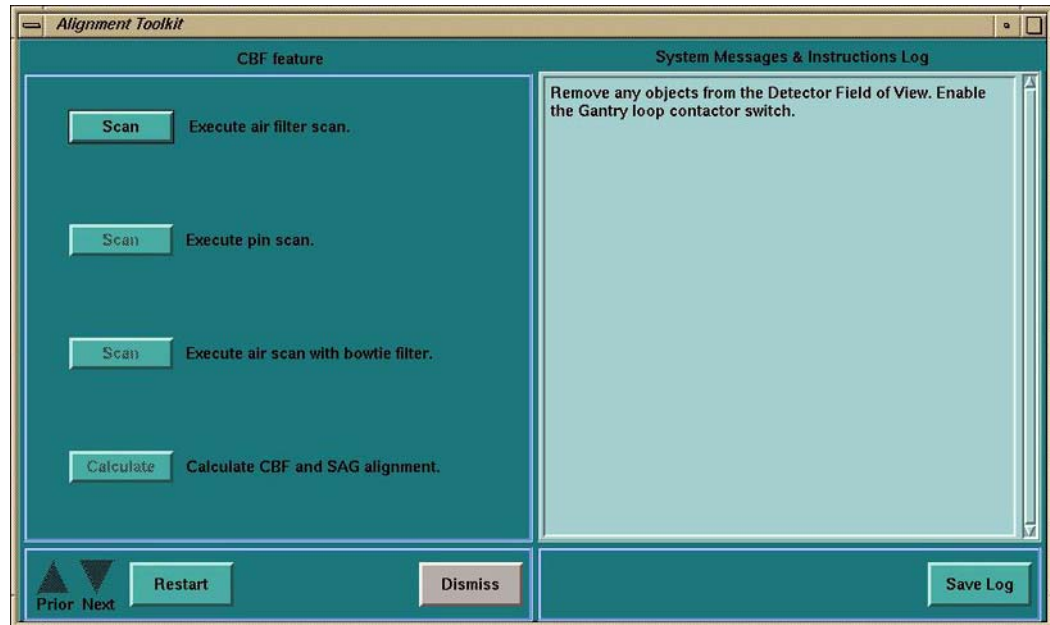


Figure 9-17 CBF/SAG Alignment Screen

4.3 Adjustment Procedure

Tools Required: 1/8 inch PIN (#0 Phillips) screw driver

Note: **Do not loosen anything until after you have run your first scan and calculated your first adjustment.**

- 1.) Press the flashing **START SCAN** button to acquire an air filter scan.
- 2.) Position the 1/8 inch Pin (#0 Phillips round) screw driver on the phantom holder, parallel to the Z-Axis to the end of the table nearest the gantry or on the phantom holder.
- 3.) Turn On the laser lights.
- 4.) Advance and adjust the table to position the metal shaft right of ISO center.
- 5.) Verify that the shaft remains level (perpendicular to the scan plane).
- 6.) Acquire a pin scan.
- 7.) Acquire an air scan with bowtie filter in place.
- 8.) Select **CALCULATE** to calculate the CBF and SAG alignment.
- 9.) Mount the dial indicator onto the collimator.
Remember to zero the dial indicator.
- 10.) Loosen the collimator bolts (4 in front, 2 in the rear). Use a swivel adapter on a ratchet wrench to loosen the bolt behind the cable connections.
- 11.) Adjust the collimator as indicated by the calculation results. Ignore the negative sign in front of the adjustments.
- 12.) Tighten the collimator bolts (4 in front, 2 in the rear). (Torque to 50 ft.-lb.)
- 13.) Wait 15 minutes for the tube to cool.
- 14.) Rescan and calculate.
- 15.) Repeat steps 10 – 11 until the adjustment falls within the limit, then proceed to the next section.
- 16.) Enter results in the HHS table.

Section 5.0 ISO Alignment

5.1 Overview

Never position the 1/8 inch metal shaft directly on the ISO center - always offset it. The software calculates tube movement for IOS center and looks for a sinusoidal waveform generated by the off center of the metal shaft. A metal shaft positioned at ISO center generates a straight line waveform.

5.2 Cool Tube

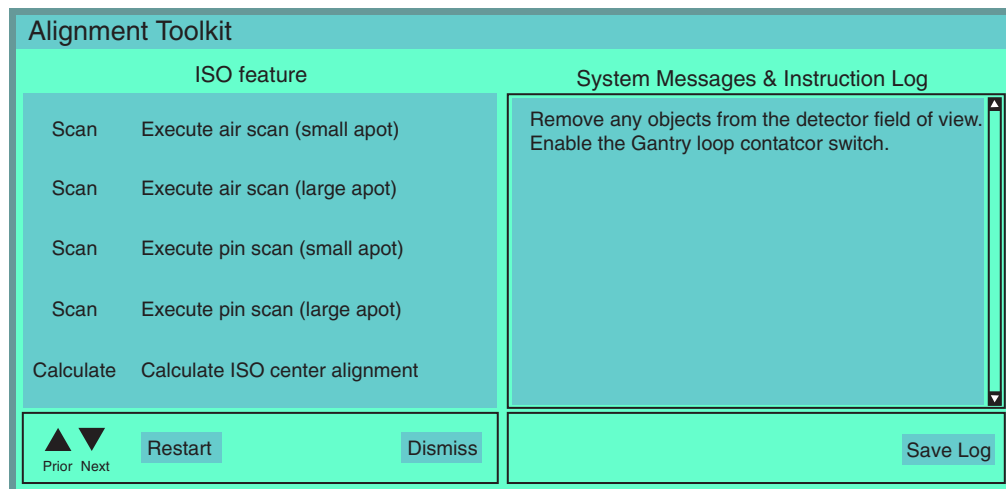
Make sure the tube is cool before continuing. Wait 10 minutes per shot taken after the completion of CBF/SAG Alignment.

5.3 Access the Software

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select SYSTEM ALIGNMENTS.
- 5.) Select ISO.



↑ Use the Prior/Next Buttons to Access the Calculate Button

Figure 9-18 ISO Alignment Menu

5.4 ISO Adjustment Procedure

Tools Required: 1/8 inch PIN (#0 Phillips) screw driver

Note: Do not loosen anything until after you have run your first scan and calculated your first adjustment.

- 1.) Acquire Air scan (small spot). (Refer to [Figure 9-18.](#))
- 2.) Acquire Air scan (large spot). (Refer to [Figure 9-18.](#))
- 3.) Position the 1/8 inch Pin (#0 Phillips round) screw driver on the phantom holder in parallel with the Z-axis to the end of the table nearest the gantry or on the phantom holder.
- 4.) Turn On the laser lights.
- 5.) Advance and adjust the table to position the metal shaft right of ISO center.
- 6.) Verify that the shaft remains level (perpendicular to the scan plane).
- 7.) Acquire a small spot pin scan.
- 8.) Acquire a large spot pin scan.
- 9.) Select CALCULATE ISO. (If ISO falls outside 389.75 channels, no wait between tube adjustment and the test scan is required. If ISO falls between +/- 0.02 and +/- 0.1 channels, wait 15 minutes after you adjusted the tube before you scan.)
 - a.) If adjustments are required, proceed to step 10
 - b.) If adjustments are not required (pass), skip to step 15
- 10.) Zero the dial indicator, and mount it on the tube assembly.
- 11.) Using a 10 mm hex, loosen the four (4) M-12 bolts on the tube assembly.
- 12.) Refer to [Figure 9-19.](#)
 - a.) Locate the adjustment bolt on the top of the tube.
 - b.) Adjust the tube UP/DOWN as indicated by the calculation.
- 13.) Repeat steps 1 through 7.
 - If adjustments fall within the acceptable range, proceed to step 13.
 - If adjustments fall outside the limits, wait 15 minutes for the tube to cool and then return to step 9.
- 14.) Tighten the 4 M-12 bolts using a 10mm hex (Torque to 50 ft.-lb.).
- 15.) Enter results in the HHS table.

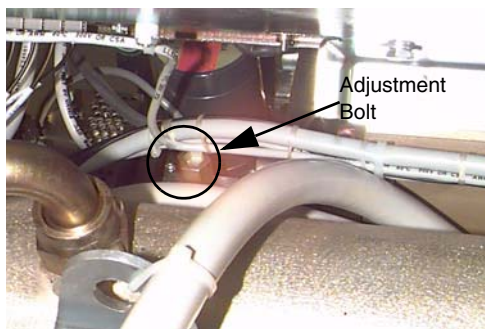


Figure 9-19 ISO Adjustment Bolt

- 16.) Wait 15 minutes, then recheck BOW to verify that the Beam-on-Window values are within the drift spec of -1.5mm +/- 0.5 mm.
 - If BOW is within spec, DONOT adjust anything
 - If BOW is out of spec, alignments must be done starting with POR and ending with BOW recheck.

Section 6.0

HOT ISO Alignment

THIS PROCEDURE SHOULD BE COMPLETED IF AND ONLY IF ANY ADJUSTMENTS WERE MADE.

6.1 Accessing the Software

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select SYSTEM ALIGNMENTS.
- 5.) Select HOT ISO ALIGNMENT.

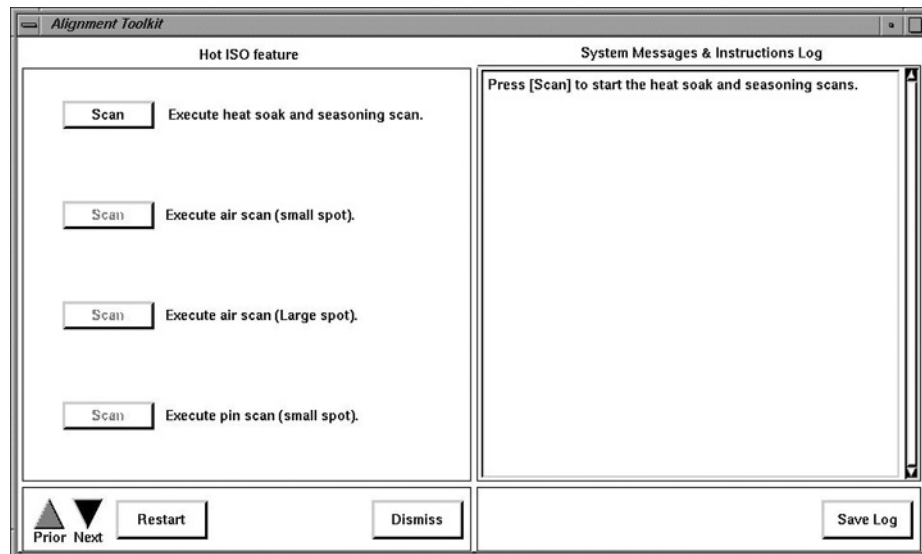


Figure 9-20 HOT ISO Screen

6.2 Adjustment Procedure

Note: No movement of the tube is required. This correction is a software correction done automatically.

- 1.) Execute the Heat Soak and Seasoning scan.
- 2.) Execute Air Scan (small spot).
- 3.) Execute Air Scan (large spot).
- 4.) Place the 1/8 inch screw driver on the phantom holder (should be pointing into the Z direction).
- 5.) Execute Pin Scan (small spot).
- 6.) Execute Pin Scan (Large spot).
- 7.) Calculate correction factor with Calculate button.

Chapter 10

Generator Calibration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 14](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

1.1 Process Overview

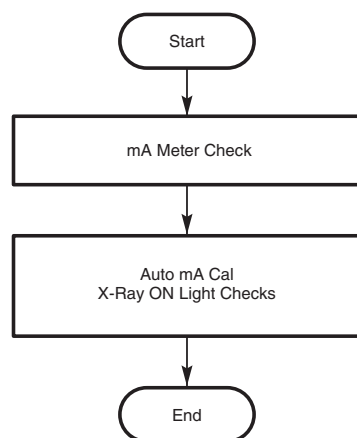


Figure 10-1 Chapter Flowchart

IMPORTANT:

- You must record the data collected in this chapter on **Form F4879**, located in Chapter 14 of this book.
- Locate the HHS data, sent from the factory, in the software collector box.
- Retain this data with the HHS data collected at the site.
- A complete generator RECAL is not required.

Note: If any one of the calibration procedures fail, data collected is different, or you replace any component in the High Voltage section, YOU MUST RECALIBRATE THE SYSTEM. To re-calibrate the system, refer to procedures in the *LightSpeed 3.X System Service Manual*.

1.2 Required Tool

DVM with meter leads

Section 2.0

mA Meter Check With Verification



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon,



2.) Click on the UTILITIES icon,

3.) Select INSTALL.

4.) Select GENERATOR CALIBRATION.

5.) Switch OFF the HVDC (550) ENABLE inside the gantry.

6.) Switch OFF the AXIAL DRIVE inside the gantry.

7.) Rotate the anode tank to the two o'clock position.

8.) Pin the gantry.

9.) Remove the OBC cover (if necessary).

10.) Use a DVM as a DC mA meter. Set the meter range to one decimal place.

11.) Connect the DVM black lead to TP8 (ACAL1) on the mA board.

12.) Connect the DVM red lead to TP11 (ACAL2) on the mA board.

Collimator
HEMRC
MA Control
KV Control
Gentry I/O
LSCOM
Heurikon (CPU)

Figure 10-2 OBC Board Slot Assignment

13.) Select MA METER CHECK.

14.) Select ACCEPT.

15.) Record the displayed and measured Anode mA values in the Circuit OFF and Circuit ON table in space provided below. Acceptable Circuit ON values: 140 to 200mA.

Note: **The cathode side reads approximately 19 mA during Circuit ON. The anode should read 140 to 200mA, as stated above.**

16.) Repeat the test for the cathode side:

a.) Use a DVM as a DC mA meter; set the meter range to one decimal place.

b.) Connect the DVM red lead to TP9 (CCAL1) on the mA board.

c.) Connect the DVM black lead to TP14 (CCAL2) on the mA board.

17.) Select ACCEPT.

- 18.) Record the displayed and measured cathode mA values in the Circuit OFF and Circuit ON table in space provided below. Acceptable Circuit ON values: 140 to 200mA.
- Note: **The anode side reads approximately 19 mA during Circuit ON. The cathode should read 140 to 200mA, as stated above.**
- 19.) Disconnect the test equipment.
- 20.) Switch ON the HVDC (550) enable.

2.1 mA Meter Verification - HHS Requirement

Circuit ON		
	A/D	Range
Anode mA		15 - 25
Cathode mA		15 - 25
DVM - A/D = Delta Compare Delta to Limits		

Table 10-1 mA Meter Verification (Table 1)

2.2 Auto mA Cal

Use this procedure to verify and complete Auto mA Cal on the system. If you have just run a Heat Soak and Seasoning, wait 30 minutes before you run the Auto mA Cal.

Note: **The system automatically warms up the tube.**

- 1.) Select AUTO MA CAL.
- 2.) Press START SCAN when it flashes. The program automatically runs Ductility and Auto mA Cal.
- 3.) During this test, verify the X-Ray on indicators, located on the gantry display and the rear gantry shroud, illuminate during X-Ray on.

If the system cannot complete the Auto mA Cal because HV aborts, run the "Tube Heat Soak and Seasoning" scan protocol (see the *LightSpeed 3.X System Service Manual* for details). Place a 48cm phantom in the X-Ray beam to protect the detector from radiation damage during this procedure. Do not worry about the image quality of the scans reconstructed during the Tube Heat Soak and Seasoning.

☐ Check box when complete Initial: _____

The MX200 tube uses a metal/ceramic frame, which means some electrons hit the target and actually bounce off, and are then captured by the frame. This causes a frame current, which results in an anode mA measurement of slightly less than the cathode mA, typically 5 to 10% less. Because the cathode mA is the true indicator of radiation output, the CT QX/i system controls cathode mA rather than anode mA.

2.3 X-ray Lights

- 1.) Perform several scans (see below).
- 2.) Verify that the X-ray ON lights are on during the scans. Check boxes in [Table10-2](#)
- 3.) Record data on **FORM 4879** (see [Chapter 14 on page 287](#)).

PERFORMING A SCAN



- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Select DIAGNOSTICS.
- 3.) Select DIAGNOSTIC DATA COLLECTION
- 4.) Set the scan time to 2.00
- 5.) Set the Kv to 80
- 6.) Set the ma to 40.

Light On?	Light Locations
☐	SCIM
☐	Gantry Front
☐	Gantry Back (use a mirror to view)
☐	Room Light

Table 10-2 X-ray light chart

Chapter 11

System Tests



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 14](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

This chapter contains descriptions of the procedures used to test system operation after installation, alignment and calibration, but prior to customer turnover and acceptance. The Image Series sections instruct you to scan system phantoms, record the means and standard deviations, and evaluate the image quality of the resulting images.

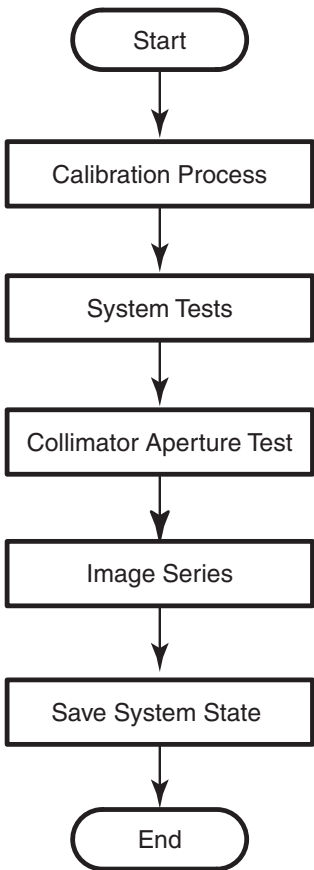


Figure 11-1 System Test Overview

Section 2.0

Calibration Process

2.1 Reference Procedures

Do NOT perform these procedures now. They are placed here as Reference for later procedures. You will be directed to follow these instructions at several points throughout this chapter.

2.1.1 Scanning with Service Protocols

Locate the Manufacturing and Installation protocols under Infant area 20.

Note: **Manufacturing and Service share this Protocol list. Different product option offerings also use this list. Carefully follow the scan section instructions, and verify you acquired the images with the correct technique before filling out the data sheet. Otherwise you may troubleshoot an image problem that only exists because you used the wrong technique.**

- 1.) Select the NEW PATIENT icon.
- 2.) Enter a Patient ID (e.g., getest).
- 3.) Click on the box labeled PEDIATRICS.
- 4.) Select a service protocol from the list, to display the corresponding view edit screen.

Optional Method: Enter the protocol number into the Protocol Number Field on the Exam Rx Screen.

2.1.2 Center Phantom

- 1.) Select SCANNER UTILITIES.
- 2.) Select CENTER PHANTOM.
- 3.) Follow the on-screen procedures.
- 4.) The phantom center spec is ± 1 mm.
- 5.) Select QUIT, when the phantom is within spec.

2.2 Prepare the QA Phantom

Note: **The QA phantom is shipped water filled.**

- 1.) Locate the multi-language sticker packet in the QA phantom shipping box.
- 2.) Attach the sticker with the customer's language to the face of the phantom hanger bracket.

2.3 Calibration Process Introduction

If your system has a factory supplied state MOD, you used it to load the system calibration files during the Restore System State ([3.1 on page 175](#) in Chapter 6 of this manual).

2.4 Fast Cal



- 1.) Select  to warm up the tube.
- 2.) Select FAST CALIBRATION from the Daily Prep menu.

Note: Use the default Fast Cal selections determined by the system configuration. (The system defaults to all four kV stations, but you can choose kV stations to calibrate during reconfig.)

- 3.) Run the selected air calibrations.
- 4.) When the calibration process completes, click on QUIT.

2.5 Table/Gantry Alignment Procedure

2.5.1 Conditions

This procedure may require the removal of the table bottom, side and rail covers.
This procedure requires that mechanical final alignment was completed and passed.

2.5.2 Procedure

- 1.) If this is a Non-Fixed Site, check the Van level. Correct, if necessary.
- 2.) Check Table level. Correct, if necessary.
- 3.) Drive the table to its highest elevation (phantom holder removed).
- 4.) Drive the table cradle in and out five times to seat the rollers.
- 5.) Turn on the alignment lights. Check the scan window for proper installation.
- 6.) Advance the end of the cradle (black dot on cradle) to the internal light.
- 7.) Complete [Section 2.1.1, "Scanning with Service Protocols"](#) now.
- 8.) Select the service protocol, PERPENDICULAR ALIGNMENT. Protocol should appear as shown in [Table 11-1](#). Add two scouts to this procedure. 120kv 40 ma tube at zero, 120kv 40 ma tube at ninety.

Scan Type	Direction	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Algorithm
Axial	Head First	120	40	Large	5mm/1i	2sec	S0	S0	Bone
Axial	Head First	120	40	Large	5mm/1i	2sec	I1000	I1000	Bone

Table 11-1 Perpendicular Alignment Scan Parameters

Set Internal Landmark. The red boxes will disappear from the screen.

- 9.) Click on CONFIRM, then press the START SCAN button when lighted, to perform the scan.
- 10.) Use over/under multi-image display to view both images at the same time:
 - a.) Click on IMAGE WORKS
 - b.) Highlight the scans just performed and click on VIEWER.
 - c.) Click on FORMAT, and select the "over/under" view (two boxes aligned vertically).
- 11.) Use the upper image (cradle front) to verify that table is centered:
 - a.) Click on the "grid" icon.
 - b.) Click on MEASURE, and select the straight (angled) line.
 - c.) Using the mouse, position one end of the line on the vertical axis.
 - d.) Position the other end at one outside edge of the cradle, forming a perfectly horizontal line (angle = 90). Read the distance displayed on screen.
 - e.) Repeat step d to measure the distance from the vertical axis to the other edge.
- 12.) Repeat step 11 for the lower image (cradle at 1000 mm).
- 13.) If the edges are off by greater than 1 mm (front to back), perform the following:

- a.) Loosen the table anchors at locations #5, #6 and #7.
 - b.) Move the table half the measured distance.
 - c.) Return to Step 8. Repeat the procedure until the table edges line up (± 1 mm). If needed a scout can be completed to check table horizontal presentation image.
Check the two scout scans
 - i.) Check the tube at zero scan for table gantry perpendicularity. Make table adjustments as needed
 - ii.) Check the tube at 90 scan for cradle level and visual irregularity with the white side molding. Molding strips should be parallel within ± 3 mm. Cradle should be level within ± 3 mm.
 - d.) Tighten the anchors and torque to 74.6 N-m (55 ft.-lb.) for fixed sites only. (75 ft.-lb.) for non-fixed systems.
 - e.) Reinstall all table parts removed to access the anchors, including covers, ground straps, elevation motor leads and the center support bar.
- 14.) Install all table and gantry covers.
- 15.) Use ERASE ALL to remove lines and grids from the images.
- 16.) Click on END EXAM to exit the program.

2.6 Auto CT N Number Adjustment

Note: **If it has been more than 30 minutes since calibration scans, rerun FASTCAL (approx 15 min). N Number Adjustment needs the tube to be warm, or CT numbers will change.**

Run the small and large phantom CT N number adjustment before you acquire the image series, or any time you doubt the N# values. The calibration procedure modifies the pcal (phantom cal) vector to assign water a CT number equal to zero. After a successful N number adjustment, any water phantom scanned against a small or large CAL has a CT number near zero.

- 1.) Place the QA Phantom in the holder.
- 2.) From the application desktop, select SCANNER UTILITIES.
- 3.) Select AUTO CT NUMBER ADJUSTMENT.
- 4.) Follow directions on the screen.
 - a.) When centering the phantom:
 - * Centering may be easier to perform from the gantry right side.
 - * Center on the water portion of the phantom (not over the internal block).
 - b.) The program will automatically make adjustments. If the program is unable to make adjustments, it will stop at the technique it is unable to adjust. Troubleshoot with manual CT # Adjust before continuing.

Section 3.0 System Tests

3.1 Introduction

Use the system tests in the following sections to exercise all aspects of the system, to assure system integrity, before turnover to the customer. Although the means, standard deviation, and resolution specifications do not apply during system functional tests, treat any artifact or image anomaly as a failure.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise, or center artifacts.
- Correct artifacts, system test, or image series failures when they occur. Any delay in repairs could increase the number of retests.

3.2 Tomographic Plane Indication

- 1.) Place the QA phantom on the phantom holder.
- 2.) Turn ON the internal alignment lights, and drive the phantom into the gantry opening, until the line on the phantom lines up with the internal laser lights.
- 3.) Verify that BOTH internal axial lasers line up along the line on the QA phantom. If not, check table/gantry, cradle, and/or laser alignment.
- 4.) Center the phantom in the scan plane with the calibration program. (See [2.1.2 on page 224](#), for details on the phantom centering procedure.)
- 5.) Select the service protocol, TOMO PLANE INDICATION. (See [2.1.1 on page 224](#), for details on scanning with service protocols.)

or

Manually select the scan parameters in [Table 11-2](#).

Scan Type	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Algorithm	Interval
Helical	120	200	Small	1.25HQ	1.0sec	I3.0	S3.0	Bone detail	0.2

Table 11-2 Tomographic Plane Indication Scan Parameters

- 6.) Display the image series, and locate the scan plane indicator, the longest bar in the bar pattern on the right side of the phantom. The right side of the phantom corresponds to the side of the image labeled **L** on the display screen.
- 7.) On the HHS Data Sheet, record the scan location (shown on the image annotation) of the image with the darkest scan plane indicator (darkest long bar).
- 8.) If your system meets all the installation and alignment specifications, the image at scan location zero (S0.0) should contain the scan plane indicator. If scan location S1.0 or scan location I1.0 has the darkest bar, the system still meets the specification. The scan plane deviation should equal $S0.0 \pm 1.0\text{mm}$. If necessary, adjust the internal alignment light position to meet the $S0.0 \pm 1.0\text{mm}$ requirement.

- 9.) Repeat the Tomographic Plane Indication test with the external alignment lights.
 - a.) Use the external alignment light, and press the external landmark.
 - b.) Verify the external light lines up along the black line on BOTH the left and right sides of the QA phantom.
 - c.) The scan plane indication must fall within the $S0.0 \pm 1.0\text{mm}$ specification.
- 10.) Check the box on **Form 4879**.

3.3 System Scanning Test

Use the System Scanning Test to verify hardware functionality. Review images for visible artifacts, and review the message log for unacceptable errors.

- 1.) Place the QA phantom on the cradle.
 - Drive the table to an elevation of 100.
 - Align the line on the phantom with the internal laser lights.

Never scan above 50mA without first placing a phantom in the field of view. Levels in excess of 50mA can cause temporary radiation damage to the detector that lasts several hours. If you acquire image series cals with a radiation damaged detector, the cals may cause artifacts in subsequent image series scans.

Note: 2.) Select the service protocol, SYSTEM SCAN/CUST QUAL RELI.
Stop the service protocol after the second helical series. It is not necessary to proceed past that point in the protocol.

-or-

Manually select the scan parameters specified in [Table 11-3](#).

Scan Type	kV	mA	SFOV	Thickness	Rotation Time	Start Loc.	End Loc.	Tilt/Table Speed
Scout	120	40	-	-	-	S200	I800	0°
Scout	120	40	-	-	-	S200	I800	90°
Axial (2 scans)	120	50	Large	10mm/1i	1.0sec	S0	S0	I30
Axial (2scans)	120	50	Large	10mm/1i	1.0sec	S0	S0	S30
Helical HQ	120	50	Large	5mm	1.0sec	S70	I75	15mm/rot
Helical HS	120	50	Large	5mm	0.8sec	S70	I75	22.5mm/rot

Table 11-3 System Scanning Test Scan Parameters

- 3.) Complete the scans.

Section 4.0

Collimator Aperture Test

4.1 Small Aperture Test

- 1.) Clear all obstructions out of the gantry SFOV.
- 2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon, .
- 3.) Click on the UTILITIES icon. .
- 4.) Select INSTALL.
- 5.) Select COLLIMATOR APERTURE TEST.
- 6.) Select SMALL APERTURE SCAN to display a scan protocol window, and start gantry activity (refer to [Figure 11-2](#)).
- 7.) Press the START SCAN key when it flashes, to initiate the first scan.
- 8.) When the test completes, the screen updates to display the Small Aperture values.

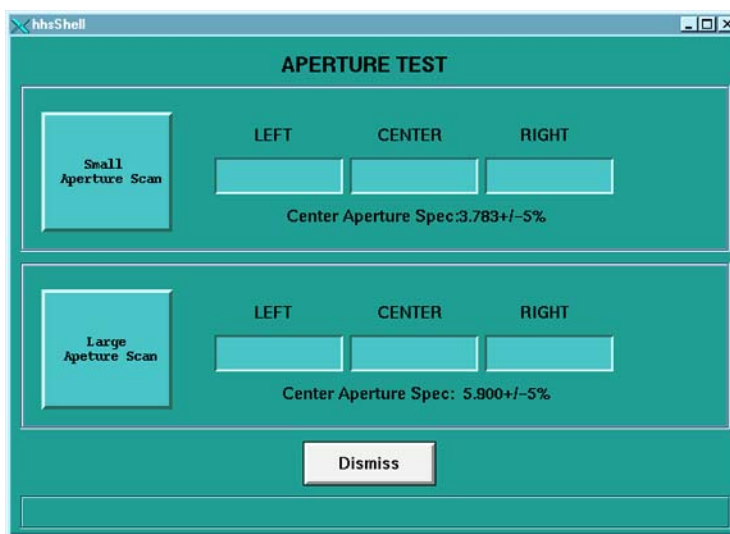


Figure 11-2 Collimator Aperture Test Screen

4.2 Large Aperture Test

Note: The center channel is the only channel with a spec. The left and right channels may or may not pass. Small aperture center spec is $3.783 \pm 5\%$. Large aperture center spec is $5.900 \pm 5\%$.

- 1.) Select LARGE APERTURE to display a scan protocol window, and start gantry activity (Refer to [Figure 11-2](#)).
- 2.) Press the START SCAN key when it flashes, to initiate the first scan.
- 3.) When the test completes, the screen updates to display the large aperture values.
- 4.) Record the large and small aperture values on Form 4879, located in [Chapter 14](#).

Section 5.0 Image Series

5.1 Scan Protocol

The person who acquires the image series has the responsibility to review the images, and verify they meet the specifications listed on data sheets. Responsibilities also include means and standard deviation measurements, and keeping a record of failures that occur during the image series.

Unless otherwise stated, use the following scan parameters during the image series acquisition:

- Scan FOV equal to display FOV (Field of View)
- 512x512 matrix size

Note: **Consider *any* image series scan that does not meet specifications as failing.**

For means and standard deviations, 90% of the slices must pass. Any failure on a particular technique requires at least ten additional slices to evaluate effectively.

Systems with metal-free cradles have a phantom holder with a perpendicular adjustment (Z-axis) knob on it. **Each time you change phantoms**, make sure you use a bubble level, and the Z-axis knob on the phantom holder, to level the phantom.

5.2 Data Recording: Means and Standard Deviation

Any failure on a particular technique requires at least a ten additional slices to evaluate effectively. For means and standard deviations, 90% of the slices must pass.

- Record means to two decimal places, and round to the nearest one-tenth, (one decimal place) when you compare the resulting values to the spec.
- Record standard deviations to two decimal places, then round off to one decimal place, to compare it to the spec.
- Average standard deviations: Use two decimal places to average the values, then round off to one place.

Before you record the means and standard deviations, check the image data sheets to determine whether to average the means and standard deviations, or record them slice by slice. Make sure you record all the required image data on the HHS data sheets.

5.3 Term Definitions & Screens

Xc - Mean CT number for the specified center coordinates of the phantom image.

AvXc - Average Mean CT number for the center of the phantom image: Average the mean CT value for all specified center coordinates of all slices in an exam.

Xo - Mean CT number for the outside of the phantom image: Average the mean CT value for all specified outside coordinates of one slice.

AvXo - Average outside mean CT number for the number of slices in an exam.

AvSDc - Average image noise about the center image coordinate (measured as the standard deviation) of all slices in an exam.

AvSDo - Average image noise (standard deviation) for the outside of a phantom: Average of all outside coordinates of all the slices in an exam.

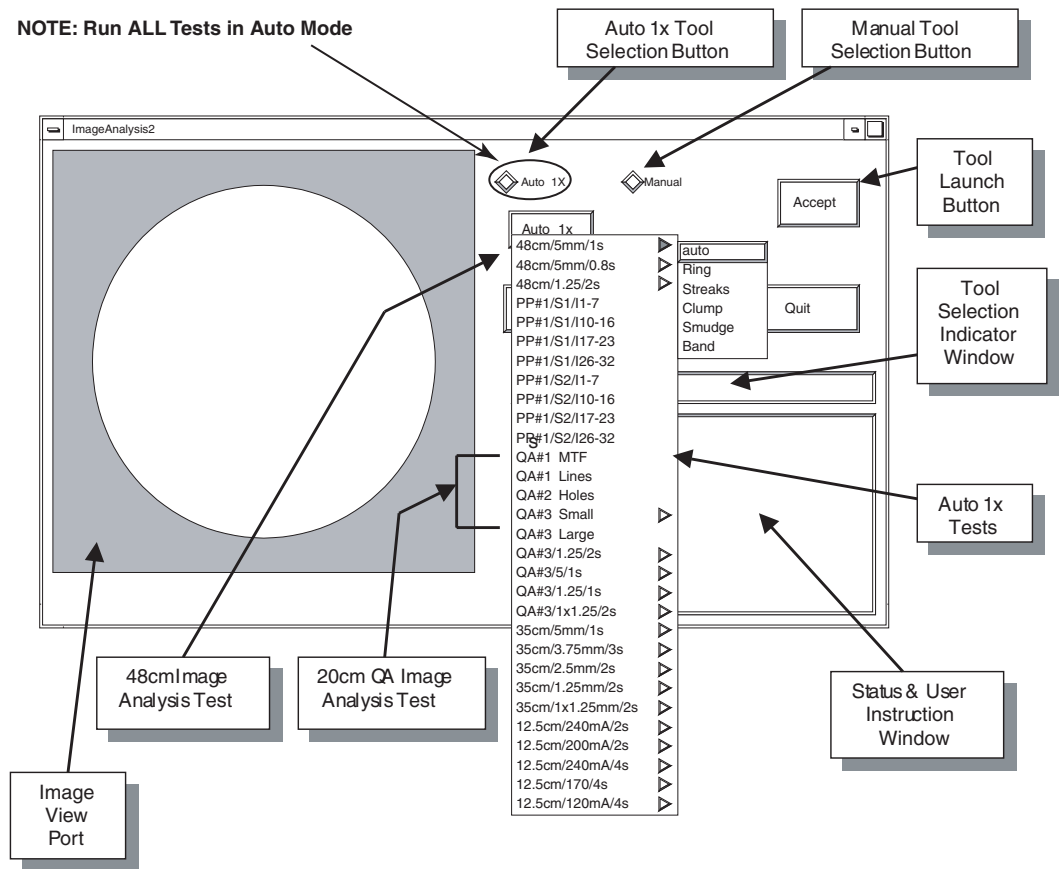


Figure 11-3 Image Analysis Tool User Interface – Auto 1x Test Pull Down Menu

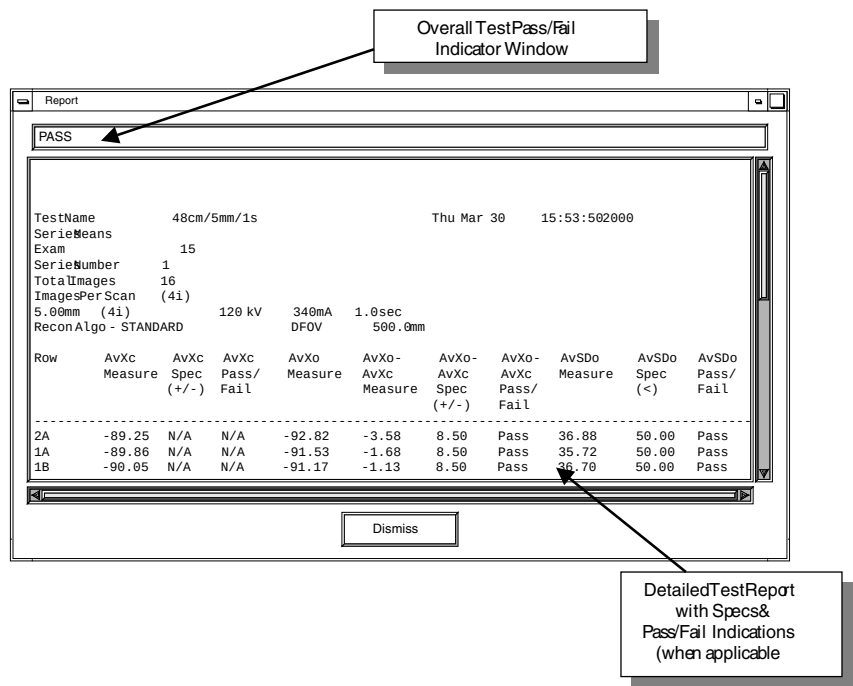


Figure 11-4 Analysis Tool User Interface – Test Results Report Window

5.4 48cm Phantom Image Series Image Performance Verification

5.4.1 Acquiring the 48cm Phantom Image Series

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 48cm Phantom on the Phantom Holder.
- 3.) Align, level, & center the 48cm Phantom.
 - Align phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the CENTER PHANTOM utility in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan a four scan, 16 image, 48cm Phantom image series.

MANUAL SCAN PROTOCOL SETUP

Refer to [Table 11-4](#) and set-up an Axial scan with the parameters as shown.

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
48 Lg Series 1	Axial Full 1.0 sec.	I7.50	S7.50	16	5.0 4i	0.00	S0.0	Large	120	340	4.0 sec.	50.0	Std

Table 11-4 48cm Phantom Image Series Scan Parameters

AUTO SCAN PROTOCOL SETUP

- a.) On the Exam Rx desktop, select NEW PATIENT.
- b.) Type the following entries in the two listed Patient Information fields:

Patient ID: Service

Name: 48cm Phantom Image Series
- c.) From the Protocol display, click on the infant box.
- d.) On the infant display window, click on the area below the infant's right foot to display the Miscellaneous menu.
- e.) Click on the 20:10 IMAGE SERIES 48 CM button.
- f.) Select the first series (Series Description filed should display 48cm Lg Series 1)
- 5.) Set internal Landmark.
- 6.) Acquire a four-scan, 16 image, 48 cm Phantom image series.

5.4.2 Brightness Uniformity and Noise

5.4.2.1 Image Performance Verification

- 1.) Select the 48cm Phantom image series exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
 - b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 45 x 45 pixel reference ROI Box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a square 45 mm x 45 mm (2025 mm²) box. Tolerance: 45 mm +/- 5 mm (1600 mm² to 2500 mm²).
If required, magnify the image to adjust to proper dimensions. See [Figure 11-5](#) for additional ROI size and placement information.
- 3.) Collect Mean and Standard deviation values for five reference ROI Box positions on the sixteen 48cm Phantom images.
If required, magnify the image to adjust to proper dimensions. See [Figure 11-5](#) for additional ROI placement information.
 - a.) Position the reference ROI Box built in step 2 directly over the center of the image using the grid cross-hairs as a guide.
 - b.) Type `prop a` in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
 - c.) Click on the MEASURE button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.
 - d.) Reposition Box # 2 to the left center portion on the first image (Box # 2 position in [Figure 11-5](#)).
 - e.) Type `prop a` in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.
 - f.) Repeat steps 3.c. through 3.e. for each of the remaining ROI box positions shown in [Figure 11-5](#) (Box # 3 through 5).
 - g.) Record the Mean and Standard Deviation values of the 16 images in the series for each of the five box positions in [Table 11-5](#).
Each image can only display text for the mean, standard deviation, and box area for three images at a time. To view the data for a particular box, select the box on the image and the system displays the data for the box number selected.

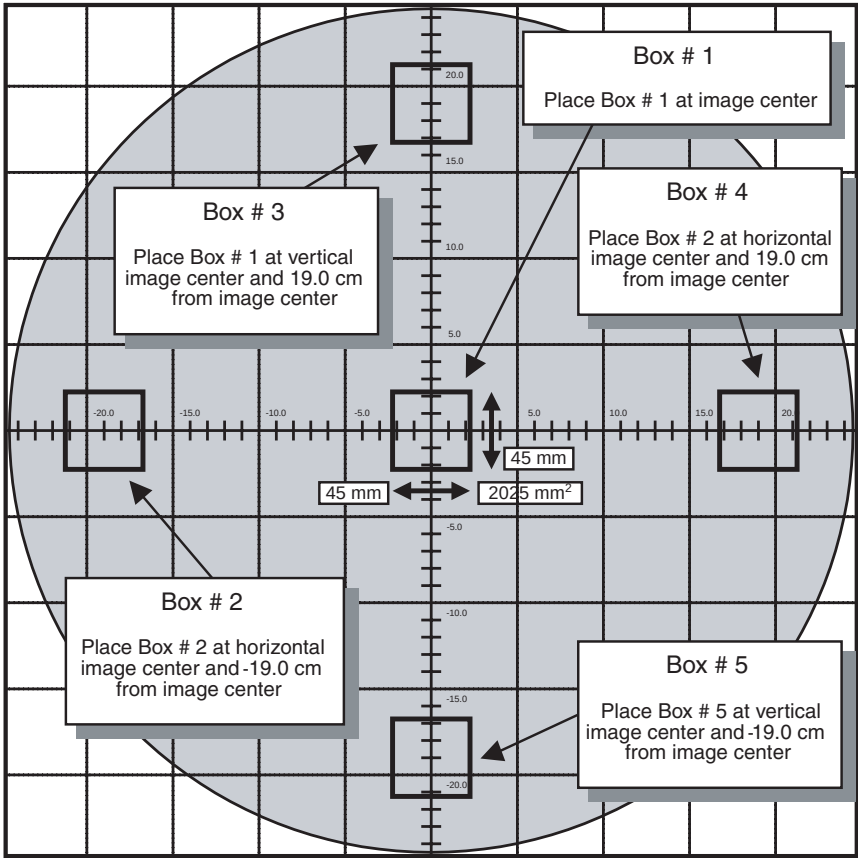


Figure 11-5 48cm Phantom Brightness Uniformity & Noise Measurement – Building Placing Reference ROI Boxes

Image	Box 1 (Center)	Box 2 (Left Center)		Box 3 (Top Center)		Box 4 (Right Center)		Box 5 (Bottom Center)		Outside Boxes Avg	
	Means (Xc)	Means	Std dev	Means	Std dev	Means	Std dev	Means	Std dev	Means (AvXo)	Std dev (AvSDo)
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											
11											
12											
13											
14											
15											
16											

Table 11-5 48cm Phantom CT# Brightness Uniformity & Noise Image Performance Worksheet

Box Size = 1600 mm² to 2500 mm²
 45 mm (+/- 5 mm) x 45 mm (+/- 5 mm)
 45 (+/- 4 pixels) x 45 (+/- 4 pixels)

Box Positions: Box 1 = 0 mm x 0 mm
 Box 2 = 0 mm x -190 mm
 Box 3 = 190 mm x 0 mm
 Box 4 = 0 mm x 190 mm
 Box 5 = -190 mm x 0 mm

Row	Images	Center Box Means (AvXc)	Outer Boxes Means Averages (AvXo)	Outer Boxes Means Averages minus Center Box Means (AvXo - AvXc)	Outer Boxes Standard Deviation Averages (AvSDo)	Comments
2A	1, 5, 9, 13					
1A	2, 6, 10, 14					
1B	3, 7, 11, 15					
2B	4, 8, 12, 16					
Specifications		n/a	n/a	± 8.5	< 70	

Table 11-6 48cm Phantom CT# Brightness Uniformity & Noise Row Performance Worksheet

- 4.) Calculate the Brightness Uniformity and Noise values for each image in the series, record values in [Table 11-6](#), and compare to specifications.
 - a.) Calculate and record the average means and average standard deviation values for the four outside Boxes (Boxes 2 through 5) for each of the images and record in [Table 11-5](#).
 - b.) Calculate and record the center box (Box 1) average means values (AvXc) for each row in [Table 11-6](#).
 - c.) Record the outside boxes (Box 2 through 5) average means values (AvXo) and average standard deviation values (AvSDo) for each row in [Table 11-6](#).
 - d.) Calculate the Brightness Uniformity (AvXo - AvXc) value for each row and record in [Table 11-6](#).
 - e.) Verify Brightness Uniformity (AvXo - AvXc) value and Noise value (AvSDo) for each row meets specifications listed in [Table 11-6](#).
 - f.) Record the 48cm Phantom Brightness Uniformity (AvXo - AvXc) value and Noise value (AvSDo) for each row in the HHS Record Tables.

5.4.2.2 Failure Recovery

Specifications

Each Row (2A, 1A, 1B, and 2B) of the series *must* pass 48cm Brightness Uniformity and Noise (for the first series scan parameters) specifications:

- AvXo - AvXc: < +/- 8.5
- AvSDo: < 70.0

Recommended Recovery

- A.) Perform DETAILED CAL.
- B.) Perform AUTO CT# ADJUST.
- C.) Repeat this procedure to verify 48cm Phantom Image Performance.

5.5 20cm QA Phantom Image Series Image Performance Verification

5.5.1 Acquiring the 20cm QA Phantom Image Series

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 20cm QA Phantom on the Phantom Holder.
- 3.) Align, level, & center the 20cm QA Phantom.
 - Align the etched line (QA#1 position) on phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the CENTER PHANTOM procedure in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan the QA#1, QA#2, and QA#3 positions on the 20cm QA Phantom.

MANUAL SCAN PROTOCOL SETUP

Refer to [Table 11-7](#) and set-up an Axial scan with the parameters shown

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
QA#1	Axial Full 1.0 sec.	I5.00	S5.00	4	10.0 2i	0.00	S0.0	Small	120	260	2.0 sec.	25.0	Std
Recon 2: Q	Recon #2 of Series #1–4 images											15.0	Bone
QA#2	Axial Full 1.0 sec.	S40.00	S50.00	8	10.0 2i	0.00	S0.0	Small	120	260	4.0 sec.	25.0	Std
QA#3	Axial Full 1.0 sec.	S55.00	S65.00	8	10.0 2i	0.00	S0.0	Small	120	260	4.0 sec.	25.0	Std

Table 11-7 20cm QA Phantom Image Series Scan Parameters

AUTO SCAN PROTOCOL SETUP

- a.) On the Exam Rx desktop, select NEW PATIENT.
- b.) Type the following entries in the listed Patient Information following fields:
 - * Patient ID: **Service**
 - * Name: **20cm QA#3 Phantom Image Series**
- c.) From the Protocol display, click on the infant box.
- d.) On the Infant display window, click on the area below the infant's right foot to display the Miscellaneous menu.
- e.) Click on the 20:12 IMAGE SERIES QA button.
- f.) When the scan prescription appears on the left monitor, select the 1st series.
- 5.) Set internal Landmark.
- 6.) Acquire the QA#1, QA#2 and QA#3 positions of the 20cm QA Phantom.

5.5.2 High Contrast Spatial Resolution

5.5.2.1 Image Performance Verification

- 1.) Select the first 20cm QA Phantom image series (QA#1) from the exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
 - b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and the first series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 17 x 17 pixel reference ROI Box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a square of approximately 8 mm x 8 mm (64 mm²) box. If required, magnify the image to adjust to proper dimensions. See [Figure 11-6](#) for additional ROI size and placement information.
- 3.) Collect Mean and Standard deviation values for three reference ROI Box positions on the four 20cm QA#1 Phantom images.

If required, magnify the image to adjust to proper dimensions. See [Figure 11-6](#) for additional ROI placement information.

 - a.) Position the reference ROI Box built in step 2 directly over the 1.6 mm Line Pair Pattern on the QA#1 phantom image. Fine adjust size of Box #1 ROI to ensure that the box does not exceed the line pair boundaries. See [Figure 11-6](#).
 - b.) Type `prop a` in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
 - c.) Click on the MEASURE button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.
 - d.) Reposition Box # 2 over the Plexiglas portion of the QA#1 phantom image (Box # 2 position in [Figure 11-6](#)).
 - e.) Type `prop a` in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.
 - f.) Click on the MEASURE button and select the box ROI icon to display Box # 3. The system places an ROI box labeled # 3 at the center of the image with the exact same dimensions as Boxes # 1 and 2.
 - g.) Reposition Box # 3 over the water portion of the QA#1 phantom image (Box # 3 position in [Figure 11-6](#)).
 - h.) Record the Mean and Standard Deviation values of the four images in the series for each of the three box positions in [Table 11-8](#).

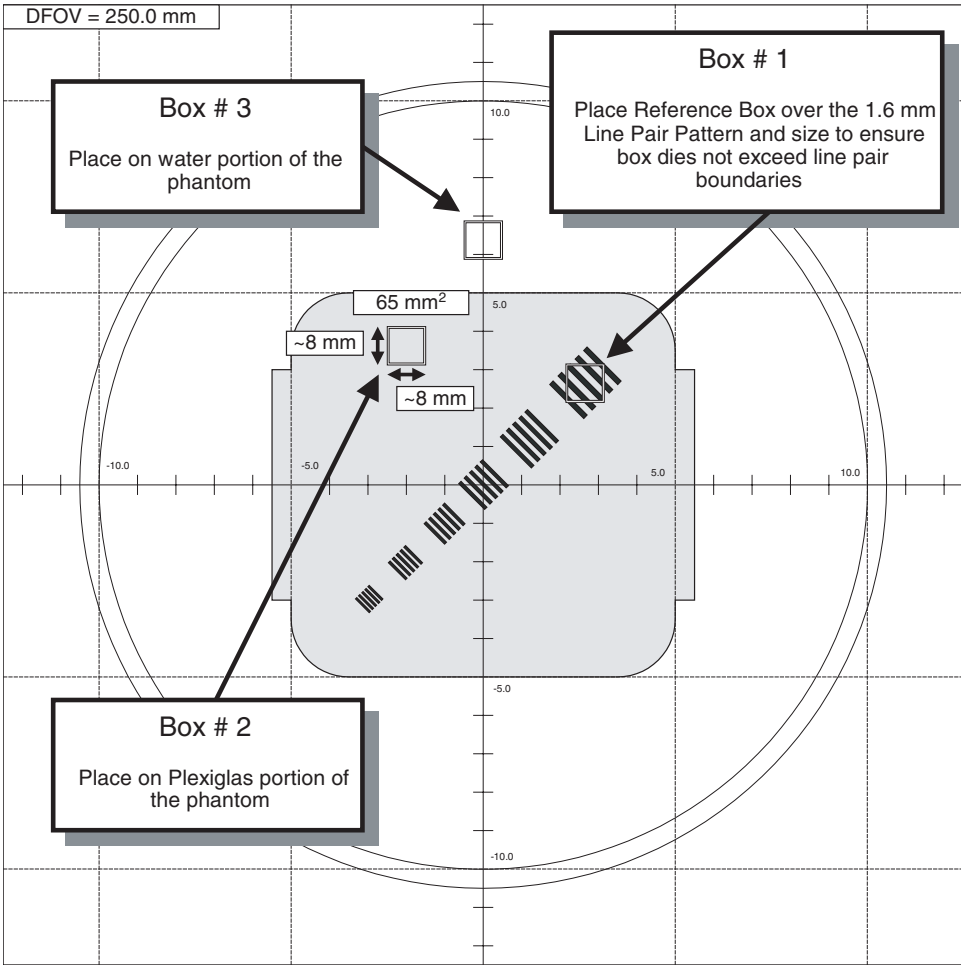


Figure 11-6 20cm QA#1 Phantom High Contrast Spatial Resolution Testing - Building And Placing Reference ROI Boxes

Image	Box 1		Box 2		Box 3		Contrast Scale (Box 2 Means – Box 3 Means)	Std Dev Average (Box 2 Std Dev – Box 3 Std Dev / 2)
	Means	Std Dev	Means	Std Dev	Means	Std Dev		
1								
2								
3								
4								

Table 11-8 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance Worksheet #1

Image	MTF	MTF 4-slice average	Contrast Scale	Comments
1		n/a		
2		n/a		
3		n/a		
4		n/a		
Specifications	n/a	0.65 to 1.0	110.0 to 130.0	n/a

Table 11-9 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance Worksheet #2

- 4.) Calculate the Contrast Scale value and four-image average MTF values for each image in the series, record values in [Table 11-9](#), and compare to specifications.
 - a.) Calculate and record the Contrast Scale value (Means value of Box 2 - Means value of Box 3) for each image and record in [Table 11-8](#) and [Table 11-9](#).
 - b.) Calculate and record the MTF value for each image in [Table 11-9](#) using the following formula. Record Standard Deviation Average (Std Dev AVE) values in both [Table 11-8](#) and [Table 11-9](#).

$$\text{Std Dev AVG} = (\text{Std Dev}_{\text{Box 2}} + \text{Std Dev}_{\text{Box 3}}) / 2$$

$$\text{Modulation} = \sqrt{\text{StdDev}_{\text{Box 1}}^2 - \text{StdDev}_{\text{AVE}}^2}$$

$$\text{MTF} = 2.2 \times (\text{Modulation} / \text{Contrast Scale})$$
 - c.) Calculate the average MTF value for each image and record in [Table 11-9](#).
 - d.) Verify Contrast value for each image and the four-image MTF value meets specifications listed in [Table 11-9](#).
 - e.) Record the 20cm QA Phantom High Contrast Spatial Resolution values (Contrast Scale for each image and four-image MTF value) in the HHS Record Tables.
- 5.) View each image of the Recon 2 Series, record all the visible Line Pair Patterns in [Table 11-10](#), and verify each image meets specifications.
 - a.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
 - b.) Select the exam and the second series (Recon 2: Q) acquired in the previous section.
 - c.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.
 - d.) While viewing each image, indicate in [Table 11-10](#) all the Line Pair Pattern groups that can be visually distinguished. See [Figure 11-7](#).
 - e.) Verify Line Pair Pattern visual check meets specifications and record in the HHS Record Tables.

Image	Line Patterns Visible	Comments
1		
2		
3		
4		
Specifications	B, C, D, E, F	n/a

Table 11-10 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance Worksheet #3

5.5.2.2 Failure Recovery

Specifications

Each image of the series must pass 20cm QA#1 High Contrast Spatial Resolution parameter (Contrast Scale, Four-Image MTF Average and Visible Line Pair (for the first and second series scan parameters) specifications:

- Contrast Scale: 110.0 to 130.0
- MTF Average: 0.65 to 1.0
- Visible Lines: B, C, D, E, F

Recommended Recovery

- A.) Check Phantom Alignment (leveling is critical) and repeat this scanning and High Contrast Spatial Resolution Performance Test.
- B.) Perform Alignment Procedures (POR Alignment, BOW Alignment, CBF/SAG Alignment, ISO Alignment, and Hot ISO Alignment), perform Full Calibration (Detailed Calibration and Auto CT# Adjust, and repeat this scanning and High Contrast Spatial Resolution Performance Test.

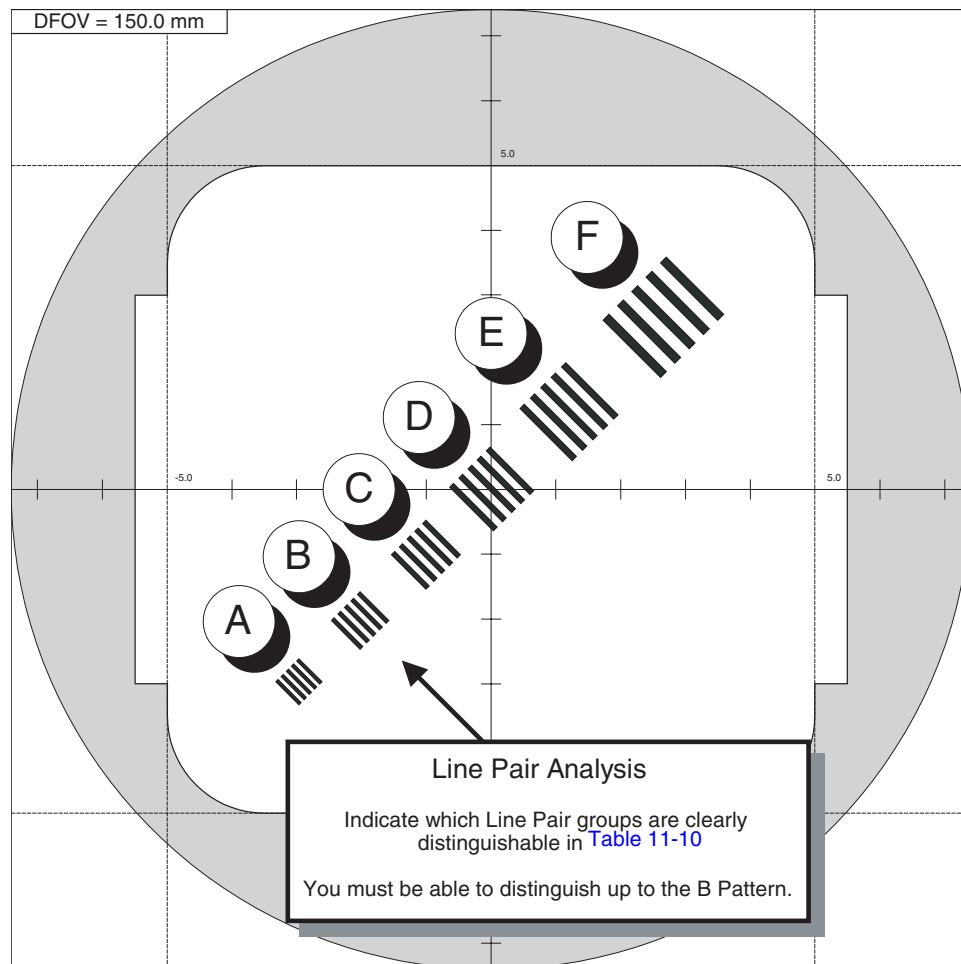


Figure 11-7 20cm QA#1 Phantom High Contrast Spatial Resolution (Visible Line Verification)

5.5.3 Low Contrast Detectability

5.5.3.1 Image Performance Verification

- 1.) Select the third 20cm QA Phantom image series (QA#2 Holes) from the exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
 - b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and the third series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 10 mm x 100 mm reference ROI Box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a rectangle of approximately 10 mm x 50 mm (500 mm²) box.

If required, magnify the image to adjust to proper dimensions. See [Figure 11-8](#) for additional ROI size and placement information.
- 3.) Collect Mean values for two reference ROI Box positions on four (1st, 3rd, 5th, and 7th) of the eight 20cm QA#2 Phantom images.

If required, magnify the image to adjust to proper dimensions. See [Figure 11-8](#) for additional ROI placement information.

 - a.) Adjust the Window setting of the image to 20 by simultaneously holding the center mouse button down while dragging the cursor to the left. Adjust level for a viewable image as shown in [Figure 11-8](#).
 - b.) Position the reference ROI Box built in step 2 on the Plexiglas portion of the QA#2 phantom image without touching the water portion or the hole pattern. See [Figure 11-8](#).
 - c.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
 - d.) Click on the MEASURE button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.
 - e.) Reposition Box # 2 over the water portion of the QA#1 phantom image and directly above Box # 1 (Box # 2 position in [Figure 11-8](#)).
 - f.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.
 - g.) Record the Mean values of the 1st, 3rd, 5th, and 7th images in the series for each of the two box positions in Table 8.

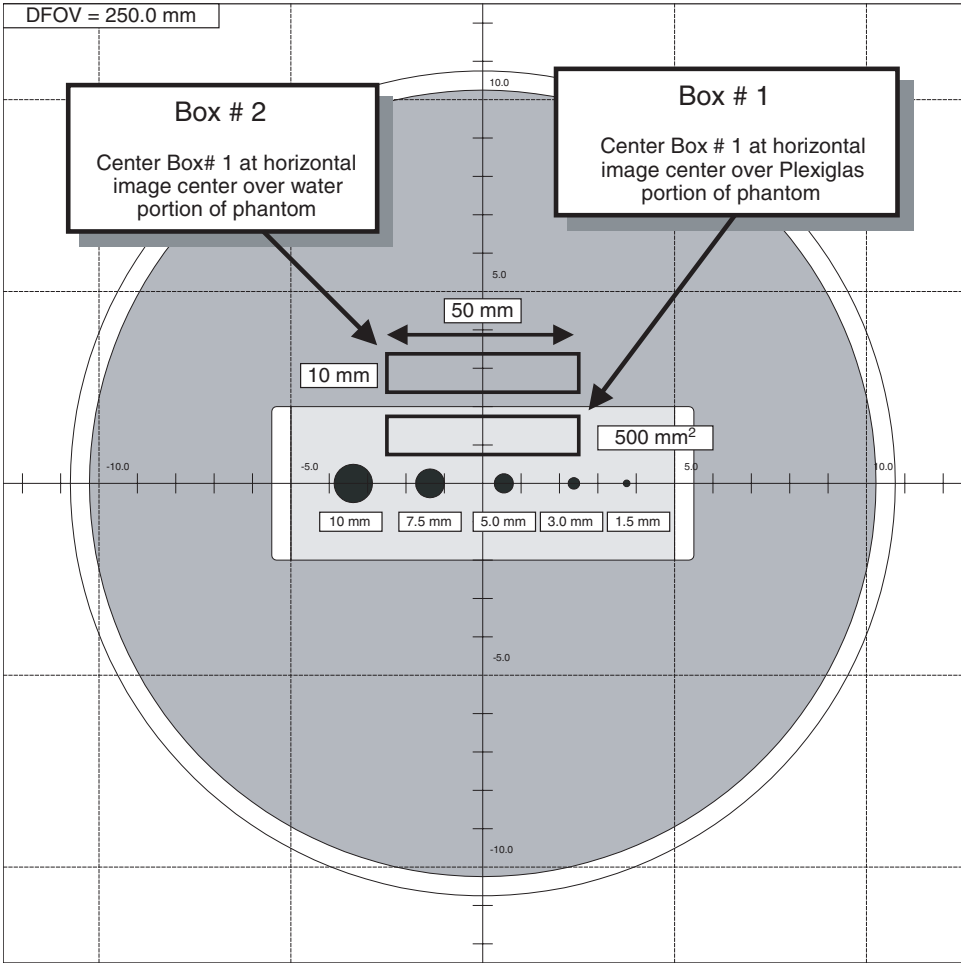


Figure 11-8 20cm QA#2 Phantom Low Contrast Detectability - Building and Placing Reference ROI Boxes

Image	Visible Holes Viewable at Window 20	Box 1 Means (Plexiglas)	Box 2 Means (Water)	Contrast Factor (Box 1 Means – Box 2 Means)	Comments
1					
3					
5					
7					
Specifications	See Table 11-12	n/a	n/a	2.0 to 12.0	n/a

Table 11-11 20cm QA#2 Phantom Low Contrast Detectability Image Performance Worksheet #1

- 4.) View the 1st, 3rd, 5th, and 7th images of the QA#2 Holes Series, record the number of visible holes in Table 11-11, and verify each image meets specifications.
- a.) While viewing the 1st, 3rd, 5th, and 7th images, indicate in Table 11-11 the number of holes that can be visually distinguished. See Figure 11-8.
 - b.) Verify Visible Hole visual check meets specifications listed in Table 11-12 for the calculated Contrast Factor and record in the HHS Record Tables.

Contrast Factor Range (Box 1 Means – Box 2 Means)	Visible Number of Holes		Smallest Visible Hole Size
	Lower Limit*	Upper Limit*	
2.00 to 3.99	2	5	7.5mm
4.00 to 7.99	3	5	5.0mm
8.00 to 12.00	4	5	3.0mm

*Required number of visible holes depends on the contrast factor.

Table 11-12 20cm QA#2 Phantom Visible Hole Specifications

5.5.3.2 Failure Recovery

Specifications

At least two out of the four images of the series must pass the 20cm QA#2 Low Contrast Detectability parameter (Visible Hole Size for a calculated Contrast Factor) for the third series scan parameters specifications.

Recommended Recovery

- A.) Check Phantom Alignment (leveling is critical) and repeat this scanning and High Contrast Spatial Resolution Performance Test.
- B.) Perform Alignment Procedures (POR Alignment, BOW Alignment, CBF/SAG Alignment, ISO Alignment, and Hot ISO Alignment), perform Full Calibration (Detailed Calibration and Auto CT# Adjust, and repeat this scanning and Low Contrast Detectability Performance Test.

5.5.4 QA#3 Phantom Brightness Uniformity and CT#

5.5.4.1 Performance Verification

- 1.) Select the fourth 20cm QA#3 Phantom exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
 - b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and fourth (QA#3) series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 31 x 31 pixel reference ROI Box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a square 15 mm x 15 mm (225 mm²) box. Tolerance: 15 mm +/- 1 mm (196 mm² to 256 mm²).

If required, magnify the image to adjust to proper dimensions. See [Figure 11-9](#) for additional ROI size and placement information.
- 3.) Collect Mean values for five reference ROI Box positions on the eight 20cm QA#3 Phantom images.

If required, magnify the image to adjust to proper dimensions. See [Figure 11-9](#) for additional ROI placement information.

 - a.) Position the reference ROI Box built in step 2 directly over the center of the image using

the grid cross-hairs as a guide.

- b.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
- c.) Click on the **MEASURE** button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.
- d.) Reposition Box # 2 to the left center portion on the first image (Box # 2 position in [Figure 11-5](#)).
- e.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.
- f.) Repeat steps 3.c. through 3.e. for each of the remaining ROI box positions shown in [Figure 11-6](#) (Box # 3 through 5).
- g.) Record the Mean values of the eight images in the series for each of the five box positions in [Table 11-13](#).

Each image can only display text for the mean, standard deviation, and box area for three images at a time. To view the data for a particular box, select the box on the image and the system displays the data for the box number selected.

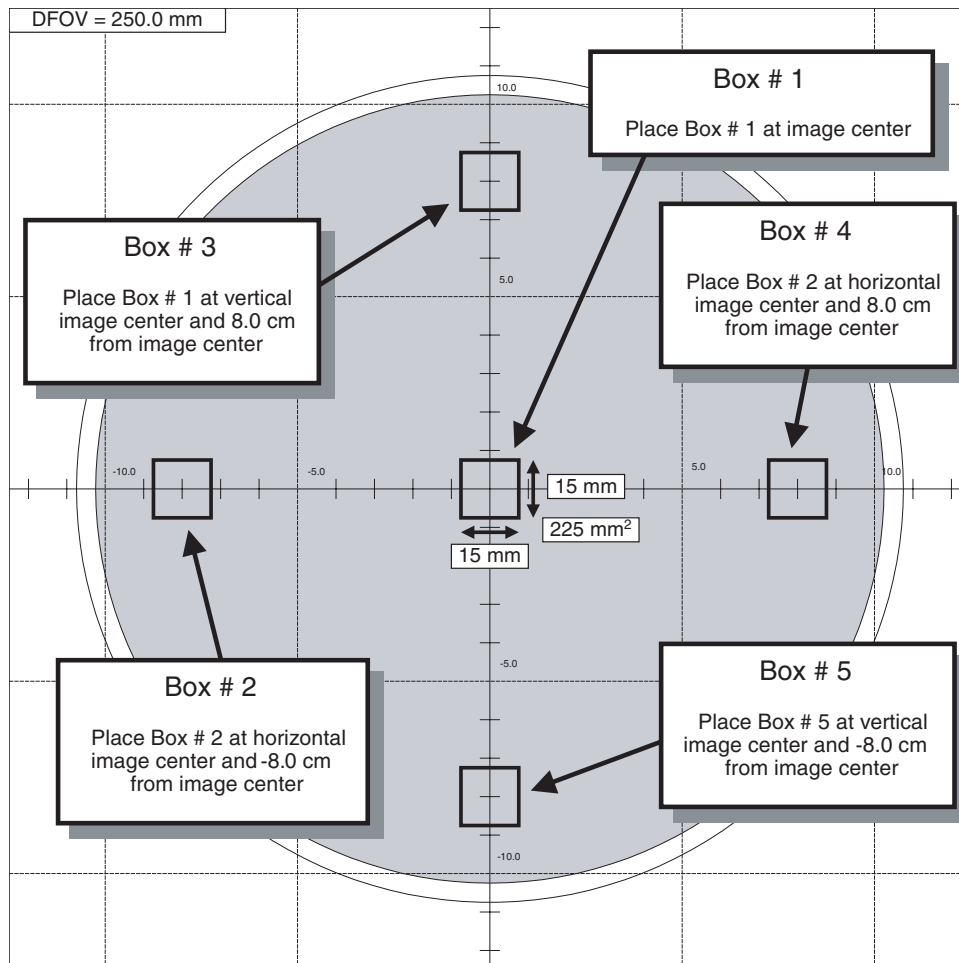


Figure 11-9 20cm QA#3 Phantom Brightness Uniformity & CT# Measurement - Building And Placing Reference ROI Boxes

Image	Box 1 (Center)	Box 2 (Left Center)		Box 3 (Top Center)		Box 4 (Right Center)		Box 5 (Bottom Center)		Outside Boxes Avg	
	Means (Xc)	Means	Std dev	Means	Std dev	Means	Std dev	Means	Std dev	Means (AvXo)	Std dev (AvSDo)
1			n/a		n/a		n/a		n/a		n/a
2			n/a		n/a		n/a		n/a		n/a
3			n/a		n/a		n/a		n/a		n/a
4			n/a		n/a		n/a		n/a		n/a
5			n/a		n/a		n/a		n/a		n/a
6			n/a		n/a		n/a		n/a		n/a
7			n/a		n/a		n/a		n/a		n/a
8			n/a		n/a		n/a		n/a		n/a

Table 11-13 20cm QA#3 Phantom CT# Brightness Uniformity & CT# Image Performance Worksheet

Box Size = 196 mm² to 256 mm²

15 mm (+/- 1 mm) x 15 mm (+/- 1 mm)

31 (+/- 2 pixels) x 31 (+/- 2 pixels)

Box Positions: Box 1 = 0 mm x 0 mm

Box 2 = 0 mm x -80 mm

Box 3 = 80 mm x 0 mm

Box 4 = 0 mm x 80 mm

Box 5 = -80 mm x 0 mm

Row	Images	Center Box Means (AvXc)	Outer Boxes Means Averages (AvXo)	Outer Boxes Means Averages minus Center Box Means (AvXo - AvXc)	Outer Boxes Standard Deviation Averages (AvSDo)	Comments
2A1A	1, 3, 5, 7				n/a	
1B2B	2, 4, 6, 8				n/a	
Specifications		0.0 ± 3.0	n/a	<± 3.0	n/a	

Table 11-14 48cm Phantom CT# Brightness Uniformity & Noise Row Performance Worksheet

- 4.) Calculate the Brightness Uniformity and CT# values for each image in the series, record values in [Table 11-14](#), and compare to specifications.
 - a.) Calculate and record the average means values (AvXo) for the four outside Boxes (Boxes 2 through 5) for each of the images and record in [Table 11-13](#).
 - b.) Calculate and record the average center box (Box 1) means values (AvXc) for each row (2A1A: Images 1, 3, 5, and 7; 1B2B: Images 2, 4, 6, 8) in [Table 11-14](#).
 - c.) Calculate and record the average outside boxes (Box 2 through 5) means values (AvXo) for each row (2A1A: Images 1, 3, 5, and 7; 1B2B: Images 2, 4, 6, 8) in [Table 11-14](#).
 - d.) Calculate the Brightness Uniformity (AvXo - AvXc) value for each row and record in [Table 11-14](#).
 - e.) Verify Brightness Uniformity (AvXo - AvXc) value and the average CT# value (AvXc) for each row meets specifications listed in [Table 11-14](#).
 - f.) Record the 20cm QA#3 Phantom Brightness Uniformity (AvXo - AvXc) value and average CT# value (AvXc) for each row in the HHS Record Tables.

5.5.4.2 Failure Recovery

Specifications

Each Row (2A1A, 1B2B) of the series must pass 20cm QA#3 Phantom Brightness Uniformity and average CT# specifications:

- AvXo - AvXc: < +/- 3.0
- AvXc: < +/- 3.0

Recommended Recovery

- A.) Perform DETAILED CAL.
- B.) Perform AUTO CT# ADJUST.
- C.) Repeat this procedure to verify 20cm QA#3 Phantom Image Performance.

5.5.5 QA#3 Phantom Noise

5.5.5.1 Performance Verification

- 1.) Select the fourth 20cm QA#3 Phantom exam acquired in the previous section.
 - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
 - b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
 - c.) Select the exam and fourth (QA#3) series acquired in the previous section.
 - d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.
- 2.) Build a 51 x 51 pixel reference ROI Box using the Image Works Viewer tools.
 - a.) Click on the grid button to place a grid on the first image.
 - b.) Click on the MEASURE button and select the box ROI icon.
 - c.) Adjust the size of the ROI box to be a square 25 mm x 25 mm (625 mm²) box. Tolerance: 25 mm +/- 1 mm (576 mm² to 676 mm²).
If required, magnify the image to adjust to proper dimensions. See [Figure 11-10](#) for additional ROI size and placement information.
- 3.) Collect Standard Deviation values for the single reference ROI Box position on the eight 20cm QA#3 Phantom images.
If required, magnify the image to adjust to proper dimensions. See [Figure 11-10](#) for additional ROI placement information.
 - a.) Position the reference ROI Box built in step 2 directly over the center of the image using the grid cross-hairs as a guide.
 - b.) Type `prop a` in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
 - c.) Record the Standard Deviation values of the eight images in the series for centered box position in [Table 11-15](#).
- 4.) Calculate the average Noise values for each image in the series, record values in [Table 11-15](#), and compare to specifications.
 - a.) Calculate and record the average Noise values (AvSDc) for the inside Boxes for each of the two rows (2A1A and 1B2B) and record in [Table 11-15](#).
 - b.) Verify Noise (AvSDc) values for each row meets specifications listed in [Table 11-15](#).
 - c.) Record the 20cm QA#3 Phantom Noise (AvSDc) for each row in the HHS Record Tables.

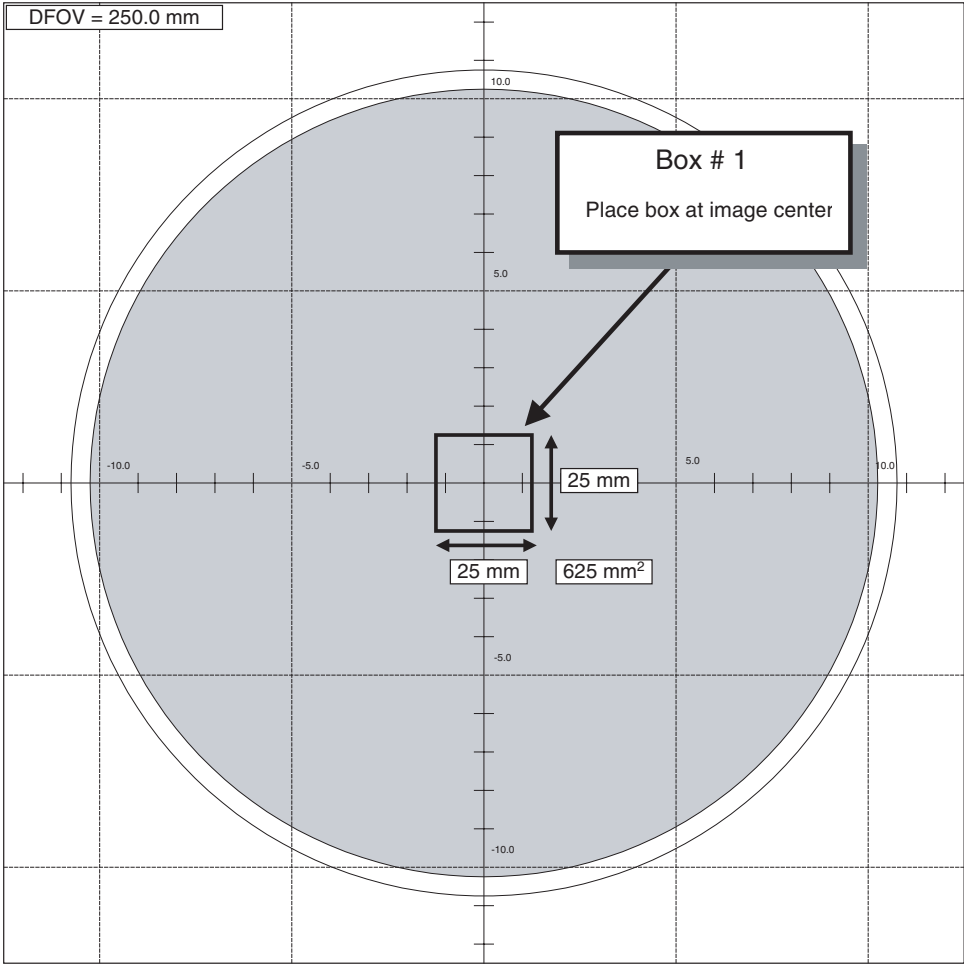


Figure 11-10 20cm QA#3 Noise Measurement - Building And Placing Reference ROI Box

Image	Row	Box 1 Std Dev (SDc)	Average Std Dev (AvSDc)	AvSDc Specifications		Comments
				Systems with less than 5000 scans	Systems with more than 5000 scans	
1	2A1A			3.2 ± 0.3	3.2 ± 0.4	
3						
5						
7						
2	1B2B					
4						
6						
8						

Table 11-15 20cm QA#3 Phantom Noise Performance Worksheet

Box Size = 576 mm2 to 676 mm2
25 mm (+/- 1 mm) x 25 mm (+/- 1 mm)
51 (+/- 2 pixels) x 51 (+/- 2 pixels)

Box Position: Box 1 = 0 mm x 0 mm

5.5.5.2 Failure Recovery

Specifications

Each Row (2A1A, 1B2B) of the series must pass 20cm QA#3 Phantom Noise specifications shown in [Table 11-15](#).

Recommended Recovery

- A.) Perform DETAILED CAL.
- B.) Perform AUTO CT# ADJUST.
- C.) Repeat this procedure to verify 20cm QA#3 Phantom Image Performance.

Section 6.0

Save System State

Use the following commands to create the System State MOD.

- 1.) Load a MOD into the mod drive on the front of the console.

- 2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon,



- 3.) Click on the PM icon.



- 4.) Select SYSTEM STATE to open the System State Save/Restore menu.

- 5.) Select ALL

- 6.) Select SAVE

- 7.) When the save operation completes, select FILE and QUIT from the pull down menu.

- 8.) Remove the MOD from the drive

Chapter 12

Hospital/System Network Integration



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 14](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Overview

LightSpeed systems support two basic Networking Protocols:

- Advantage NET (IC, Signa 4.X, CT-HLA, CT/I ...)
- DICOM (CT/I, CT Synergy, Advantage Workstations, ...)

DICOM NETWORK INTRODUCTION

DICOM networks basically operate on the **tasks** or services that various devices on the network use or provide. These services are labeled as Application Entity Titles (AE Titles). The LightSpeed 3.X system is a **user** of six DICOM Network Services and is a **provider** of two DICOM Services:

AS A DICOM SERVICE USER:

- Send or **Push** images to another network device.
- Send or **Push** images to a DICOM Printer.
- Review image database on another device and retrieve or **Pull** selected images from that device (Query/Retrieve User).
- Send or **Push** images to a an image storage device and obtain confirmation that the images have been archived (Storage Commitment).
- Obtain Patient Worklist Information from the Hospital HIS/RIS System.
- Store images on MOD media.

AS A DICOM SERVICE PROVIDER:

- Receive **Pushed** images from another network device
- Allow another network device to review the LightSpeed image database and to retrieve or **Pull** selected images (Query/Retrieve Provider)

For each DICOM Service that the LightSpeed system will be a **User** (except for storing images on MOD media), you must **declare** this device on the LightSpeed system using three menu selections. For some devices, you must declare not only the device, but each service (AE Title) that the device provides. For example, you may be required to declare a PACS System twice on the LightSpeed system: once as a destination to **push** images and, second, as destination that provides storage commitment capability after images have been **pushed**.

For each DICOM Service that the LightSpeed system will be a **Provider**, you must **declare** the LightSpeed system on the network device that will be using these services.

Information required to complete configuring a hospital DICOM network is provided by the hospital network administrator (hostnames, IP Addresses) and the DICOM Conformance Statement document (AE Titles, Port Numbers) provided with each DICOM compatible network device on the network.

Section 2.0 Before You Start

2.1 Network Physical Requirements

Before setting up the LightSpeed 3.X system on the hospital network, verify the following physical items are complete:

- LightSpeed console, monitor, keyboard, and mouse are installed and connected.
- LightSpeed system power is on
- Hospital Ethernet network RJ45 Class IV twisted pair cable is connected to the LightSpeed console network receptacle.
- Hospital network connection is operational and is running 10baseT or 100baseT.

2.2 LightSpeed 3.X Network Identity Information

To declare the LightSpeed 3.X system on the network, ensure the following network identity information is available from the Hospital Network Administrator:

- Hostname (No more than 16 Characters)
- Internet Protocol (IP) Address
- Subnet Net Mask IP Address (if applicable)
- Broadcast Address (if applicable)

2.3 LightSpeed to DICOM Remote Hosts Network Information

To declare DICOM remote hosts (PACS systems, archival devices, review workstations) on the LightSpeed 3.X system, ensure the following information is available for each remote host:

From the Hospital Network Administrator:

- Hostname
- Internet Protocol (IP) Address
- Network Protocol (DICOM for LightSpeed Systems)

From the Remote Host Device DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
- DICOM Listening Port Number

2.4 LightSpeed to DICOM HIS/RIS Interface Network Information

To declare DICOM Hospital HIS/RIS Interface devices (Mitra and others) on the LightSpeed 3.X System, ensure the following information is available:

From the Hospital Network Administrator: *Internet Protocol (IP) Address*

From the HIS/RIS Interface Device DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM Service that the HIS/RIS interface provides)
- DICOM Listening Port Number

2.5 LightSpeed to DICOM Printers Network Information

To declare DICOM on the LightSpeed 3.X System, ensure the following information is available for each printer:

From the Hospital Network Administrator:

- Hostname
- Internet Protocol (IP) Address

From the Printer DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
- DICOM Listening Port Number

Section 3.0

Declaring the System on the Hospital Network

3.1 Enter Configuration Routine

- 1.) On the OC, open a Unix shell window.
- 2.) Enter root as a superuser:
Type: `su -` **ENTER** at the prompt.
Type: `#bigguy` **ENTER** at the password prompt.
- 3.) Change directory to scripts:
Type: `cd /user/g/scripts` **ENTER** at the root prompt.
- 4.) Launch the Install Utility:
Type: `reconfig` **ENTER** at the prompt.
The OC displays the Install Utility Window as shown in [Figure 12-1](#).



Figure 12-1 Install Utility Window

- 5.) Enter the Configuration Routine:
Using the mouse, click on the CONFIG button.
The OC displays the System Configuration - System Settings screen, as shown in [Figure 12-2](#).

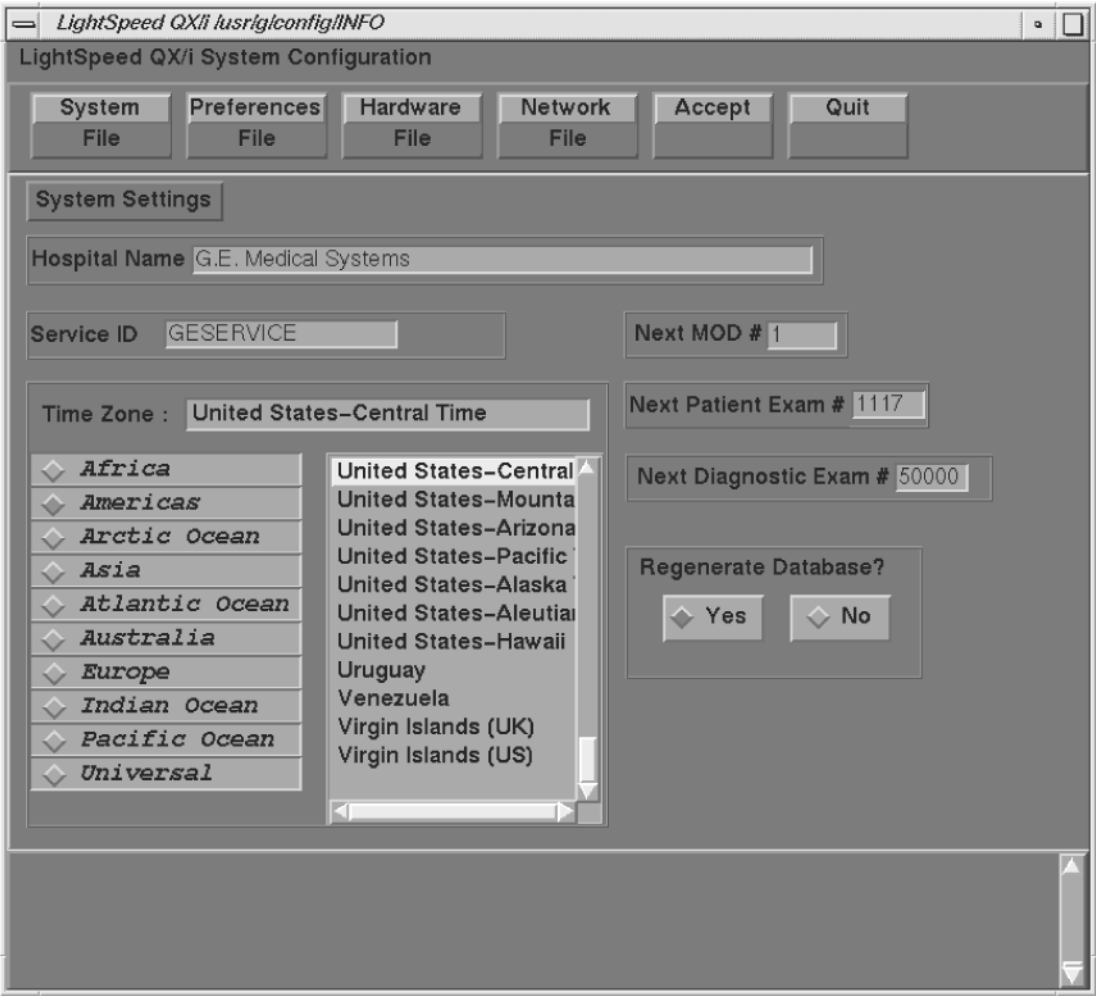


Figure 12-2 System Settings Screen

3.2 Configure Network Settings

This screen provides the ability to declare the system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for systems on a subnet) must be obtained from the hospital network administrator.

- 1.) Select the NETWORK button to display the Network Settings screen as shown in [Figure 12-3](#).
2.) Enter the Suite Name.

The Suite Name is a means of identifying this particular LightSpeed system as a part of a group of CT Systems in a suite configuration. This Suite Name will appear on all image headers.
The Suite Name must start with a letter, followed by three alphanumeric characters (total MUST be four characters long). The name of the OC interface will be <Suite Name>_oc and the SBC interface will be <Suite Name>_sbc.

Figure 12-3 Networks Settings Screen

- 3.) Enter the hospital provided Host Name.
The Host Name identifies the network hostname and AE Title of the LightSpeed system.
The Host Name:
 - **MUST NOT** be <Suite Name>_oc or <SUITE NAME>_OC.
 - **MUST NOT** exceed 16 Characters.
 - **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _
- 4.) Enter the hospital provided IP Address.
- 5.) Enter the hospital provided Net Mask (if the system is on a subnet).
- 6.) Enter the Broadcast Address:
The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1s or 0s depending on the configuration of the network. The standard default is 1s but older SunOS machines used 0s.
For example:
If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the broadcast address.
If the network contains genesis based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.
- 7.) Enter the hospital provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.
- 8.) Select NIS (Yellow Pages database) Advanced Option only if requested by the hospital network administrator as follows:
 - a.) Select ADVANCED OPTIONS button on the Network Settings screen.
 - b.) Select Use NIS? button.

- c.) Enter the hospital provided Domain Name.
- 9.) Record all the Network parameters in the *LightSpeed 3.X Software Installation Procedures* Document, or on the worksheet in Chapter 6.

3.3 Initiate System Reconfiguration

- 1.) Select ACCEPT on the System Configuration Screen.
The system loads the application software, OS patches, and kernal changes, and configures the system on both the OC and the SBC.
This loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window, which closes when the loading process is complete. All the window output is logged to a file named:
`/var/adm/install.log.YYYYMMDDWWWHMMSS.`
(Where `YYYMMDDWWWHMMSS` is the Date/Time that the loading process was started.)
- 2.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES.
- 3.) The system will automatically login as ctuser after the reboot. Select OK on the Autostart Disabled popup message.
- 4.) To startup Applications, in the console shell window, type **startup** ENTER.

Section 4.0

Declaring Remote Hosts on the System

4.1 Enter Remote Host Configuration Screen

- 1.) On the OC, select the IMAGE WORKS icon.
- 2.) Select NETWORK.



4.2 Declaring Advantage NET Remote Hosts on the LightSpeed

Use Advantage NET Protocol networks to communicate with older CT or MR Systems (MR Signa 4.x, CT-HLA, CT/I Systems, and Workstations that support the Advantage NET protocol). Advantage NET Protocol does not offer full compatibility with LightSpeed DICOM formats.

Repeat the following procedure for each Advantage NET Remote Host device that the customer expects to have communicating with this system.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host Parameter Screen as shown in [Figure 12-4](#).

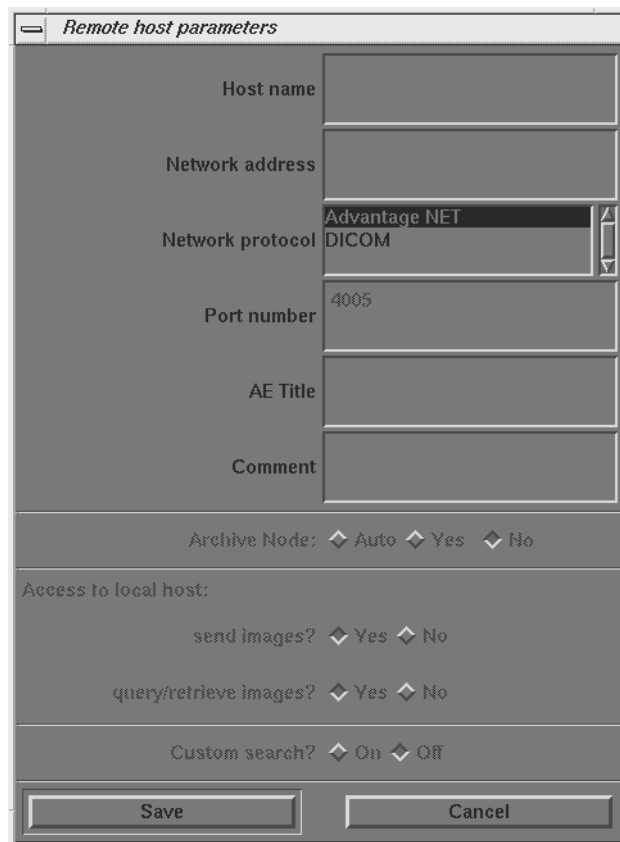


Figure 12-4 Advantage Net Network Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).

- 4.) Select ADVANTAGE NET as the Network Protocol.
The systems automatically de-highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM protocol parameters and do not apply to Advantage NET type devices.
- 5.) Select SAVE to store the parameter settings of the remote host.

4.3 Declaring DICOM Remote Hosts on the LightSpeed

Use DICOM protocol networks to communicate to DICOM devices such as CT/i, CT Synergy, DLX, MR Lx, and third party hosts.

Repeat the following procedure for each DICOM remote host device that the customer expects to have communicating with this system.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host Parameter screen as shown in [Figure 12-5](#).

Figure 12-5 DICOM Network Setting Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).
- 4.) Select DICOM as the Network Protocol.
The system automatically highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM Protocol parameters.
- 5.) Enter the TCP/IP Listening Port Number from the DICOM Conformance Statement provided with the device.

- 6.) Enter the AE Title from the DICOM Conformance Statement provided with the device.
Application Entity Titles (also known as ACR-Nema or Dicom Name) refer to the DICOM Network Services that a device provides to the LightSpeed System. For most devices, the AE Title is the same as the hostname (LightSpeed systems are equipped with this feature).
However, some devices such as PACS systems may have separate AE Titles and port numbers for each of the services that the PACS system provides. In these cases, you must enter a separate remote host (same hostname and IP Address) for each of the independent AE Title Services that the host provides (one host as an image **push-to** destination, another host as a **query/retrieve** provider, and another host as a **storage/commitment** provider).
Be sure to review the DICOM Conformance Statement for each device that will provide a remote host network service for the LightSpeed system (image **push-to** or store destination, Query/Retrieve, and Storage Commitment) to ensure that each service is correctly configured.
- 7.) Select the correct Archive Node choice for the device. The Archive Node selection field defines the ability of the remote host to act as a DICOM Storage/Commitment provider and indicate to the operator that a study/series/image was archived. Select:
 - AUTO to have the LightSpeed system automatically check to see if the designated remote host is a DICOM Storage/Commitment Provider.
 - YES if the device is the hospital designated DICOM Storage/Commitment Provider. During an Application Study Archive process, the local browser screen will indicate Archive Status = Y to the operator.
 - NO if the device is not a DICOM Storage/Commitment Provider.
- 8.) Select the correct Access to local host: settings. These two selections allow you to selectively block the remote host from using the LightSpeed DICOM services as a provider (image **push-to** destination, and a **Query/Retrieve** provider).
 - Send Images? Set to YES if the customer wants the LightSpeed system to be able to have images **pushed** to the system from the applicable remote host. Set to NO if the customer wants to block an image **push** from the applicable remote host.
 - Query/retrieve images? Set to YES if the customer wants the remote host to be able to review the LightSpeed image database (query) and **pull** selected images from the database. Set to NO if the customer does not want the remote host to have this ability.
- 9.) Select the correct Custom search? setting. This selection allows the LightSpeed to selectively search through the remote host's image database when the operator is using remote browser screen to **query** the remote host. The search parameters that the LightSpeed system allows the customer to use are: last name contains, patient ID, exam number, accession number, and exam date.
 - Select ON if the device supports custom searches as part of the devices **Query/Retrieve** DICOM Provider service.
 - Select OFF if the device does not support custom searches.
- 10.) Record all the remote host network parameters for each remote host in the *LightSpeed 3.X Software Installation Procedures* Document.
- 11.) Select SAVE to store the parameter settings of the remote host.

Section 5.0

Declaring the System on Remote Hosts

5.1 Declaring LightSpeed on Advantage NET Protocol Devices/Systems

Refer to the appropriate service manual provided with the Advantage NET Protocol device or system to find instructions how to declare the System as an Advantage NET remote host.

5.2 Declaring LightSpeed on DICOM Protocol Devices/Systems

Refer to the appropriate Service Manual provided with the DICOM protocol device or system to find instructions how to declare the System as a DICOM remote host.

The LightSpeed System provides two DICOM Services as a provider to remote hosts:

- A remote host can **push** images to the LightSpeed image database.
- A remote host can review the LightSpeed image database (query) and **pull** selected images (retrieve).

Use the following LightSpeed parameter information to configure the DICOM device/system to either **push** images to the LightSpeed and/or perform a **Query/Retrieve** operation:

- Hostname: Provided by the Hospital Network Administrator. Exactly the same LightSpeed assigned hostname entered in the Network Configuration Screen.
- Application Entity Title: Exactly the same entry as the Hostname.
- Network Address: Provided by the Hospital Network Administrator. Exactly the same LightSpeed assigned IP Address entered in the Network Configuration Screen.
- Network Protocol: DICOM 3.0.
- Port Number: For all DICOM service that the LightSpeed System provides, use 4006.
- Provider Type: This field concerns the LightSpeed DICOM **Query/Retrieve** provider capability. All LightSpeed systems are wstudy root systems, which allow queries at the exam, series, and image level.
- Support Worklist: This field concerns whether a DICOM **Query/Retrieve** provider capable device or system supports a filter search of the image database. All LightSpeed systems support a filtered search of the image database as part of the LightSpeed DICOM **Query/Retrieve** provider capability.

Section 6.0

Declaring DICOM HIS/RIS Interfaces on the System

6.1 Prerequisites

Most hospital HIS/RIS systems are not DICOM compatible and require a DICOM HIS/RIS Worklist Interface to provide patient scheduling information to the LightSpeed system. Contact your local HNS support engineer to determine exactly what DICOM HIS/RIS Interface is appropriate for the customer.

In addition, the LightSpeed system must have the ConnectPRO software option installed to utilize the DICOM Protocol Worklist capability.

6.2 Loading ConnectPRO Software Option on the System

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.



- 3.) Click on INSTALL.

- 4.) Select INSTALL OPTIONS and click START. The console displays the Software Options window as shown in Figure 12-6.

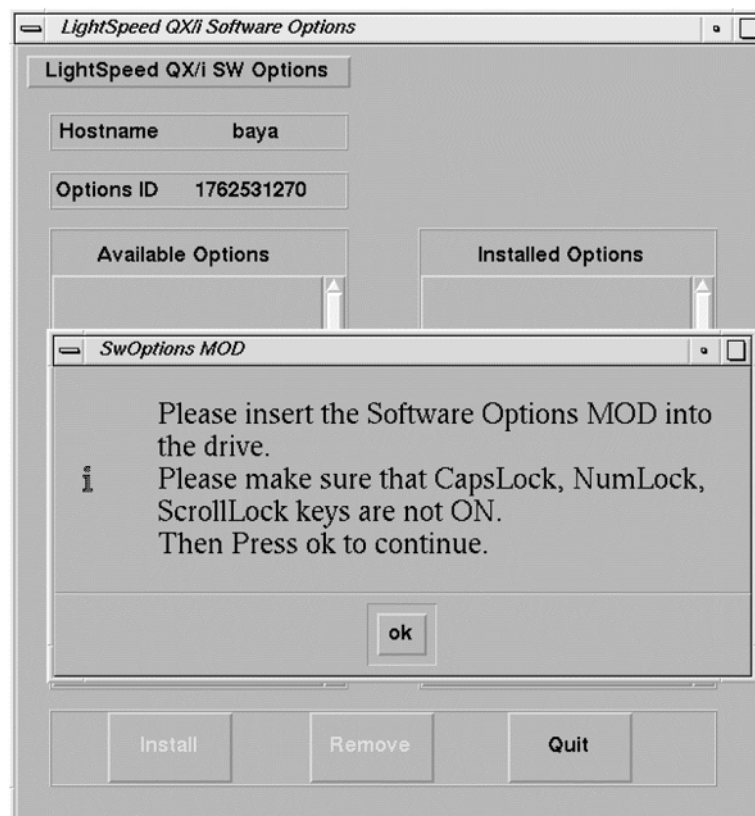


Figure 12-6 Options Window when First Selected

- 5.) Insert the ConnectPRO Options MOD into the MOD drive.
- 6.) Click on OK to continue.
- 7.) On the Software Option window, select the CONNECTPRO option in the Available Options window.
- 8.) Click on the INSTALL button.

The console may display a message box during the software loading operation. When the system has completed loading the software:

- the Installed Options window displays the ConnectPRO option, and
- the console displays the ConnectPro Setup window as shown in [Figure 12-7](#).

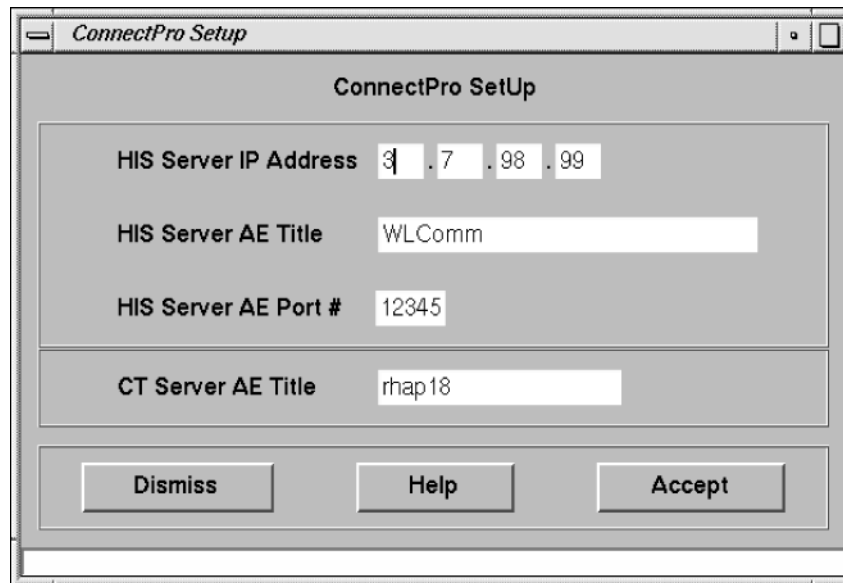


Figure 12-7 ConnectPro Setup Screen

- 9.) Enter the hospital provided HIS Server IP Address.
- 10.) Enter the HIS Server AE Title from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 11.) Enter the HIS Server AE Port # from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 12.) Record all the ConnectPRO HIS/RIS Network parameters in the *LightSpeed 3.X Software Installation Procedures* Document.
The system will automatically load the LightSpeed CT Server AE Title (Hostname of the system).
- 13.) Click on the ACCEPT button. The system will check valid setup entries, execute a perl script that makes changes to the configuration files for HIS/RIS, and return to the Software Option window shown in [Figure 12-6](#).
- 14.) Click on the QUIT button and the subsequent OK button in the message window.
- 15.) Remove the MOD from the MOD drive, and write protect the side containing the ConnectPRO option.

Section 7.0

Declaring DICOM Filming Devices on the System

7.1 Prerequisites

Before configuring DICOM filming devices (cameras, printers) on the LightSpeed System, ensure the following are complete:

- Filming Device Service Representative to assist in camera/printer setup for best image quality film presentation.
- Hospital DICOM network is operational.
- Filming device is connected to the DICOM network with the correct filmer DICOM interface.
- Filming device is DICOM protocol compatible.
- Filming device DICOM Conformance Statement document is available.

Note: **Filmer DICOM Application Entity Titles may be site specific. Make sure that you check with the Filming Device Service Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination filming device.**

7.2 Declaring DICOM Filming Devices on the System



NOTICE Potential For Data Loss

Empty all filming queues before modifying camera parameters.

This section contains procedures for recording important Camera setup information. Use the table(s) at the end of this section to record information from the setup screens.

- 1.) Click on the SERVICE DESKTOP button.
- 2.) On the Desktop Toolbar select UTILITIES -> INSTALL -> INSTALL CAMERA. The Install Camera window appears, along with a warning message pop-up box, to remind you that all filming queues must be empty before you begin to update or delete a camera.
- 3.) The Graphical User Interface displayed shows a list of cameras installed (See Figure 12-9). First, you must click OK in the warning message box. See Figure 12-8.

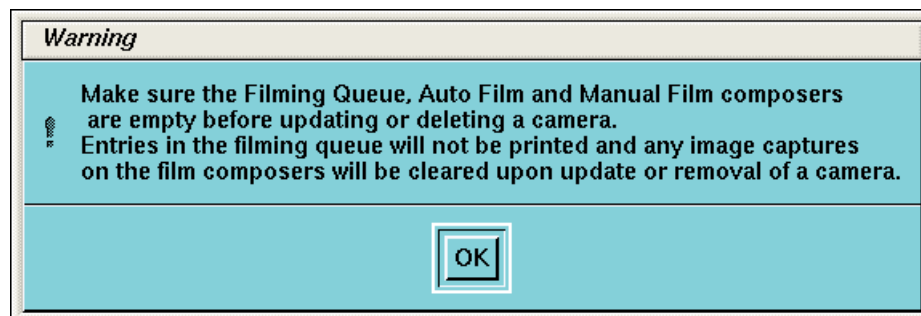


Figure 12-8 Warning Screen

- 4.) Now you are asked a series of questions.
 - a.) To add a new camera, click the ADD button (See Figure 12-9).
 - b.) A dialog window for the camera type (DASM/DICOM) appears. If no DASM is detected during the OC boot, the DASM button will be disabled (Figure 12-10). If a DASM is present

and has not been detected, reboot the OC and run the camera configuration tool again.

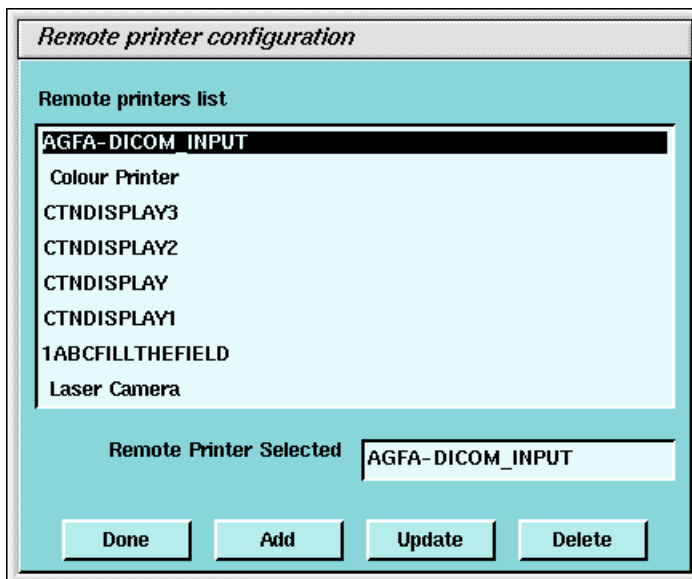


Figure 12-9 Printer Configuration GUI

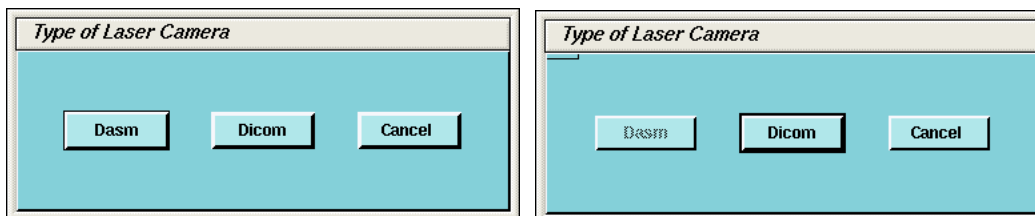


Figure 12-10 Dialog Box for camera Type (with and without DASM)

- 5.) To add a new laser camera, click DASM in the camera type dialog box. This brings up a list of available camera models. Select the appropriate model form the list and click SELECT (See Figure 12-11). Now you must configure it.

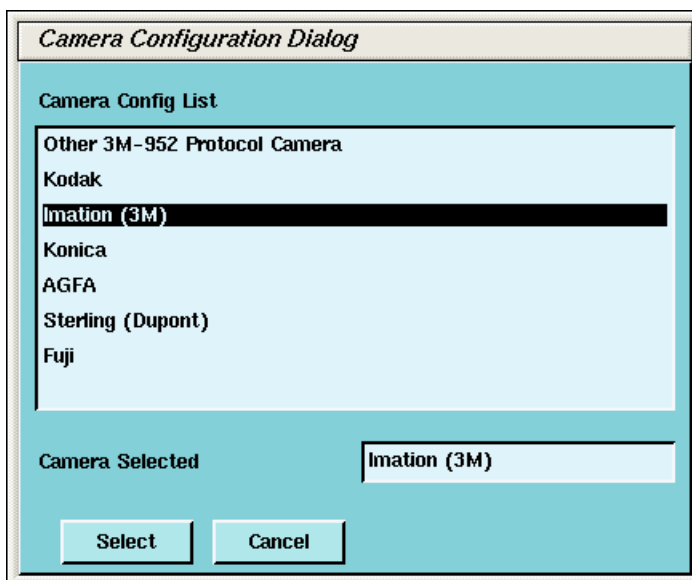


Figure 12-11 Camera Model Dialog (with DASM)

- a.) DASM Interface is automatically detected as either Analogue or Digital
- b.) Two Laser Options are available for laser cameras: SLIDES and ZOOM. Set this option only if the camera being installed supports slides and zoom. Setting the option allows it to be enabled or disabled at the application level.
- c.) Camera manufactures provide two (2) Magnification Type options for cameras. The SMOOTH resolution blurs the image, while the SHARP resolution makes the image pixels more pronounced. The default is smooth.

Comment:

To film good images, and have them look like images filmed by other GE CT products, the following camera settings are suggested:

Kodak: SMOOTH
 Dupont/Sterling: SMOOTH
 3M/Imation (Laser Camera): SHARP
 3M/Imation (Dry View): SMOOTH
 Agfa: SMOOTH

- d.) Select the appropriate File Format. Select ON from the drop down list boxes on the menu. Valid film formats are determined by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4). When finished setting parameters, click on DONE and proceed to step 8.

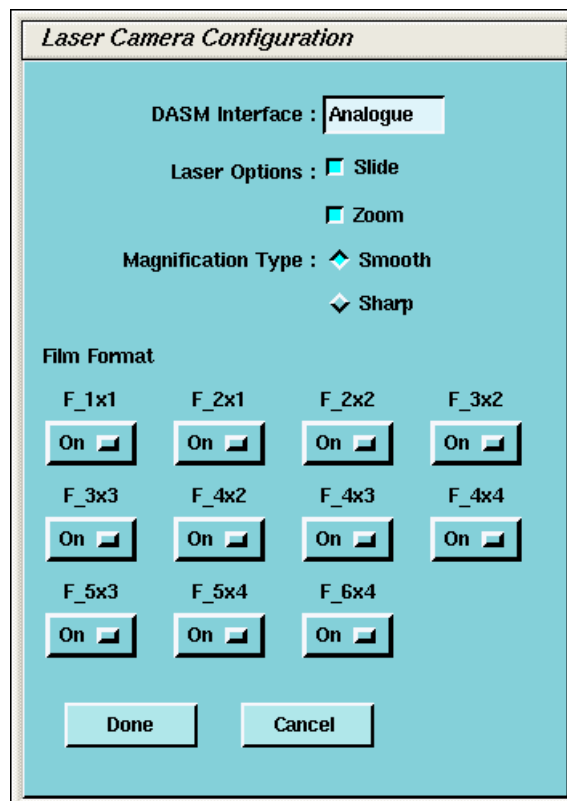


Figure 12-12 Laser Camera Configuration

- 6.) To add a new DICOM camera, click on ADD and then DICOM in the dialog box that appears.
 - a.) A list of camera models appears (See Figure 12-13). Select the appropriate model from the list and click SELECT. Clicking SELECT presets all the parameters to that models

except the Network parameters.

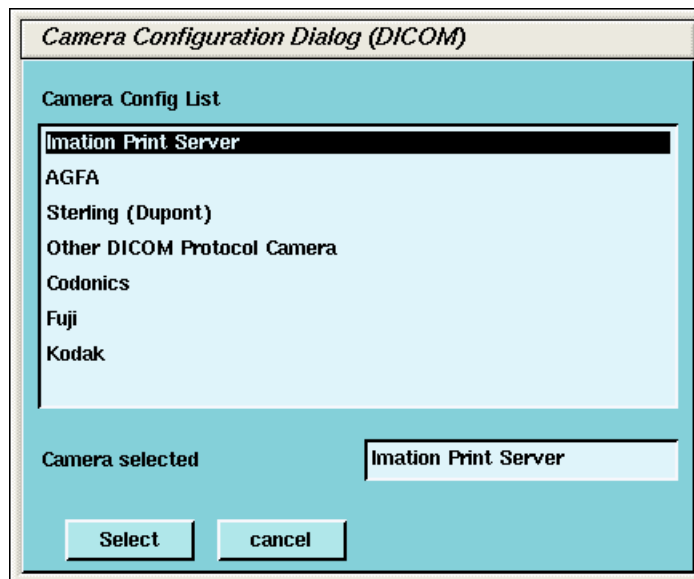


Figure 12-13 Camera Model Dialog (DICOM)

*Comment:
It's advised to
recheck the
preset
information with
the camera
manufacturer's
representative.*

Selection of a different camera model clears the Image Quality parameters, because these are camera manufacture dependent.

b.) Enter the Network Parameters (See Figure 12-14)

- > Device Name A unique name used to identify the camera.
- > Host Name DICOM print server host name, as defined by the hospital.
- > IP Address DICOM print server IP address, as defined by the hospital.
- > AE Title DICOM print server application entity title, as defined by the print server. *You should consult the manufacturers DICOM Conformance Statement.*

Note: The Application Entity Title for the Camera may be site specific. Make sure that you check with the Camera Manufacturer's Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination DICOM Print Camera.

- > TCP/IP Listen Port DICOM print server TCP/IP listen port, as defined by the server. *You should consult the manufacturers DICOM Conformance Statement.*
- > Comments Optional comments used by the DICOM print server.

Figure 12-14 DICOM Camera Configuration

- c.) Medium Type specifies the type of film being used. Currently, only BLUE FILM and CLEAR FILM are supported.
- d.) Set Destination to the final location for film output: either MAGAZINE or PROCESSOR.
- e.) Orientation selects film orientation. Only PORTRAIT is currently supported.
- f.) Set the Magnification Type. This parameter selects the algorithm used to interpolate pixels for proper film resolution. Set this parameter after consulting the camera manufacture to ensure optimal image quality. Choices are describe below:
 - > None No interpolation. This option is not supported by all cameras.
 - > Replicate Adjacent pixels are interpolated. This can result in "pixelized" images. *This algorithm is not normal preferred.*
 - > Bilinear A 1st order interpolation of pixels. Results in images usually described as blurred. *This algorithm is not normal preferred.*
 - > Cubic A 3rd order interpolation. Used with a large number of possible formulations. Camera manufactures define parameters called "smoothing type" to set coefficients used in this algorithm. The implementation of these "smoothing soefficients" is camera dependent.
- g.) Select the Standard Film Formats. Select the film format by choosing ON in the pull-down menu box located below each selection. See Figure 12-14. Valid film formats are set by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and

Comment: For most Camera Manufacturers, the preferred selection is CUBIC.

AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).

- h.) After the camera has been configured, click the **ADVANCED** button. This creates the camera device file for you automatically and pops up the Advanced Parameters screen. See Figure 12-15.

Figure 12-15 Advanced Parameters (Camera Attribute Dialog)

- 7.) Advanced camera parameters control the image quality of films.

Note: For more information on the proper settings for these parameters, please see the Camera's DICOM Print Device Conformance Statement or the Camera Manufacturer Representative. You may need to refer to a copy of the Conformance Statement as you are working with the Camera Manufacturer's Representative, to correctly set up the DICOM Print Camera I/Q and Time-out Settings.

- a.) Configuration - This parameter is camera manufacturer dependent as is typically used to specify the image contrast. The Configuration field may be up to 1024 characters long. The field will scroll automatically as text is entered. To review your entry, simply click and hold the middle mouse button, while the cursor is in the field, and drag the mouse towards the right (or left) as needed.

Note: **Recommended Configuration Setting Values:**

Agfa Drystar (MG3000)	PERCEPTION_LUT=KANAMORI (100)
Imation Dryview (8700)	LUT=0,7
Kodak Laser Printer 190	CS434\CN0\PD1.20

- b.) Smoothing Type - Set Smoothing Type to **ON**, and input the selected value. This parameter is used when the magnification type is CUBIC. It represents the coefficient for the image resolution algorithm. This parameter is camera manufacturer dependent, and should be re-verified with your radiology department.

Note: **Recommended Smoothing Type Starting Values and Ranges:**

Agfa DryStar (MG3000)	Start Value: 140	Range: 137-150
Imation Dryview (8700)	Start Value: 3	Range: 3-13
Kodak Laser Printer 190	Start Value: Enhanced	Range: Normal

- c.) Minimum and Maximum Density - Used to set brightness of the images on film. The range of values is 0-4095, although the valid range for a specific camera is manufacture dependent. For Maximum Density, input the correct value into the text box. For Minimum Density, set it to ON and input the correct value in the text box.

Note:

Recommended Minimum and Maximum Density Starting Values:

Agfa Drystar (MG3000)	Min.: 20 or 23	Max: 300
Imation Dryview (8700)	Min.: (Blank)	Max: 300
Kodak Laser Printer 190	Min.: 20	Max: 300

- d.) Empty Image Density - This parameter sets the density for empty film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- e.) Border Density - This sets the density for the border used around the film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- f.) Film Size - Allows the system to specify a particular film size, if selected.
- g.) Trim - YES produces a white (clear) box surrounding each image.
- h.) Priority - This sets the print priority.
- i.) If you have completed entry of advanced parameters, click DONE.
- 8.) **Camera Data Tables:** To locate Install Camera Information: Click on the SERVICE DESKTOP button. On the Desktop Toolbar select UTILITIES -> INSTALL -> INSTALL CAMERA. The Install Camera window appears. Select each of the cameras that are installed from the list of installed cameras, and click on UPDATE to view the camera's settings. Record the values used to set up each camera in the tables that follow. Extra tables are provided for multiple cameras

Note: You can determine this information by looking at the contents of the following files:

- For a DASM Camera: `/usr/g/ctuser/app-defaults/devices/camera.dev`
- For a DICOM Print Camera: `/usr/g/ctuser/app-defaults/devices/name.cfg`
where, *name.cfg* is the camera device name from the printer configuration GUI.

Example: `more <filename from above> ENTER`

DASM CAMERA #1

GUI SETTING	SELECTIONS	VALUE
Camera Type	Model Type of Camera	
DASM Type	Digital or Analog	☐ Digital ☐ Analog
Options	Slides or Zoom	☐ Slides ☐ Analog
Film	Smooth or Sharp	☐ Smooth ☐ Sharp
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Film Format Default	1x1, 2x1, 2x2, 3x2, etc.	

Table 12-1 DASM Camera #1

DICOM CAMERA #1

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 12-2 DICOM Camera #1

DICOM CAMERA #2

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 12-3 DICOM Camera #2

DICOM CAMERA #3

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 12-4 DICOM Camera #3

7.3 Troubleshooting Tips

The following is a summary of troubleshooting information for DICOM print that was gathered during software testing and validation of the DICOM print feature.

There is also a significant amount of additional troubleshooting procedures, and the theory of the DICOM print feature in the System Service Manual. Should you have problems installing a DICOM print camera, first read the information in System Service Manual.

ERROR BRINGING UP THE CAMERA INSTALLATION/FILMING APPLICATION

- **Symptom:** After creating/modifying the DICOM print camera the install camera interface does not come up and the filming application indicates it cannot bring up the film composer.
Cause: The configuration field contains invalid information such as a backslash (\) as the final character in the field or brackets ({}).
- **Solution:** The camera.dev file in ~ctuser/app-defaults/devices must be manually edited to remove the offending characters in the set configuration line. Invalid characters include \{ }

CONFIGURATION INFORMATION FIELD

- **Symptom:** Cannot view the entire configuration field (> 25 characters)
- **Solution:** Hold down the middle mouse button and move the field contents

NEED TO SET DICOM PRINT ATTRIBUTES NOT SUPPORTED BY SOFTWARE

- **Symptom:** User wants the white border around each image box ON/OFF permanently for this system and it cannot be set as the default for the camera.
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.
For Trim Off - set TRIM NO
For Trim On - set TRIM YES
- **Symptom:** DICOM print camera supports multiple film sizes and the user only wants to print if the film size is correct for LightSpeed 3.X (14x17). [Otherwise the camera will queue the films or return an error causing the queue to pause (based upon the DICOM print camera specifications).]
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.
To force a 14x17 film size - set filmSize 14INX17IN

NEED TO PREVENT DICOM PRINT ATTRIBUTES FROM BEING SENT TO DICOM PRINT CAMERA

- **Symptom:** Some DICOM print attributes are optional, and may result in fatal errors. For example, the Fuji camera does not support the Empty Image Density parameter for the film box.
- **Solution:** Using your favorite editor, add the following line(s) to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.
 - To prevent sending the Smoothing Parameter set FB_Smooth FALSE
 - To prevent sending the Border Density set FB_Border FALSE
 - To prevent sending the Empty Image Density set FB_EID FALSE
 - To prevent sending the Minimum Density set FB_MinD FALSE

- To prevent sending the Trim Parameter set FB_Trim FALSE

ERROR TRYING TO CONNECT TO THE DICOM PRINT CAMERA

- **Symptom:** DICOM print server can be reached (ping), but Application error indicates "Unable to start filming interface" and the help message talks about running the install.dasm (Association Error)
- **Solution:** The system is unable to complete the association. Check the AE Title and the Port number of the DICOM print server and correct them through the Install Camera procedure.

FILM COMPOSER ERROR NOT USABLE

- **Symptom:** Film composer error says "unrecognized status - code 0"
- **Solution:** Review the log file, the attention and status windows. These areas have the correct filming status (for example, **film jam** and **supply empty**).

DEBUGGING CONNECTION ISSUES DIFFICULT

- **Symptom:** The timeouts for the DICOM print are very long, which means one needs to wait a long time before you know the application is not working.
- **Solution:** The timeouts for the DICOM print were setup to ensure that the system would work regardless of whether the DICOM print camera was on a LAN or a WAN halfway around the world. The DICOM print timeouts for the association and DIMSE classes (for example, **N-GET**, **N-DELETE**) can be modified within the DICOM print camera installation. They can be reduced down to 90 seconds.

DICOM PRINT ERROR ON N-GET TIMEOUT CONFUSING

- **Symptom:** When the N-GET timeout goes off, the error message in the prslog file will be "Could not get printer status, invalid command sequence for N-GET".
- **Solution:** When the user sees the above error they may want to consider that the issue may be an inactivity timer on the N-GET DIMSE service.

DICOM PRINT CAMERA SLIDE SUPPORT

- **Symptom:** Current implementation of DICOM print does not allow selection of slide format.
- **Solution:** Feature not currently supported. Possibly in future releases.

CONFUSION ON FILM FORMAT NOTATION

- **Symptom:** GEMS Laser Camera and DICOM Print film format notations are opposite.
- **Solution:**
 - GEMS Laser Camera film format notation has always been row x col (for example, 12 on 1 = 4x3)
 - DICOM Print Standard film format notation is col x row (for example, 12 on 1 = 3x4)

Section 8.0

Teleradiology (Framegrabber Type) Systems & Aux. Monitors

8.1 Introduction

Some Genesis based systems have teleradiology (TR) systems that framegrab the Genesis GFB video (512 x 512 50/60Hz). LightSpeed DOES NOT directly support this type of TR. The LightSpeed RGB color display video is a much larger format at a much higher pixel frequency. GE does not promise any direct compatibility with framegrabbing TR systems (DICOM 3.0 TR systems may work depending on the DICOM implementation but GEMS does not and cannot validate all the various TRs.)

In the framegrabber case, a high quality (300Mhz bandwidth) video splitter/amplifier (as listed above) is needed to intercept and re-drive the display CRT RGB video. Composite greyscale would then be available on Green #2 (1280x1024 pixels at 72Hz). Any framegrabber hardware attempting to capture this signal must be capable of a 140Mhz pixel rate. This also involves TR system configuration parameters. The TR capture software may also need upgrading to deal with 1280x1024 and/or **crop** the signal. The TR remote display software may need upgrading to view the larger format. The image transmission times to the remote TR may be up to 4 times as long. GEMS will supply all technical information necessary to assist TR suppliers in making their systems work with LightSpeed, but GEMS cannot be responsible for this third party TR equipment, software, or compatibility with LightSpeed.

The following common parts are available from GE Medical Systems:

PART NUMBER	DESCRIPTION
2237018-2	Four-way Video Splitter (BNC Converter)
2154425	BNC to BNC 7' Splitter Cable
2142221	BNC to Octane 7' video Cable
2256482	DB15 to Octane 7' Video Cable
2256485	DB15 to BNC 7' Adapter Cable

Table 12-5 Common Parts Available From GEMs

8.2 Auxiliary Monitors

Figure 12-16 illustrates the **only approved GE CT configuration** for remote monitors, including the part numbers for the only approved components. The use of other configurations and/or components should be strictly avoided.

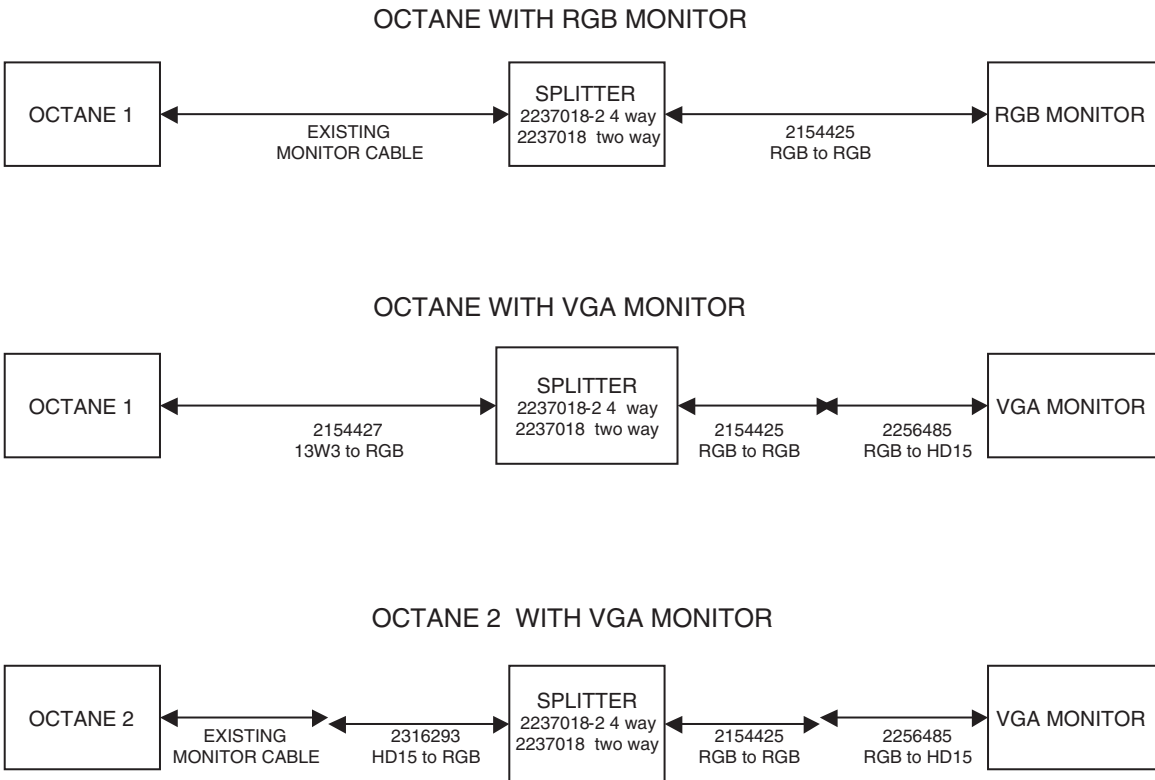


Figure 12-16 Video Splitter / Monitor Cabling

8.3 GEMS Analog Filming Interface Specs (Video & Serial)

Note: The analog (VDB) video output specs and serial interface were specified from, and are the same as, the Genesis interface output specs. This does NOT apply to the digital (LCAM) filming interface.

OUTPUT	VALUE
amplitude	1 volt peak-to-peak
video	0.643V $\pm$ 10%
setup	0.071V $\pm$ 10%
sync	0.286V $\pm$ 10%
DAC resolution	8 bits
diff. linearity	$\pm$ 1 LSB max
glitch area	80 picovolt seconds max, for any step size
rise/fall times	> 10 nsec, $\pm$ 10-90%
FS settling time	7.5 nsec typical to 1 LSB
transfer function	guaranteed monotonic
noise level	> 5.0 millivolt peak-t~peak, combined sync/async noise
DC offset	$\pm$ 1 VDC referenced to ground

Table 12-6 Video Output (Measured Into 75Q at BNC Output)

OUTPUT	VALUE
logic family	F series TTL
output low level	0.8VDC max
output high level	2.0VDC max
output period	41.336 nsecs $\pm$ 10%
transition times	10 nsec max, $\pm$ 10-90%

Table 12-7 Pixel Clock Output

VIDEO TIMING CHARACTERISTIC	60HZ	50 HZ
pixel frequency	24.192 Mhz	24.192 Mhz
pixel period	41.336 nsec	41.336 nsec
horizontal line frequency	33.6 Khz	33.6 khz
horizontal line width	720 pixels	720 pixels
horizontal active	544 pixels	544 pixels
horizontal blanking	176 pixels	176 pixels
horizontal front porch	26 pixels	26 pixels
horizontal sync	76 pixels	76 pixels
horizontal back porch	74 pixels	74 pixels

Table 12-8 Video Timing Characteristics

VIDEO TIMING CHARACTERISTIC	60HZ	50 HZ
vertical frame frequency	60Hz	50Hz
vertical frame time	560 lines	672 lines
vertical active	524 lines	524 lines
vertical blanking	36 lines	148 lines
vertical sync	3 lines	3 lines
vertical back porch	30 lines	86 lines
vertical front porch	3 lines	59 lines
scanning format	non-interlaced	non-interlaced

Table 12-8 Video Timing Characteristics (Continued)

FORMAT	VALUE
visible field	544 pixels by 524 lines
image field	512 pixels by 512 lines
greyscale field	32 pixels by 16 level grey bar on left side of image
greyscale	software selectable on/off
greyscale off value	0 (black)
initial greyscale	255 (white) at upper left corner
border field	12 lines at bottom of visible field
border field value	any 8bit value, software programmable

Table 12-9 Video Display Format

ATTRIBUTE	VALUE
interface	R5422
25D conn pinout	pin 8 (RX+), pin 21 (RX-), pin 9 (TX+), pin 22 (TX-), pin 7 (GND)
baud rate	1200 baud
word length	8 bit, 1 start bit, 1 stop bit
parity	even
type	asynchronous

Table 12-10 Host Communications/Control Serial Port (analog interface only)

Note: The digital DASM/LCAM serial control has standard R5232 on pins 2, 3, & 7.

Chapter 13

Customer Touch Leakage Test

Section 1.0 Personnel Requirements

REQUIRED PERSONS	PRELIMINARY REQS	PROCEDURE	FINALIZATION
1	mins	1 hrs	mins

Section 2.0 Overview

This procedure must be completed after all options have been installed. It covers three safety and leakage current checks:

- Patient Touch current test (completed after installation)
- System Ground resistance measurement (completed during installation)
- Ground current typical (completed after installation - optional)

Section 3.0 Preliminary Requirements

3.1 Tools and Test Equipment

ITEM	QTY	PART#
Standard service tool kit	1	
Fluke 87	1	
AMP Clamp-on AC Probe	1	2338920
Dale 600 Leakage Current Meter	1	46-328406G1
Dale 600 Leakage Current Meter Ultra Sound Kit	1	46-285652G1
Dale 600 Leakage Current Meter European 220 VA	1	46-328406G2

3.2 Consumables

No consumables required.

3.3 Replacement Parts

No replacement parts required.

3.4 Safety



WARNING



POTENTIAL FOR SHOCK

**GROUND WIRES WILL HAVE GROUND CURRENT PRESENT WITH POWER “ON”.
FOLLOW APPROPRIATE SAFETY PROCEDURES FOR WORKING WITH AN
ENERGIZED SYSTEM.**

3.5 Required Conditions

No required conditions.

Section 4.0 Procedure

4.1 Patient Touch Leakage Current Test Procedure

- 1.) See [Figure 13-1](#) for the usage of the Dale 600 meter to measure leakage current.

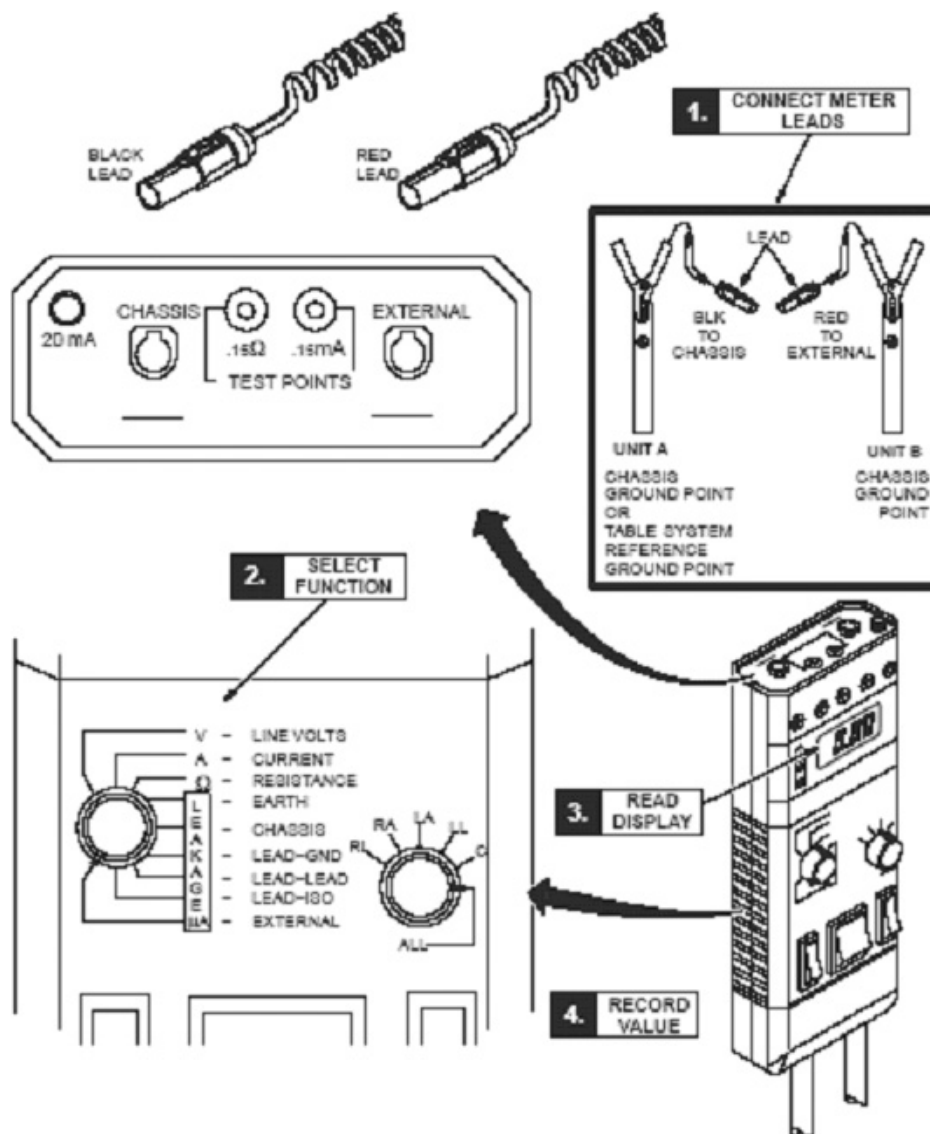


Figure 13-1 Using Dale 600 to Measure Leakage Current

- 2.) Plug the Dale 600 Meter into the outlet nearest the Table. (Table base covers must be removed to use table outlet.) Confirm the outlet is correctly wired per the three LED indicators on the Meter.
- 3.) Connect the black lead to the chassis plug and the red lead to the external plug on the top of the Dale 600 meter. Set the function switch to external and display readings will be in micro-amps.

The black lead will be connected to the table base ground buss and the red lead is then connected to the devices (components) under test.

- 4.) Leakage current cannot exceed the following and is tested with power on:
 - a.) **Critical care areas (invasive) - 10 μ a**
 - b.) General care areas - 20 μ a
 - c.) Not intended for patient area - 50 μ a
- 5.) Testing must be completed between the system reference ground point (table base) to unit reference ground points (i.e. gantry and table, see illustration 1).
- 6.) Test all conductive surfaces and components within patient reach or within 6 feet or 1.9m around table and 7.5 feet or 2.3m above. Measure at table maximum travels. On some site wall outlet cover plates and sinks may become an issue.
- 7.) Test all Optional components such as in room monitors, injector, overhead monitor suspensions, and table options.

Record these results in the data sheet. To be completed at install and during PM inspections.

Gnd Bus to	Install <10 μ A	6-month PM <10 μ A	6-month PM <10 μ A
Table gnd points			
Gantry gnd points			
Injector gnd points			
Boom in Room			
Monitors			
Sinks			

- 8.) Test conditions: Table max elevation Min - Max travel; Patient face up and face down, and head first feet first. Table min elevation 7.2' patient access to all conductive surfaces Including: IV poles, any smart step monitors and stands, table bearing rails and all other conductive surfaces. Be aware of Static discharge: from scan window, keypads, display, and touch pads or other plastic surfaces.
- 9.) Procedure Hints:
 - a.) Look for items that have abnormal measurements, high or low.
 - b.) These could indicate mis-wiring, loose connections, or poor connections due to corrosion, painted surfaces, etc.
 - c.) High leakage could indicate a wiring error such as a neutral connected to the ground.
 - d.) Fluctuating ground currents could indicate a short, poor connection, or facilities ground problem causing leakage currents from other areas of the facility to flow through our system grounds.

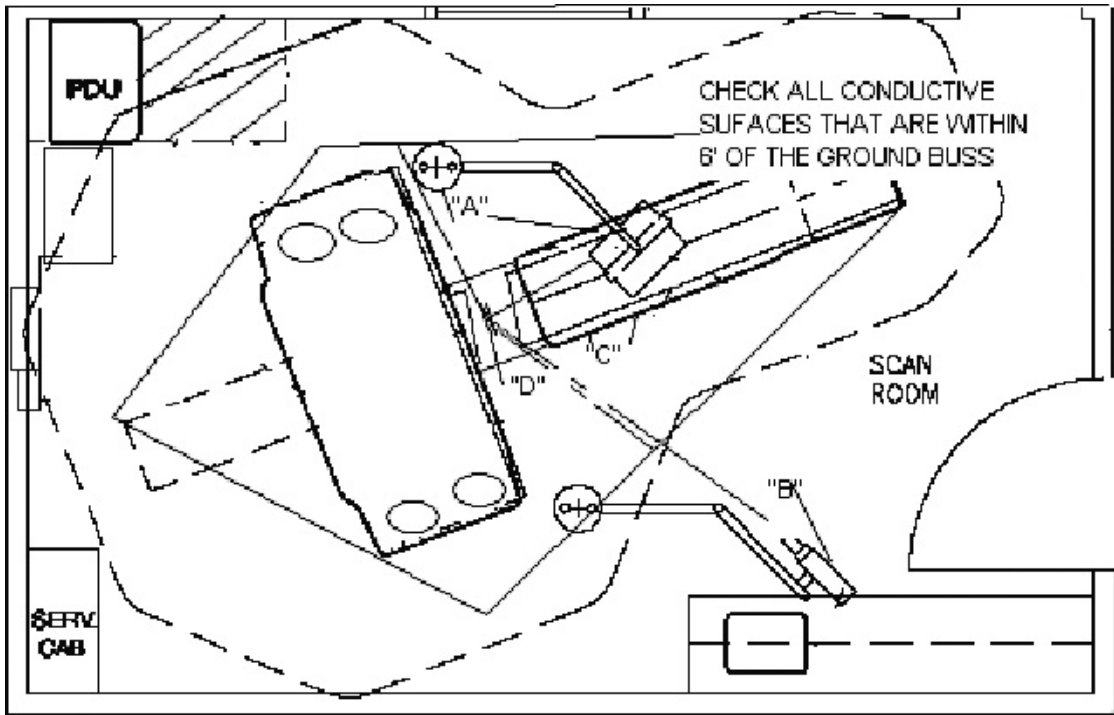


Figure 13-2 CT Leakage Current Map

4.2 Ground Resistance Checks Procedure (done during mechanical install)

Use Fluke 87 to measure ground resistance.

- 1.) Use a Fluke 87 or equivalent RMS Meter to verify the presence of less than 0.5 ohm of resistance between each of the following points:

FROM	TO	
PDU Ground Bus	Vault Ground	☐ Check box when complete
PDU Ground Bus	Table/Gantry raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Table Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console Chassis	☐ Check box when complete
All Display or Computing Options (if any)	Operator Console Chassis	☐ Check box when complete

Table 13-1 Resistance Verification - Site Ground

- 2.) Connect the meter leads to the Fluke 87 meter.
3.) Set the function switch to resistance ohms.
4.) The black and red leads will be connected per Table 13-1.

- 5.) All ground resistance testing is done with power off. Ground current will cause errors in the resistance readings.
- 6.) Use lockout/tag out procedures.
- 7.) Maximum ground resistance is 0.5 ohms impedance.
- 8.) Using the table footswitch ground bar as a reference. Check the ground resistance to all 5 system grounds.

Record the measurements in the datasheet:

SYSTEM GROUND BUS	TYPICAL READING	SITE DATA
Gantry Ground Stud	0.5 Ohms	
Console Ground Stud	0.5 Ohms	
PDU Ground Stud	0.5 Ohms	
Table Ground Stud	0.5 Ohms	

- Hints:
- Look for undersized ground wires and repair.

- Ground resistance readings should not fluctuate. If so this indicates an intermittent

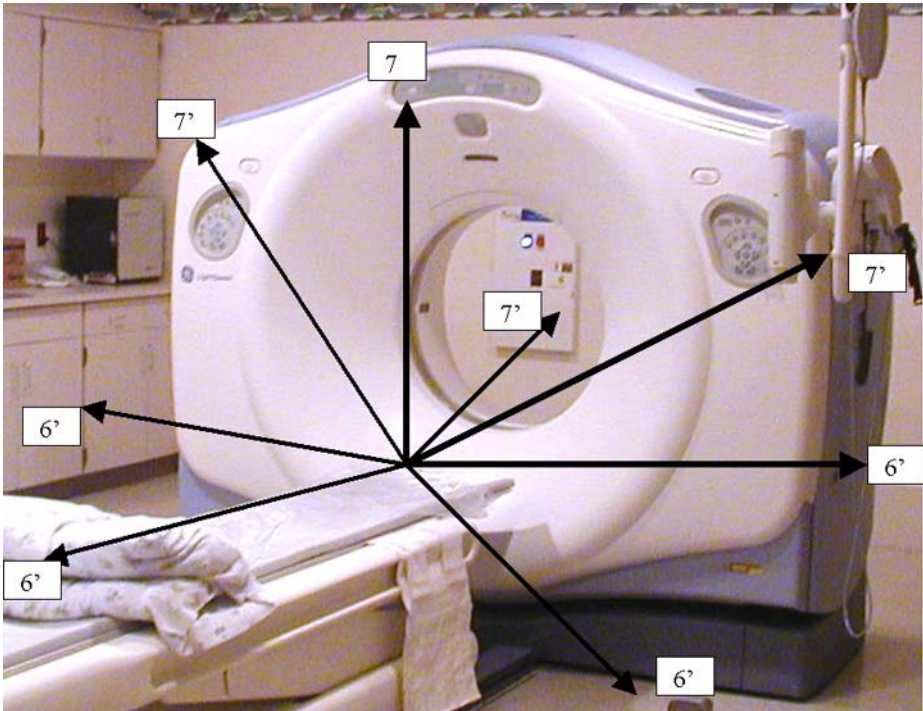


Figure 13-3 Measurement Points

Section 5.0

Finalization

No finalization required.

Chapter 14

FORM 4879 (Data Record for LightSpeed 3.X)



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss, please do the following:

- Record data collected from procedures into Form F4879 if directed, located in this chapter.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Site History

SYSTEM HISTORY

Where Installed _____

City _____ State _____

Customer's Telephone Number _____ Room No. _____

System I.D. No. _____

Operator's Console (Master Control)

Model Number _____ Serial Number _____

System Gantry

Model Number _____ Serial Number _____

Dates of Tests _____ Performed by _____

Table 14-1 Data Record

TEST EQUIPMENT HISTORY

Test Equipment Used (If Applicable)				
Name	Manufacturer	Model Number	Serial Number	Calib. Due
HV Bleeder				
Oscilloscope				
DMM				

Table 14-2 Test Equipment Used

COMPONENT TRACEABILITY

The Component Locator Installation Report has been completed and sent to the Product Locator File.

Certified Component Changes				
Equipment	Model Number	Serial Number	Install Date	Exp. Ctr. Reading
Old Tube Casing				
Old Tube Insert				
New Tube Casing				
New Tube Insert				
Old Detector				
New Detector				
Old Collimator				
New Collimator				
Old Anode Tank				
New Anode Tank				
Old Cathode Tank				
New Cathode Tank				

Table 14-3 Certified Component Changes

Section 2.0 System Tests

MA METER VERIFICATION - HHS REQUIREMENT

Initials: _____

Circuit ON		
	A/D	Range
Anode mA		15 - 25
Cathode mA		15 - 25
DVM - A/D = Delta Compare Delta to Limits		

Table 14-4 mA Meter Verification (Table 1)

AUTO MA CAL

☐ Check box when complete

The MX200 tube uses a metal/ceramic frame, which means some electrons hit the target and actually bounce off, and are then captured by the frame. This causes a frame current, which results in an anode mA measurement of slightly less than the cathode mA, typically 5 to 10% less. Because the cathode mA is the true indicator of radiation output, the CT QX/i system controls cathode mA rather than anode mA.

OPERATOR INDICATORS - HHS REQUIREMENT

Initials: _____

X-Ray Indicators on Front and Back of Gantry Illuminate During X-Ray

SIGNAL TO NOISE TEST (X-RAY VERIFICATION)

Initials: _____ ☐ Check box when complete

CT NUMBER CHECK

All CT N numbers for all calibrated techniques fall within 0 ± 1.5 HU. All Values 0 ± 1.5 HU? Yes: ☐

TUBE Z-AXIS ALIGNMENT (FORMERLY TUBE PLANE OF ROTATION (POR))

☐ Check box when complete

Z-Align Used: Adjusted In Specification. (Go to Detector Z-Axis, step A),

-OR-

Tube Plane of Rotation (POR) using Film. XF - XR = $1.8\text{mm} \pm 0.3\text{mm}$

BEAM ON WINDOW

☐ Check box when complete

- 1.) If you used Z-Align (**Future**) for the Z-Axis Alignment, no additional adjustment is necessary
-OR-

Run BOW on a cold tube. If you have to check BOW after completing the system alignments, let at least three hours elapse from the last scan, to let the tube completely cool.

- The channel should equal $-1.5\text{mm} \pm 0.25\text{mm}$ on initial checks.
- The channel should equal $-1.5\text{mm} \pm 0.50\text{mm}$ on verification checks.
- This program computes the position of the detector in three places: Right, Center and Left

- 2.) Record the displayed position values in the following table:.

Right/Low	Center/Medium	Left/High

Table 14-5: Beam On Window

ISOCENTER

☐ Check box when complete

Average of Large and Small Focal Spots. Spec: 389.75 ± 0.02 channels

CENTER BODY FILTER (CBF)

☐ Check box when complete

Adjust CBF Value. Spec: Target Value ± 0.2 channels

COLLIMATOR APERTURE CHECK - HHS REQUIREMENT

Initials: _____ ☐ Check box when complete

Aperture	Center	Spec - Center only
Small		$3.783 \pm 5\%$
Large		$5.9 \pm 5\%$

Table 14-6 Collimator Aperture Check

X-RAY BEAM LOCATION - HHS REQUIREMENT

Initials: _____

TOMO PLANE INDICATION ACCURACY

Refer to the system service manual, and verify that the relationship of the alignment lights to the x-ray beam meets compliance.

- Internal Lights (1mm)
- External Lights (1mm)
- All alignment lights remain visible with room lights at maximum settings.

IMAGE CALIBRATION

☐ Check box when complete

Detailed Calibrations completed and CT N Number adjustment completed. The Number of Phantom Cals vary with site-specific scanning protocols.

IMAGE QUALITY CHECK - HHS REQUIREMENT

Initials: _____

48CM POLY PHANTOM

48cm/L Scan parameters: 5mm (4i)/ 340mA /1 sec/16 Images

Exam/Series/ Image	AvXc	AvXo	AvXo - AvXc	AvSDo	Comments	Scan Tech Signature/ Date
___/___/ 1, 5, 9, 13						
2, 6, 10, 14						
3, 7, 11, 15						
4, 8, 12, 16						
SPECS	---	---	±8.5	<70.0	---	---

Table 14-7 48cm Poly Phantom

*If your AvSDo fails, check to make sure you did not scan this technique at 1mm.

Box Size: 45 x 45 pixels

Center Coordinates: 256, 256

Outside Coordinates: 256, 60 452, 256256, 45260, 256

Image Acceptance/Date: _____

Certified Image Reviewer: _____

ARTIFACT LIMITS:

Rings:	Smudge:
48/L; 30 to 36 counts	48/L; 6.8 counts
1.25mm	1.25mm; 12.0 counts
Band:	Streaks:
8.0 counts	4.0 counts
Band Radius:	Center Artifact:
0 to 23.5cm	N/A
Clump:	Center Artifact:
48/L; 3.0 SIGMA	1.25mm; 14.0 counts
1.25mm; N/A	
Center Spot: N/A	

Table 14-8 Artifact Limits

IMAGE QUALITY CHECK - HHS REQUIREMENT

Initials: _____

20CM QA#1 PHANTOM

QA#1 scan parameters: S/10mm (2i)/120kV/260mA/1 sec/4 images. Use this scan data for Bone Retro. Record data on **Form 4879**.

Exam/Series/Image	MTF	MTF 2 slice avg.	Contrast Scale	Comments /Artifacts	Scan Tech Signature/Date
____/____/1		MTF1 + MTF3 ÷ 2 =			
3					
2		MTF2 + MTF4 ÷ 2 =			
4					
SPECS	---	0.65 to 1.0	110.0 to 130.0	---	---

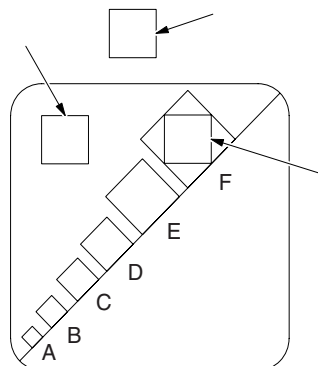
Table 14-9 QA#1 Data (Prospective Images)

QA#1 scan parameters: S/10mm (2i)/120KV/260mA/1 sec/4 images Bone Retro/Display FOV = 15cm Record data on **Form 4879**

Exam/Series/Image	Line Patterns Visible	Comments/Artifacts	Scan Tech Signature/Date
____/____/1			
2			
3			
4			
SPECS	B, C, D, E, F	---	---

Table 14-10 QA#1 Data (Retrospective Images)

Box Size: 17 x 17 pixels



Contrast Scale = mean of B—mean of C.

SD = SD of ROI Box over F slits.

SD = (SD + SC) /2

Modulation = _____

MTF = 2.2 (Modulation/Contrast Scale)

IMAGE QUALITY CHECK - HHS REQUIREMENT

Initials: _____

20CM QA#2 PHANTOM

QA#2 scan parameters: S/10mm (2i)/120KV/260mA/1 sec/Peristaltic OFF

Scan from S40.0 to S50.0 - Record results on **Form 4879**

Exam/Series/ Image at location S40.0	Visible Hole View w/ Window 20	Contrast Factor	Comments/ Artifacts	Scan Tech Signature/ Date
____/____/1				
3				
5				
7				
SPECS	See Table, below	2.0 to 12.0	- - -	- - -

Table 14-11 QA#2 Data (Part 1)

Box size: 49 x 9 pixels

Contrast Factor	Visible Number of Holes: Lower Limit*	Visible Number of Holes: Upper Limit*	Smallest Visible Hole: Size
2.00 to 3.99	2	5	7.5mm
4.00 to 7.99	3	5	5.0mm
8.00 to 12.00	4	5	3.0mm

Table 14-12 QA#2 Data (Part 2)

*Required number of visible holes depends on the contrast factor.

2 out of the 4 scans taken must meet this specification.

Image Acceptance/Date: _____

Certified Image Reviewer: _____

FOR REFERENCE ONLY:

Use the manual method for calculating Contrast Factor.

ROI D

ROI E				
10mm	7.5mm	5.0mm	3.0mm	1.5mm

ROI D 49 X 9 pixels x=256, y=200
ROI E 49 X 9 pixels x= 256, y=235
Contrast Factor = Mean E – Mean D

IMAGE QUALITY CHECK - HHS REQUIREMENT

Initials: _____

20CM QA#3 PHANTOM

QA#3 scan parameters: S/10mm 2i/120KV/260mA/1 sec/8 Images

Scan from S55.0 to S65.0

Record data on **Form 4879**.

Exam/Series/ Image	AVXc 4 scan avg.	Std Dev 4 scan avg.	AvXo - AvXc	Comments/ Artifacts	Scan Tech Signature/ Date
____/____/ 1,3,5,7					
2,4,6,8					
SPECS	0 ±3.0	2.6 to 3.4	±3.0	---	---

Table 14-13 QA#3 Data

Note: **The average value of each row in [Table14-13](#) must meet the specifications technique listed at the bottom of the corresponding column.**

Box Size: 31 x 31 pixels at Coordinate (256, 256)

Center Coordinates: (256, 256)

Outside Coordinates: (256, 95), (417, 257), (256, 417), (95, 256)

Visually Inspect images for rings and bands

Visual results acceptable?. Yes: ☐

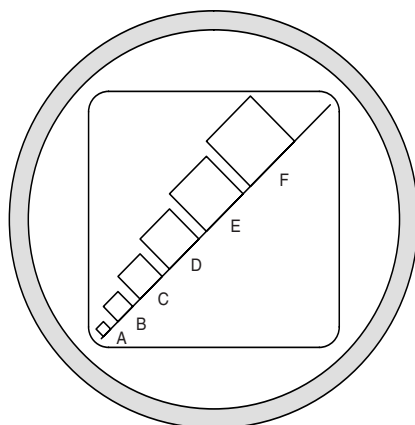
Artifact Limits:

- **Rings:** 4.0/4.8 cts
- **Band:** 2.8 cts
- **Band Radius:** 0-8.5 cm
- **Clump:** N/A
- **Center Spot:** N/A
- **Center Artifact:** 3.5 Std. Dev.
- **Smudge:** 2.2 counts 1mm; 4.0 counts
- **Streaks:** 4.0 cts

Image Acceptance/Date: _____

Certified Image Reviewer: _____

UNUSUAL ALIASING, OTHER ARTIFACTS



Visual inspection for unusual aliasing, and other artifacts. Visual results acceptable? Yes: ☐

Chapter 15

Install Completion Checklist

Complete the following checklist:

Section 1.0 CT System

- ☐ Functional tests completed.
- ☐ Power and ground audit completed.
- ☐ Seismic kit installed (if needed in your area).

Section 2.0 CT Options

- ☐ Injector functional tests completed.
- ☐ AWW functional tests completed.
- ☐ Filming/Camera/DASM functional tests completed.
- ☐ UPS functional tests completed.
- ☐ Network items installed and functional tests completed.
- ☐ Verify that all customer software options are installed and functional.
- ☐ Teleradiology Connections (Refer to [Section 8.0 on page 276](#))

Section 3.0 System Cleanup

- ☐ Clean system with supplies shipped with your system.
- ☐ Touch-up nicks and scratches (as needed).
- ☐ Install pads and stickers (as needed).
- ☐ Store customer materials appropriately.
- ☐ Store service documents, tools, and materials appropriately.

Section 4.0 Dollies Return

- ☐ Confirm that dollies have been picked up. If not, complete the necessary documentation.

Section 5.0

Install Completion Material Checks

- ☐ Complete all HHS documentation. Complete Install CQAs as needed.
- ☐ Complete all dispatching activities (03-04-10 codes).
- ☐ Complete customer acceptance checks/test.
- ☐ Complete system transfer and notify appropriate GE personnel.
- ☐ Communicate with the customer and address installation issues.

Appendix C

Additional Characterization Procedures

Section 1.0 Tilt Characterization

Tools Required: Hex (metric) set
Flat blade screw driver



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



2.) Click on the UTILITIES icon.

3.) Select INSTALL.

4.) Select TABLE GANTRY INTEGRATION.

5.) Select CHARACTERIZATION.

6.) Select TILT.

7.) Follow the Tilt Characterization instructions on the screen.

The screenshot shows the 'Mechanical Characterization Tool' window. The title bar reads 'Mechanical Characterization Tool'. The menu bar has 'File'. The main window is titled 'Tilt Characterization: Pot Input'. On the left, there is a sidebar with buttons: 'Hardware Reset', 'Characterize' (with a sub-menu showing 'Tilt', 'Table Elevation', and 'Cradle'), 'Data View' (with sub-menus 'Charact Limits', 'View Values', and 'View Log'). The main area is titled 'Tilt Characterization' and contains the following text: 'Read the Tilt Limit Label inside the Gantry and type the tilt limits in degrees in the fields below exactly as they appear on the label. Example: Backward(-) Position -30.20, Mid Position 3.45, Forward(+) Position 29.56. Double check the values you typed then press Accept.' Below this text are three input fields: 'Backward(-) Position', 'Mid Position', and 'Forward(+) Position'. At the bottom right is an 'Accept' button.

Figure C-1 Gantry Tilt Characterization Screen

Section 2.0

Elevation Characterization

Tools Required: None

2.1 Prerequisite

The table to gantry MUST be level before performing this procedure. The following tools are required:

- Metric tape measure
- Flat blade screwdriver
- Metric hex key set
- 9" level

2.2 Procedure

- 1.) Remove upper right and lower right table covers.
- 2.) Locate the small L-shaped measurement block, stored on top of the bigger measurement block.
 - a.) Remove the small L-shaped measurement block from its storage position.
 - b.) Install it in its characterization position, at the right rear of the table.
 - c.) Move the measurement plate, on the right side of table, to the **out** position.
 - d.) Tighten the mounting screws for both plates.

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the UTILITIES icon.
- 3.) Select INSTALL.
- 4.) Select TABLE GANTRY INTEGRATION.
- 5.) Select CHARACTERIZATION.
- 6.) Select TABLE ELEVATION.
- 7.) Follow the Elevation Characterization instructions displayed on the screen. *Important - Take care to measure the exact elevation distances. Incorrect measurement results in Elevation Characterization failure.* Table Low = 374.1 mm. Table High = 849.1 mm.

Refer to [Figure C-2](#). Measure the elevation distance as vertically as possible. Measure between the bottom surface of the each measurement block and the center of the rear leg upper pivot pin.

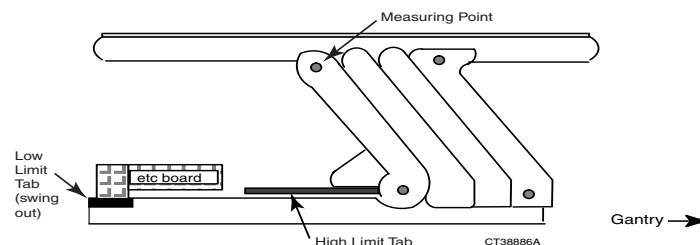


Figure C-2 Measure Elevation Distance

- 8.) Return the measurement plates to the storage position.

Section 3.0

Cradle Characterization

Tools Required: Hex (metric) set
Metric tape measure



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



2.) Click on the UTILITIES icon.

3.) Select INSTALL.

4.) Select TABLE GANTRY INTEGRATION.

5.) Select CHARACTERIZATION.

6.) Select CRADLE.

7.) Follow the cradle characterization instructions displayed on the screen.

Note:
A table/gantry
reset may be
required.

If the display flashes "888", then you must reset the table before proceeding. To reset the table, lower it to its lowest position and then raise it to its maximum position.

a.) In the alignment light procedure (3rd and 4th screens), use the BLACK DOT on the cradle right-side bumper.

Elevate the table to at least 40mm from isocenter, so the black dot reaches the external scan plane alignment light *before* the cradle reaches its home position.

b.) In the end-of-cradle position procedure (5th screen), use the same BLACK DOT on the cradle right-side bumper.

Note: Do not use buttons to drive table cradle.

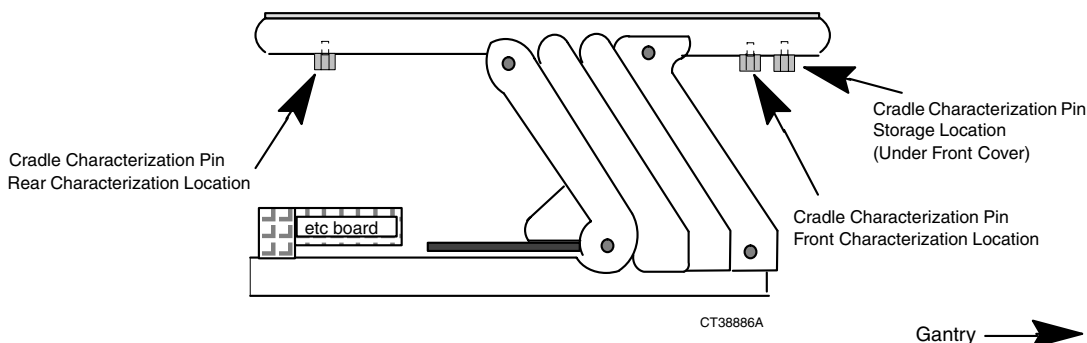


Figure C-3 Cradle Characterization Pin Locations



GE MEDICAL SYSTEMS

**GE MEDICAL SYSTEMS-AMERICAS: FAX 262.312.7434
3000 N GRANDVIEW, WAUKESHA, WI, 53188 U.S.A.**

**GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE**

**GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006
SINGAPORE**